

Mathematical Thinking

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Contents

Worksheet 1.	Fibonacci Numbers	5
Worksheet 2.	The Thue–Morse Sequence	7
Worksheet 3.	Impartial Combinatorial Games	11
Worksheet 4.	Symmetries	15
Worksheet 5.	Platonic Solids	19
Worksheet 6.	Prime Numbers	23
Worksheet 7.	The Euclidean Algorithm	25
Worksheet 8.	Pascal’s Triangle	29
Worksheet 9.	Counting Using Binomial Coefficients	33
Worksheet 10.	The Pigeonhole Principle	35
Worksheet 11.	Permutations	37
Worksheet 12.	Permutation Statistics	41
Worksheet 13.	Invariants and Monovariants	43
Worksheet 14.	Eulerian Paths and Circuits	47
Worksheet 15.	Vertices, Edges, and Faces	51
Worksheet 16.	Identification Spaces and Surfaces	55
Worksheet 17.	Modular Arithmetic	59
Worksheet 18.	Vieta’s Formulae	63
Worksheet 19.	Arithmetic Functions	65
Worksheet 20.	The Peano Axioms	67
Worksheet 21.	Basic Infinite Set Theory	71

Worksheet 22.	The Game of Set	75
Worksheet 23.	A Taste of Knot Theory	79
Worksheet 24.	Introduction to Game Theory	85
Worksheet 25.	Gerrymandering	89
Worksheet 26.	Probability Paradoxes	93
Worksheet 27.	Inequalities of Means	97
Worksheet 28.	Deterministic Finite Automata and Regular Languages	101
Worksheet 29.	Voting Systems	105
Worksheet 30.	The Koch Curve and Its Friends	111
Worksheet 31.	Finite Differences	117
Worksheet 32.	The Stern–Brocot Tree	119
Worksheet 33.	Chromatic Polynomials	121
Worksheet 34.	Rotation Numbers and the Doubling Map	125
Worksheet 35.	Sharkovskii’s Theorem	129
Worksheet 36.	Billiards	133
Worksheet 37.	The Cubic and Quartic Formula	137
Worksheet 38.	When Unique Factorization Fails	141
Worksheet 39.	Tropical Algebra	143
Worksheet 40.	Complementary Sequences	147
Worksheet 41.	Inversion	149
Worksheet 42.	Mascheroni Constructions	153
Worksheet 43.	Information Entropy	157
Worksheet 44.	Large Numbers	161
Bibliography		163

WORKSHEET 1

Fibonacci Numbers

The *Fibonacci numbers* are the sequence of numbers starting

$$0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, \dots$$

Each one is the sum of the previous two numbers. The sequence starts with the 0th Fibonacci number, and we write F_n for the n^{th} Fibonacci number. Thus we have $F_0 = 0$, $F_1 = 1$, $F_2 = 1$, $F_3 = 2$, and so on. The rule that each one is the sum of the previous two can be expressed by saying that

$$(1.1) \quad F_n = F_{n-1} + F_{n-2}$$

for all $n \geq 2$.

PROBLEM 1.1. What is the first Fibonacci number that is greater than 1000?

PROBLEM 1.2. If you know F_n and F_{n+1} , how do you use the rule (1.1) to compute F_{n-1} ? Compute F_{-n} for $1 \leq n \leq 10$. What do you notice?

PROBLEM 1.3. For which values of n is F_n even? Can you explain why?

PROBLEM 1.4. For which values of n is F_n a multiple of 3? Can you explain why?

PROBLEM 1.5. Show that for every positive integer m , there are infinitely many Fibonacci numbers that are divisible by m .

The *Lucas numbers* are defined similarly to the Fibonacci numbers, with each Lucas number being the sum of the previous two. However, the initial conditions are different: we have $L_0 = 2$ and $L_1 = 1$. The first few Lucas numbers, starting with L_0 , are

$$2, 1, 3, 4, 7, 11, 18, 29, 47, 76, \dots$$

PROBLEM 1.6. Show that no Lucas number is divisible by 5. What makes the Lucas numbers behave differently from the Fibonacci numbers, as in problem 1.5?

PROBLEM 1.7. How many ways are there to tile a $1 \times n$ rectangle using 1×1 and 1×2 rectangles? All the possibilities for $n = 4$ are shown in Figure 1. (Start by making a table of the answers for small values of n until you recognize the pattern.)



Figure 1. All ways of tiling a 1×4 rectangle with 1×1 and 1×2 rectangles.

PROBLEM 1.8. How many sequences of 0's and 1's are there of length n , such that there are no two consecutive 1's? When $n = 4$, these sequences are

0000, 0001, 0010, 0100, 0101, 1000, 1001, 1010.

PROBLEM 1.9. For $1 \leq n \leq 10$, compute $\frac{F_{n+1}}{F_n}$. What is going on with these ratios? Do they appear to be approaching a limit? Can you explain why? What about $\frac{L_{n+1}}{L_n}$, the ratio of consecutive Lucas numbers?

PROBLEM 1.10. Let $G_n = F_1 + F_2 + F_3 + \cdots + F_n$. Can you find a formula for G_n ? Explain why this formula works. What about $M_n = L_1 + L_2 + L_3 + \cdots + L_n$?

PROBLEM 1.11. What if we add up only every other Fibonacci number? That is, let $H_n = F_1 + F_3 + F_5 + \cdots + F_{2n-1}$. What is H_n ?

PROBLEM 1.12. What about $F_1^2 + F_2^2 + F_3^2 + \cdots + F_n^2$?

PROBLEM 1.13. What about $F_{n-1}F_{n+1} - F_n^2$?

PROBLEM 1.14. Explain what's going on in Figure 2: where did the missing square go?

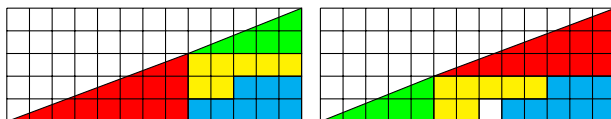


Figure 2. Where did the missing square go?

Further reading. For more combinatorial identities involving Fibonacci numbers, see [BQ03, Chapters 1 and 2]. For even more about the Fibonacci numbers, including instances of Fibonacci numbers in nature, see [Kos01, Chapters 1–3]. For number-theoretic properties of the Fibonacci numbers, see [Sil13, Chapter 39].

WORKSHEET 2

The Thue–Morse Sequence

The Thue–Morse sequence is a remarkable sequence of 0's and 1's that arises in many apparently quite different ways. Let's start with the definition, in terms of binary numbers. Given a nonnegative integer n , we can write it in binary, in a unique way. For instance, the binary representation of 22 is 10110, which means that $22 = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$. Given a nonnegative integer n , we define t_n to be

$$t_n = \begin{cases} 1 & \text{the binary representation of } n \text{ has an odd number of 1's,} \\ 0 & \text{the binary representation of } n \text{ has an even number of 1's.} \end{cases}$$

The sequence t_n , starting with $n = 0$, begins

$$(2.1) \quad 0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 0, 1, 1, 0, \dots$$

The sequence t_n is called the *Thue–Morse sequence*.

DEFINITION. We say that n is an *evil* number if $t_n = 0$, and n is an *odious* number if $t_n = 1$.

Let's eliminate the commas from (2.1) for typographical purposes. If we cut off the sequence after 1, 2, 4, 8, 16, ... terms, we get

$$\begin{aligned} S_0 &= 0 \\ S_1 &= 01 \\ S_2 &= 0110 \\ S_3 &= 01101001 \\ S_4 &= 0110100110010110, \end{aligned}$$

and so on.

Given a sequence S of 0's and 1's, let $\overline{S}$ be the sequence obtained from S by replacing all the 0's with 1's, and all the 1's with 0's. For instance, we have

$$\overline{011010111} = 100101000.$$

Given two binary sequences S and T , let ST denote the concatenation: write down all the digits of S , followed by all the digits of T .

PROBLEM 2.1. Show that, in the decomposition above, we have $S_{n+1} = S_n \overline{S_n}$, the concatenation of S_n and $\overline{S_n}$. Write this relation in terms of various values of t_n .

PROBLEM 2.2. A sequence $S = s_0 s_1 \cdots s_n$ is called a *palindrome* if $s_j = s_{n-j}$ for all j , i.e. it reads the same forward and backward. Is S_n sometimes a palindrome? For which values of n ? Determine all values of n for which S_n is a palindrome, and explain why your answer is correct.

PROBLEM 2.3. Start with a 0. Then, at every stage, replace each 0 with a 01 and each 1 with a 10. Write out the first five stages of this process. What do you notice? Can you explain what is going on?

PROBLEM 2.4. Suppose that, in the previous problem, you instead replace each 0 with 01, and each 1 with 0. What do you get? How many digits are there in the n^{th} stage of this process?

PROBLEM 2.5. Express t_{2n} in terms of t_n .

PROBLEM 2.6. Express t_{2n+1} in terms of t_n .

PROBLEM 2.7. What happens when we look at only the even terms of the Thue–Morse sequence? How about only the odd terms?

PROBLEM 2.8. Suppose $0 \leq j < 2^n$. Express t_{2^n+j} in terms of t_j .

PROBLEM 2.9. Show that if $t_n = t_{n+1}$, then n must be odd.

PROBLEM 2.10. Show that there is no nonnegative integer n such that $t_n = t_{n+1} = t_{n+2}$.

PROBLEM 2.11. Show that among any five consecutive terms of the Thue–Morse sequence, there must be two consecutive terms that are equal.

We say that a sequence S is *overlap-free* if there is no consecutive subsequence of S of the form $cxcxc$, where c and x are nonempty sequences. For instance, the sequence

01001001010111

is not overlap-free: for instance, we can let c and x be the red and blue subsequences, respectively:

01**00**1**00**1**0**10111.

In the next few problems, we will show that the Thue–Morse sequence is overlap-free.

PROBLEM 2.12. Show that if a binary sequence contains an overlap $cxcxc$, then it contains an overlap of the form $ayaya$, where a is a single digit.

If we have an overlap $ayaya$, where ay has length p for some positive integer p , then we have $ayaya = t_n t_{n+1} \cdots t_{n+2p}$. We say that the overlap has *period* p .

PROBLEM 2.13. Show that the Thue–Morse sequence cannot have an overlap with odd period. (You will need to use several of the previous problems.)

PROBLEM 2.14. Show that if the Thue–Morse sequence were to have an overlap with even period $p = 2q$, then it must also have an overlap with period q . (What happens when we take every other term of the overlap?)

PROBLEM 2.15. Show that the Thue–Morse sequence is overlap-free.

PROBLEM 2.16. In CHESS, if a position is repeated three times, then either player can declare the game to be drawn. A previous version of this rule is that if a sequence of moves is repeated three times in a row, then the game can be declared drawn. Show that this previous version allows games to last forever. (We ignore the possibility of draw by 50-move rule in this problem.)

Further reading. See [AS99] for a survey paper including many properties of the Thue–Morse sequence.

WORKSHEET 3

Impartial Combinatorial Games

Consider the game of MATCHES.¹ In this game, there is a pile of matches, with n matches in the pile initially. Two players take turns removing some matches from the pile, and the player to move can remove 1, 2, or 3 matches on a turn. The player who is not able to make a move (because the pile is gone) loses the game.

PROBLEM 3.1. For $1 \leq n \leq 20$, determine whether the first or second player has a winning strategy at MATCHES, assuming that both players play optimally. What about for all n ? Can you explain your observation?

PROBLEM 3.2. What happens if the person who takes the last match loses rather than wins?

REMARK. In combinatorial game theory, the rule that the player who is not able to make a move loses is called the *normal-play convention*, whereas the rule that the player who is not able to make a move wins is called the *misère-play convention*. Most of the time, the misère-play games are much more difficult to study than normal-play games, but in the case of MATCHES, both have simple analyses.

PROBLEM 3.3. Consider a generalization of MATCHES, where a player can remove up to k matches on each turn. Now, for which values of n does the first player have a winning strategy in normal play? What about in misère play? Explain why your answers are correct.

We call these types of games *subtraction games*. If S is a set of positive integers, we write SUBTRACTION_S for the subtraction game where the next player can remove any number of matches from S . For instance, the original game is $\text{SUBTRACTION}_{\{1,2,3\}}$.

PROBLEM 3.4. Analyze the games of $\text{SUBTRACTION}_{\{1,2,4\}}$ and $\text{SUBTRACTION}_{\{1,3,4\}}$.

Suppose we fix a subtraction set S and consider the game SUBTRACTION_S . For a pile size of n , we say that n is an $\mathcal{N}$ position if

¹It is a common convention to use SMALL CAPS for the names of games in combinatorial game theory.

the next player to move has a winning strategy with optimal play, and a $\mathcal{P}$ position if not. (The $\mathcal{P}$ stands for “previous,” since we imagine the game has been going on for a while, so the player other than the next player must have been the previous player.) We can consider a sequence $a_0, a_1, a_2, \dots$, where $a_n = \mathcal{N}$ or $\mathcal{P}$, depending on whether n is an $\mathcal{N}$ position or a $\mathcal{P}$ position, respectively.

DEFINITION. Given a sequence $a_0, a_1, a_2, \dots$, we say that it is *periodic* if it repeats indefinitely, i.e. there exists a positive integer p such that $a_{n+p} = a_n$ for all $n \geq 0$. We say that it is *preperiodic* if it eventually repeats indefinitely, i.e. there exists a positive integer p and a nonnegative integer n_0 such that $a_{n+p} = a_n$ for all $n \geq n_0$.

PROBLEM 3.5. Show that for any finite set S of positive integers, the sequence a_n corresponding to SUBTRACTION_S is preperiodic.

PROBLEM 3.6. Find a finite subtraction set S for which the sequence a_n is not periodic.

For a preperiodic sequence a_n , the *period* is the smallest positive integer p such that there exists a nonnegative integer n_0 such that $a_{n+p} = a_n$ for all $n \geq n_0$.

PROBLEM 3.7. Suppose $S = \{a, b\}$. In terms of a and b , determine the period of SUBTRACTION_S . (Try a bunch of examples before trying to guess the answer!)

REMARK. The answer to the previous problem is unknown in general if $S = \{a, b, c\}$ contains three numbers.

PROBLEM 3.8. Consider the following game between two players:² we start with a pile of n stones. The first player can remove as many stones as desired, but at least one and not the entire pile. On subsequent turns, the next player can only take at most as many stones as the previous player took. The player who is unable to make a move (usually because there are no stones left, but also at the beginning when $n = 1$) loses. For which values of n does the first player have a winning strategy?

PROBLEM 3.9. Answer the same question as in the previous problem, but now each player can take up to twice as many stones as the previous player took.

PROBLEM 3.10. Consider the following game, known as WYTHOFF’S GAME: there are two piles of stones. The player to move can either remove as many stones as desired from one pile, or can remove an equal number of stones from both piles. What are the $\mathcal{N}$ and $\mathcal{P}$ positions of

²I’m not going to tell you the name of this game, because the name gives away the answer.

this game? (You might not be able to give a complete answer to this problem, but figure out as much as you can!) Sequences arising from this game will show up again in Worksheet 40.

The game of NIM is played as follows: there are several piles of stones, and the player to move can take as many stones as desired, but only from one pile per turn. We represent a position with k piles, of size $b_1, b_2, \dots, b_k$ as $(b_1, b_2, \dots, b_k)$. While there is a beautiful general winning strategy for NIM, coming up with it on your own is not an easy task. So instead, we'll look at a few special cases.

PROBLEM 3.11. If there is only one pile of size b_1 , when is this game an $\mathcal{N}$ position, and when is it a $\mathcal{P}$ position?

PROBLEM 3.12. Consider the NIM game (b_1, b_2) with two piles. When is this an $\mathcal{N}$ position? When is it a $\mathcal{P}$ position?

PROBLEM 3.13. Find at least 10 triples (b_1, b_2, b_3) of positive integers with $b_1 < b_2 < b_3$ such that (b_1, b_2, b_3) is a $\mathcal{P}$ position. Can you find triples of many different types? What are some general patterns you can come up with?

Further reading. See [ANW19, §2.3 and Chapter 7] to see more examples of impartial games and the general theory for how to play them well.

WORKSHEET 4

Symmetries

Consider a square with vertices at $(\pm 1, \pm 1)$. If we reflect it about the x -axis, we get the same square again. Similarly, if we rotate it by 90° counterclockwise, we get the same square again. However, if we rotate it by 45° , we get a different square. See Figure 1.

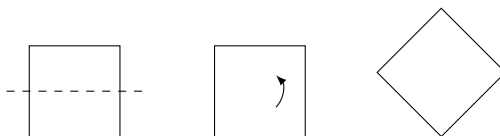


Figure 1. If we reflect the square across the horizontal line (Left) or rotate it 90° counterclockwise (Center), we get the same square, whereas if we rotate it by 45° counterclockwise (Right), we get a different square.

We call a rotation or reflection of the square a *symmetry* of the square if, after performing it, we get the same square we started with. We include the “identity” symmetry, that does nothing. Each symmetry of the square can be considered as a function. For instance, the reflection about the x -axis can be described as $f(x, y) = (x, -y)$. We consider two symmetries to be the same if they are the same function. For instance, if we reflect about the x -axis twice, then we get back to the identity function $f(x, y) = (x, y)$, so it is considered to be the same as the identity function.

PROBLEM 4.1. How many symmetries are there of the square?

PROBLEM 4.2. More generally, let $n \geq 3$ be an integer. How many symmetries are there of a regular n -gon?

PROBLEM 4.3. Find a polygon with exactly 3 symmetries. More generally, for every positive integer n , find a polygon with exactly n symmetries.

PROBLEM 4.4. Let P be a polygon. Show that if there is a line ℓ such that reflection about ℓ is a symmetry of P , then P has an even number of symmetries.

PROBLEM 4.5. Find a polygon that isn't a square, but that has exactly the same symmetries as a square.

PROBLEM 4.6. We can extend the notion of symmetries to 3-dimensional polyhedra. How many symmetries does a cube have?

PROBLEM 4.7. Let D_4 be the set of symmetries of the square.¹ Let τ be the reflection about the x -axis, and let ρ be the rotation by 90° counterclockwise. What is $\tau \circ \rho \circ \tau$? That is, what happens when you perform τ , then ρ , then τ ?

PROBLEM 4.8. Suppose α and β are two symmetries of a polygon P . Show that $\alpha \circ \beta$ is another symmetry of P . (Note that $\alpha \circ \beta$ means that you perform β first, then α . For instance if α is a rotation and β is a reflection, then $\alpha \circ \beta$ means that you perform the reflection first, then the rotation.)

PROBLEM 4.9. Show that if α is any symmetry of the square, then we can express α as a rotation (by some amount, possibly 0), optionally followed by a reflection about the x -axis.

PROBLEM 4.10. Express a reflection about the y -axis in terms of rotations and perhaps a reflection about the x -axis, as in Problem 4.9.

PROBLEM 4.11. Express a reflection about the y -axis followed by a reflection about the line $y = x$ as in Problem 4.9.

PROBLEM 4.12. Generalize Problem 4.9 to symmetries of a regular n -gon.

Extending the notation from Problem 4.7, let ρ^{-1} be a rotation by 90° clockwise. Say that two elements α and β of D_4 are *directly conjugate* if $\alpha = \tau \circ \beta \circ \tau$ or $\alpha = \rho \circ \beta \circ \rho^{-1}$, or vice versa. Say that α and β are *conjugate* if there exist elements $\alpha_1 = \alpha, \alpha_2, \dots, \alpha_k = \beta$ such that for each i with $1 \leq i \leq k-1$, the elements α_i and α_{i+1} are directly conjugate. Write $\alpha \sim \beta$ if α and β are conjugate.

PROBLEM 4.13. Show that if α is in D_4 , then $\alpha \sim \alpha$.

PROBLEM 4.14. Show that if α and β are in D_4 , and $\alpha \sim \beta$, then $\beta \sim \alpha$.

PROBLEM 4.15. Show that if α , β , and γ are in D_4 , and $\alpha \sim \beta$ and $\beta \sim \gamma$, then $\alpha \sim \gamma$.

The last three problems give us the notion of an *equivalence relation*. Here is the full definition:

DEFINITION. Let X be a set, and let $\approx$ be a relation on X , i.e. we declare that $x \approx y$ for certain pairs of elements $x, y \in X$. We say that $\approx$ is an *equivalence relation* if the following properties hold:

¹The D stands for "dihedral."

- For all $x \in X$, we have $x \approx x$.
- If $x \approx y$, then $y \approx x$.
- If $x \approx y$ and $y \approx z$, then $x \approx z$.

Thus the last three problems show that the conjugacy relation $\sim$ is an equivalence relation. Whenever we have an equivalence relation $\approx$ on a set X , the equivalence relation partitions the set into equivalence classes $[x] = \{y \in X : x \approx y\}$, in the sense that every element is contained in exactly one equivalence class. It follows from the last three problems that the elements of D_4 split up into equivalence classes, known as *conjugacy classes*. Given α in D_4 , write $[\alpha]$ for the set of all β in D_4 such that $\alpha \sim \beta$, including α itself.

PROBLEM 4.16. Find all the elements in $[\rho]$, $[\rho \circ \rho]$, and $[\tau]$.

Further reading. To learn more about symmetries of shapes in the plane, see [Arm88, Chapters 1–4].

WORKSHEET 5

Platonic Solids

DEFINITION. A shape S is said to be *convex* if for any two points a and b in S , the line segment connecting a and b is entirely contained in S .

See the example and nonexample in Figure 1.

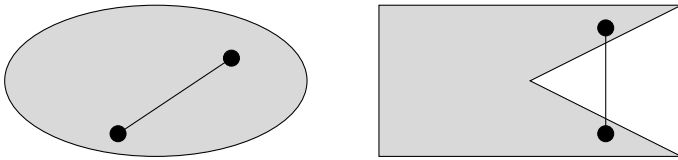


Figure 1. Left: a convex shape. Right: a nonconvex shape.

DEFINITION. A *Platonic solid* is a convex polyhedron, all of whose faces are congruent regular polygons, with the same number of faces meeting at each vertex.

Two Platonic solids, namely the cube and the tetrahedron, are shown in Figure 2.¹

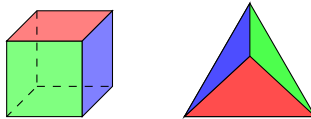


Figure 2. Two Platonic solids: the cube and the tetrahedron.

PROBLEM 5.1. In the cube, the faces are squares, and there are three of them meeting at each vertex. Can there be another Platonic solid with three squares meeting at each vertex? (We consider two Platonic solids to be the same if they are similar to each other.)

¹Cubes don't look like this, in the sense that there is no angle at which you can look at a cube and see what is shown in Figure 2. However, I used this image here because it is the most familiar 2-dimensional representation of a cube.

PROBLEM 5.2. More generally, suppose p and q are positive integers. Can there exist two different Platonic solids, each of whose faces are regular p -gons, with q faces meeting at each vertex?

The *Schläfli symbol* $\{p, q\}$ denotes a Platonic solid whose faces are regular p -gons, with q faces meeting at each vertex, if such a Platonic solid exists.

PROBLEM 5.3. Is there a Platonic solid whose Schläfli symbol is $\{p, 2\}$ for some positive integer p ?

PROBLEM 5.4. Consider a vertex v of a convex polyhedron. Show that the sum of the angles of all the faces meeting at v is less than 360° .

PROBLEM 5.5. Using the result of the previous problem, determine all possible Schläfli symbols of Platonic solids. (You don't have to show that they all actually exist, but they do exist. You may use this fact from now on.)

PROBLEM 5.6. For each Platonic solid, count the number of vertices, edges, and faces. (Try making them out of paper!)

Given a convex polyhedron P , we can form a new polyhedron Q , called its *dual*, by putting a vertex of Q at the center of each face of P , and then putting an edge between two vertices of Q if the corresponding faces in P share an edge.

PROBLEM 5.7. What are the duals of all the Platonic solids? Show that they are again Platonic solids.

PROBLEM 5.8. If P is a Platonic solid, what is the dual of the dual of P ?

PROBLEM 5.9. Suppose P is a Platonic solid with Schläfli symbol $\{p, q\}$. What is the Schläfli symbol of the dual of P ?

The *dodecahedron* is the Platonic solid with Schläfli symbol $\{5, 3\}$.

PROBLEM 5.10. Is it possible to find four vertices of a dodecahedron that form the vertices of a regular tetrahedron? How many such inscribed tetrahedra are there in a dodecahedron?

PROBLEM 5.11. Is it possible to find eight vertices of a dodecahedron that form the vertices of a cube? How many such inscribed cubes are there in a dodecahedron?

PROBLEM 5.12. Show that the number of symmetries of a Platonic solid is the same as the number of symmetries of its dual.

PROBLEM 5.13. For each Platonic solid, determine the number of symmetries it has.

Further reading. [Cro97] is a good introductory book on polyhedra, together with their history. See [Ric08] for a popular account of polyhedra and their vertices, edges, and faces.

WORKSHEET 6

Prime Numbers

DEFINITION. A positive integer $p \geq 2$ is said to be *prime* if it is not divisible by any positive integer other than 1 and itself. A positive integer $n \geq 2$ that is not prime is said to be *composite*.

PROBLEM 6.1. Write down the first 20 primes.

PROBLEM 6.2. Show that if m is a positive integer greater than 1, then two consecutive positive integers n and $n + 1$ cannot both be divisible by m .

PROBLEM 6.3. Show that if $m_1, m_2, \dots, m_r$ are finitely many positive integers, then there is a positive integer divisible by all of $m_1, m_2, \dots, m_r$.

PROBLEM 6.4. Show that if $p_1, p_2, \dots, p_r$ is any finite list of primes, then there is a positive integer not divisible by any of $p_1, \dots, p_r$.

PROBLEM 6.5. Show that there are infinitely many primes.

PROBLEM 6.6. Show that there are infinitely many primes that leave a remainder of 3 when divided by 4.

PROBLEM 6.7. Show that there are infinitely many primes that leave a remainder of 2 when divided by 3.

PROBLEM 6.8. Show that for any positive integer n , there are n consecutive composite numbers.

PROBLEM 6.9. Let $f(n) = n^2 - n + 41$. Compute $f(0), f(1), f(2), \dots, f(10)$. Observe that all these numbers are primes. Does there exist a positive integer n such that $f(n)$ is composite?

PROBLEM 6.10. Does there exist a nonconstant polynomial $f(n)$ with integer coefficients such that $|f(n)|$ is never composite for any integer n ?

PROBLEM 6.11. Find all the prime factors of the numbers $2^2 - 1$, $2^4 - 1$, $2^6 - 1$, $2^8 - 1$, and $2^{10} - 1$.

PROBLEM 6.12. Find all the prime factors of the numbers $2^3 - 1$, $2^6 - 1$, $2^9 - 1$, $2^{12} - 1$, and $2^{15} - 1$.

PROBLEM 6.13. Show that if n is a positive integer and $2^n - 1$ is prime, then n is prime.

Numbers of the form $2^n - 1$ are called *Mersenne numbers*, and primes of this form are called *Mersenne primes*.

PROBLEM 6.14. Does there exist a prime number p such that $2^p - 1$ is composite?

PROBLEM 6.15. Show that if n is a positive integer and $2^n + 1$ is prime, then n is a power of 2.

Numbers of the form $2^{2^n} + 1$ are called *Fermat numbers*. Fermat thought that $2^{2^n} + 1$ was prime for all nonnegative integers n , but $n = 5$ is a counterexample: we have $2^{2^5} + 1 = 641 \times 6700417$.

PROBLEM 6.16. Suppose you wish to check if 1003 is prime. What do you need to check? If you're going to check this by testing if it's divisible by each number for a while, do you have to go all the way up to 1002, or can you guarantee that it's prime by stopping at some earlier point? Generalize your findings.

Further reading. See [Sil13, Chapters 12–15] to learn more about elementary properties of prime numbers. See [AZ18, Chapter 1] for several proofs of the infinitude of primes, and [AZ18, Chapter 2] for Bertrand's Postulate, which says that there is always a prime between n and $2n$.

WORKSHEET 7

The Euclidean Algorithm

Given a collection $a_1, a_2, \dots, a_n$ of positive integers, their *greatest common divisor*, or *gcd*, is the largest positive number d dividing all of them. For instance, $\gcd(105, 140) = 35$ because $105 = 35 \times 3$ and $140 = 35 \times 4$, and no number larger than 35 divides both 105 and 140. If a is a positive integer, then we define $\gcd(a, 0) = a$.

DEFINITION. If $\gcd(a, b) = 1$, then we say that a and b are *relatively prime*.

PROBLEM 7.1. Suppose we know the prime factorizations of a and b : $a = 2^{e_2} 3^{e_3} 5^{e_5} \dots$ and $b = 2^{f_2} 3^{f_3} 5^{f_5} \dots$. Express $\gcd(a, b)$ in terms of the e_i 's and f_i 's.

Closely related to the gcd is the *least common multiple*, or *lcm*. The lcm of $a_1, \dots, a_n$ is the smallest positive integer divisible by each of the a_i 's.

PROBLEM 7.2. Show that if a and b are positive integers, then $ab = \gcd(a, b) \operatorname{lcm}(a, b)$.

Based on Problem 7.1, you know how to compute the gcd of a and b provided that you are given the prime factorizations of a and b . But if a and b are large, then computing the prime factorizations of a and b might be a difficult problem. So, it would be nice if there were a way of computing the gcd without knowing the prime factorizations. This seems unlikely at first, because $\gcd(a, b)$ seems to encode some facts about the prime factorizations, but remarkably, this is possible, and this is the content of the Euclidean algorithm.

PROBLEM 7.3. Show that if a and b are positive integers with $b \geq a$, then $\gcd(a, b) = \gcd(a, b - a)$.

PROBLEM 7.4. More generally, suppose that $b \geq ak$. Show that $\gcd(a, b) = \gcd(a, b - ak)$.

PROBLEM 7.5. Use the result of the previous problem to compute $\gcd(590413, 830581)$ by hand, in a reasonable amount of time.

What you presumably did in the previous problem was to run the following algorithm, known as the *Euclidean algorithm*. The notation $\lfloor x \rfloor$ means the greatest integer $\leq x$, called the *floor* of x .

Euclidean algorithm

```

function gcd( $a, b$ )
  if  $a > b$  then
    swap  $a$  and  $b$ 
  if  $a = 0$  then
    return  $b$ 
   $k \leftarrow \lfloor \frac{b}{a} \rfloor$ 
  return gcd( $a, b - ak$ )

```

PROBLEM 7.6. Use the Euclidean algorithm to compute $\gcd(233, 377)$. Why does it take so long?

PROBLEM 7.7. Show that if x and y are integers, then $ax + by$ is a multiple of $\gcd(a, b)$.

We would now like to show the reverse: that any multiple of $\gcd(a, b)$ can be written as $ax + by$ for some (not necessarily positive) integers x and y .

PROBLEM 7.8. Show that it suffices to check that $\gcd(a, b)$ can be expressed in the form $ax + by$, where x and y are integers.

In order to show that $\gcd(a, b)$ can be written as $ax + by$, we need to take a more careful look at the Euclidean algorithm. Let us assume that $a \leq b$. Let $b_0 = b$ and $b_1 = a$. If, for some n , we have $b_n = 0$, then $\gcd(a, b) = b_{n-1}$. If $b_n \neq 0$, then let $k = \lfloor \frac{b_{n-1}}{b_n} \rfloor$, so that $0 \leq b_{n-1} - kb_n < b_n$. Let $b_{n+1} = b_{n-1} - kb_n$.

PROBLEM 7.9. For $a = 35$ and $b = 91$, compute all the b_n 's.

PROBLEM 7.10. Show that each b_n can be written as $b_n = ax + by$ for some integers x and y .

PROBLEM 7.11. Conclude that there exist integers x and y such that $\gcd(a, b) = ax + by$.

This result is a very important fact in number theory, called *Bézout's Lemma*. An interesting application is a new categorization of prime numbers.

PROBLEM 7.12. Show that if p is prime, and a and b are integers such that ab is a multiple of p , then at least one of a and b is already a multiple of p .

PROBLEM 7.13. Show that if n is composite, then there exist integers a and b such that neither a nor b is a multiple of n , but ab is a multiple of n .

Further reading. See [Sil13, Chapters 5 and 6] for more on the gcd and the Euclidean algorithm. See [GKP94, §4.5] for the connection with the Stern–Brocot tree, which we will see in Worksheet 32.

WORKSHEET 8

Pascal's Triangle

Pascal's triangle is a triangular array of numbers with 1's down the sides, and each number in the middle is the sum of the two numbers above it. The first few rows of Pascal's triangle are shown in Figure 1. Observe that the number in the red cell is equal to the sum of the numbers in the two blue cells.

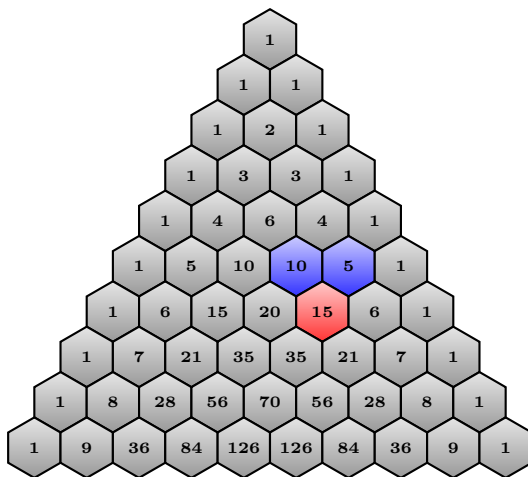


Figure 1. Pascal's Triangle.

PROBLEM 8.1. The numbers in the leftmost diagonal are all clearly equal to 1, and the numbers in the next diagonal are just the positive integers: 1, 2, 3, 4, 5, ... Can you find a formula for the numbers in the next diagonal? (That is, the one beginning 1, 3, 6, 10, 15, ...) What about the next diagonal, beginning 1, 4, 10, 20, 35, ...?

PROBLEM 8.2. What is the sum of the numbers in the n^{th} row of Pascal's triangle? (We consider the top row, consisting of just a 1, to be row 0, and then the row with two 1's is row 1, and so forth.)

If n is a positive integer, we define $n!$, pronounced " n factorial," to be the product of all the positive integers up to n , so $1! = 1$, $2! = 2$,

$3! = 3 \times 2 \times 1 = 6$, $4! = 4 \times 3 \times 2 \times 1 = 24$, and so on. We also define $0! = 1$.

PROBLEM 8.3. Show that the k^{th} entry in the n^{th} row of Pascal's triangle is $\frac{n!}{k!(n-k)!}$. (Both rows and diagonals are indexed starting from 0, so the red cell is in row 6 and diagonal 4, for instance.)

DEFINITION. The numbers $\frac{n!}{k!(n-k)!}$ are very important. We call them *binomial coefficients* and write them as $\binom{n}{k}$.

PROBLEM 8.4. Find the sum of the first m numbers in the r^{th} diagonal of Pascal's triangle, starting from the top, as a function of m and r .

PROBLEM 8.5. What is the *alternating sum* of the numbers in the n^{th} row of Pascal's triangle? The alternating sum is the same as the sum, but with every other term subtracted. For instance, the alternating sum in row 4 is $1 - 4 + 6 - 4 + 1$. Explain your answer.

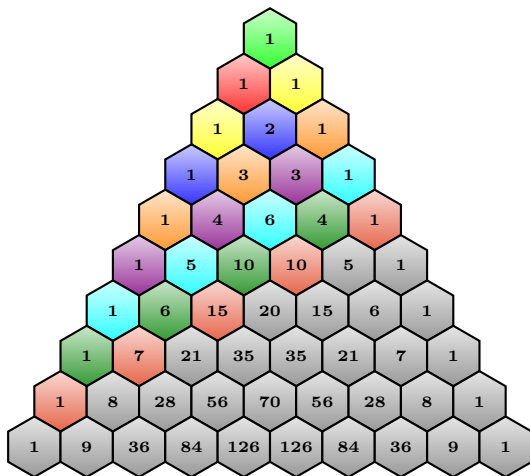


Figure 2. Shallow diagonals in Pascal's triangle.

PROBLEM 8.6. Sum the numbers in each monochromatic shallow diagonal in Figure 2. What familiar sequence of numbers do you get? Express this as a formula, using either factorials or the binomial coefficient notation. Why is your formula correct? Why do these numbers appear in Pascal's triangle?

PROBLEM 8.7. Expand $(x + y)^4$ and $(x + y)^5$. Do you recognize these coefficients? Why is this happening? Does it work in general?

PROBLEM 8.8. What are 11^3 and 11^4 ? What's going on? Why does the pattern break down when you compute 11^5 ? What happens when you compute 101^5 and 101^6 ? Explain why this happens. Where does this pattern break down, and how can you get a longer-lasting version of this phenomenon?

PROBLEM 8.9. Start at the top cell of Pascal's triangle, and at every second, move either one cell southwest or one cell southeast. How many paths are there that end up at the k^{th} diagonal in the n^{th} row?

PROBLEM 8.10. Draw a large Pascal's triangle, and shade in all the cells containing odd numbers. What pattern do you get? Can you explain what's going on?

PROBLEM 8.11. The number 1 clearly appears infinitely many times in Pascal's triangle, but every other number only appears finitely many times. The number 10 appears 4 times. Can you find a number other than 1 that appears more than 4 times? More than 6 times? What is the largest number of occurrences you can find?

Further reading. See [Edw87] for a history and elementary properties of Pascal's triangle.

WORKSHEET 9

Counting Using Binomial Coefficients

We met the binomial coefficients $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ in the previous worksheet, thanks to their appearance in Pascal's triangle. But they also show up in many other places.

PROBLEM 9.1. How many ways are there to select a k -member committee from among n people?

PROBLEM 9.2. Let m and n be positive integers. How many paths are there from $(0, 0)$ to (m, n) taking steps that go either one unit to the right or one unit up?

PROBLEM 9.3. How many ways are there to choose a committee (of any size) from among n people, then choosing one member of the committee to be the president? Can you solve this problem in two different ways and obtain an interesting identity involving binomial coefficients?

We now turn to a very important technique involving binomial coefficients, used to solve problems that initially do not appear to be well-suited to binomial coefficients. This method, called the *stars and bars technique*, involves setting up several objects, which are initially vertical bars, and then converting some of the bars into stars. Another variation involves setting up a bunch of stars, then placing bars in some of the gaps between consecutive stars.

PROBLEM 9.4. Suppose we start with n bars in a row. How many ways are there to convert k of the bars into stars?

PROBLEM 9.5. Suppose we start with n stars in a row. How many ways are there to place k bars in some of the gaps between adjacent stars? Each gap is only allowed to contain at most one bar.

PROBLEM 9.6. How many ways are there to write a positive integer n as a sum of k positive integers, where order matters? For instance, the representations $1 + 1 + 3$ and $1 + 3 + 1$ for 5 are considered to be different representations.

PROBLEM 9.7. A bagel shop sells 8 different types of bagels. How many ways are there to select a box of a baker's dozen (i.e. 13) bagels?

Two boxes are considered the same if they contain the same number of each type of bagel.

PROBLEM 9.8. How many ways are there to write a nonnegative integer n as a sum of k nonnegative integers, where order matters?

PROBLEM 9.9. How many n -digit numbers are there whose digits are strictly increasing from left to right?

PROBLEM 9.10. How many n -digit numbers are there whose digits are nondecreasing from left to right? (In other words, this is the same as the previous problem, but now consecutive digits can be equal.) Numbers are not allowed to have leading zeros.

PROBLEM 9.11. Suppose you have m cats and n dogs. How many ways are there to select a total of r animals? Can you solve this problem in two ways and obtain an interesting identity involving binomial coefficients? The resulting identity is known as *Vandermonde's identity*.

PROBLEM 9.12. Can you express the sum

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \binom{n}{2}^2 + \cdots + \binom{n}{n}^2$$

in closed form, i.e. without using a sum over a variable number of terms?

Further reading. See [AF04, Chapters 1–3] for many problems involving binomial coefficients and [BQ03, Chapter 5] for elegant bijective proofs of binomial identities.

WORKSHEET 10

The Pigeonhole Principle

PROBLEM 10.1. What is the minimum number of people needed such that you can be *certain* that at least four of them were born on the same day of the week?

PROBLEM 10.2. Show that, among any five points in a square of sidelength 1, there must exist two of distance at most $\frac{1}{\sqrt{2}}$.

PROBLEM 10.3. Show that among 26 points in an equilateral triangle of sidelength 1, there are two whose distance is at most $\frac{1}{5}$.

PROBLEM 10.4. Show that among any 51 integers from 1 to 100, we must have at least one pair of consecutive numbers.

PROBLEM 10.5. What is the maximum size of a set of positive integers up to 100 such that no integer in the set is a multiple of another one?

PROBLEM 10.6. Take an 8×8 chessboard, and remove two opposite corner squares. Is it possible to cover the remaining 62 squares using 31 dominoes? (Each domino covers exactly two adjacent squares.)

PROBLEM 10.7. What is the maximum number of knights we can place on an 8×8 chessboard such that no two attack each other?

PROBLEM 10.8. What is the maximum number of bishops we can place on an 8×8 chessboard such that no two attack each other?

PROBLEM 10.9. Show that among $n \geq 2$ people, there must be two people who know the same number of other people. (We assume that knowledge is mutual: if person A knows person B , then person B knows person A .)

PROBLEM 10.10. What is the smallest value of n satisfying the following property: given any n points on the plane such that no three are collinear, there exist four of them forming the vertices of a convex quadrilateral?

Given a sequence of numbers $a_1, a_2, a_3, \dots, a_k$, a *subsequence* is the sequence formed by taking some collection of the a_i 's, in order. For instance, 5, 1, 2 is a subsequence of 7, 3, 5, 4, 1, 2, 8, 0, 6.

PROBLEM 10.11. Show that if we have a sequence of $mn+1$ distinct positive integers, we must have an increasing subsequence of length $m+1$ or a decreasing subsequence of length $n+1$.

For instance, suppose $m = 3$ and $n = 2$. In the sequence 4, 1, 6, 7, 3, 5, 2, the numbers 7, 3, 2 form a decreasing subsequence of length 3.

Further reading. See [Zei16, §3.3] and [Eng98, Chapter 4] for many more problems involving the pigeonhole principle.

WORKSHEET 11

Permutations

A *permutation* of n is an ordering of the numbers from 1 to n . For instance, 6253174 is a permutation of 7.

PROBLEM 11.1. How many permutations are there of n ?

We write S_n for the set of all permutations of n .

Given any permutation, we can reach it starting from the “identity” permutation $123\cdots n$ by doing a bunch of swaps. For instance, for the permutation above, we can reach it by doing the following swaps (marked in red):

1234567

5234167

5243167

5273164

7253164

6253174

At the end, we reached the desired permutation.

PROBLEM 11.2. In the above example, not all the swaps were of numbers in consecutive positions. Show that it is possible to reach any permutation by means of a sequence of consecutive swaps.

Suppose τ is a permutation of n . For $1 \leq i \leq n-1$, let $\sigma_i(\tau)$ be the permutation obtained from τ by swapping the numbers in the i^{th} and $(i+1)^{\text{st}}$ positions in τ . For instance, we have

$$\sigma_3(5146372) = 5164372.$$

PROBLEM 11.3. Show that for any permutation τ of n and any i with $1 \leq i \leq n-1$, we have $\sigma_i(\sigma_i(\tau)) = \tau$.

PROBLEM 11.4. Show that if τ is a permutation of n and $1 \leq i, j \leq n-1$ with $|i-j| \geq 2$, then $\sigma_i(\sigma_j(\tau)) = \sigma_j(\sigma_i(\tau))$. Show that this is not necessarily true if $|i-j| = 1$.

PROBLEM 11.5. Show that if τ is a permutation of n and $1 \leq i \leq n-2$, then $\sigma_i(\sigma_{i+1}(\sigma_i(\tau))) = \sigma_{i+1}(\sigma_i(\sigma_{i+1}(\tau)))$.

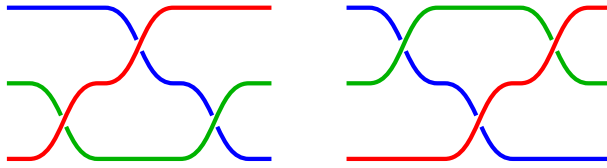


Figure 1. A graphical representation of the identity $\sigma_i \circ \sigma_{i+1} \circ \sigma_i = \sigma_{i+1} \circ \sigma_i \circ \sigma_{i+1}$.

We can draw a picture of this; see Figure 1.

PROBLEM 11.6. For any swap ρ of two numbers in a permutation, we can undo the swap by doing it again; that is, $\rho(\rho(\tau)) = \tau$. But what happens if we do a sequence of swaps, say $\rho_1 \circ \rho_2 \circ \cdots \circ \rho_k$, where each ρ_i is a swap? What do we have to do to undo this entire sequence of swaps?

PROBLEM 11.7. Suppose that $\rho_1, \dots, \rho_k$ and $\pi_1, \dots, \pi_m$ are swaps such that $\rho_1 \circ \cdots \circ \rho_k(\tau) = \pi_1 \circ \cdots \circ \pi_m(\tau)$. Show that $\pi_m \circ \pi_{m-1} \circ \cdots \circ \pi_1 \circ \rho_1 \circ \cdots \circ \rho_k(\tau) = \tau$.

In the preceding problems, there's no meaningful dependence on τ , since the same identities hold for all τ . Thus we'll write things like $\rho_1 \circ \cdots \circ \rho_k = \pi_1 \circ \cdots \circ \pi_m$.

PROBLEM 11.8. Explain how to write an arbitrary swap in terms of a sequence of consecutive swaps. If we wish to swap the terms in positions i and j , how many consecutive swaps does it take?

Given a permutation τ , a pair (i, j) with $1 \leq i < j \leq n$ such that $\tau(i) > \tau(j)$ is called an *inversion*; that is, it's a pair that's out of order. Let $\text{inv}(\tau)$ denote the number of inversions of τ , i.e. $\text{inv}(\tau) = \#\{(i, j) : 1 \leq i < j \leq n : \tau(i) > \tau(j)\}$.

PROBLEM 11.9. Compute $\text{inv}(\tau)$ for the following permutations τ :

$$12354, \quad 32145, \quad 6253174.$$

What about $\text{inv}(123 \cdots n)$ and $\text{inv}(n(n-1)(n-2) \cdots 321)$?

PROBLEM 11.10. Show that $|\text{inv}(\tau) - \text{inv}(\sigma_i \circ \tau)| = 1$.

PROBLEM 11.11. Suppose that we have two sequences of swaps $\rho_1, \dots, \rho_k$ and $\pi_1, \dots, \pi_m$ such that $\rho_1 \circ \cdots \circ \rho_k = \pi_1 \circ \cdots \circ \pi_m$. Show that if k is even, then m is as well, and vice versa. (Hint: compare the length of a sequence of swaps to the number of inversions.)

We write A_n for the set of permutations that are obtainable from the identity permutation by means of an even number of swaps in some (hence any) way. The set A_n is called the *alternating group*.

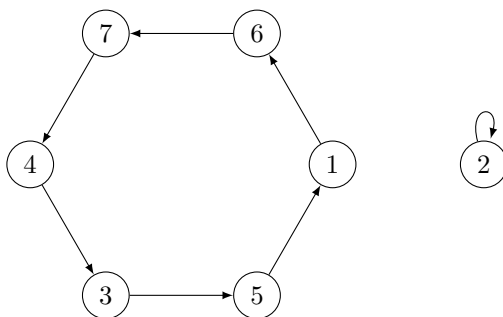


Figure 2. A picture of the permutation 6253174.

PROBLEM 11.12. How many elements does A_n have?

There is another very important way of representing permutations. Given a permutation τ written out as a sequence of numbers, we can consider it as a function $\tau : \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, n\}$ by letting the first number denote $\tau(1)$, the second number $\tau(2)$, and so on. For instance, the permutation 6253174 is the function τ such that $\tau(1) = 6$, $\tau(2) = 2$, $\tau(3) = 5$, $\tau(4) = 3$, $\tau(5) = 1$, $\tau(6) = 7$, and $\tau(7) = 4$.

It is natural to represent a function by means of a picture, where for each i , we draw an arrow from i to $\tau(i)$. The picture for this permutation τ is shown in Figure 2.

We can represent this picture more succinctly, by writing each cycle in parentheses in order. The permutation 6253174 is then $(167435)(2)$.

PROBLEM 11.13. If you are given the cycle decomposition of a permutation, how can you tell quickly whether or not it lies in the alternating group?

Further reading. For much more about permutations, see [Bon12].

WORKSHEET 12

Permutation Statistics

Let τ be a permutation of n , thought of as a function from $\{1, 2, \dots, n\}$ to itself. A number i with $1 \leq i \leq n$ is said to be a *fixed point* of τ if $\tau(i) = i$.

PROBLEM 12.1. How many permutations of n are there such that 1 is a fixed point? Such that 2 is a fixed point? Generalize.

PROBLEM 12.2. What is the total number of fixed points among all permutations of n ? What is the average number of fixed points?

In the cycle decomposition of τ , τ splits up into some number of cycles of length 1, i.e. fixed points, some number of cycles of length 2, some number of cycles of length 3, and so on. From the previous problem, we now know the average number of fixed points across all permutations. Now we would like to know the averages of other cycle lengths as well.

PROBLEM 12.3. How many permutations τ of n are there such that (12) is a cycle of length 2 in τ ? What about (13)? Generalize.

PROBLEM 12.4. What is the average number of cycles of length 2 among all permutations of n ?

PROBLEM 12.5. What is the average number of cycles of length k among all permutations of n ?

PROBLEM 12.6. What is the average number of cycles (of any length) among all permutations of n ?

PROBLEM 12.7. Show that a permutation of n cannot have more than one cycle of length $> \frac{n}{2}$, and more generally that it cannot have more than k cycles of length $> \frac{n}{k+1}$.

Here is a very neat and surprising application of our results so far on permutation statistics:

A warden in charge of 100 prisoners plays the following game with the prisoners. There are 100 boxes labeled 1–100 and 100 pieces of paper, each containing a distinct number from 1–100. The warden randomly places the papers into the boxes, with one paper going into each box. Each prisoner is also assigned a distinct number from 1–100.

One by one, each prisoner is brought into the room with the boxes and is allowed to open 50 boxes. If all the prisoners manage to find their own number among the 50 boxes they open, then they are all free, but if at least one prisoner fails to find eir¹ number, then all the prisoners have to remain in prison.

PROBLEM 12.8. Suppose each prisoner chooses 50 boxes randomly. What is the probability that they go free?

PROBLEM 12.9. Clearly, random choice is not a good strategy for success. Come up with a better strategy for them to go free. How well can you do? What is the approximate probability of freedom? (You may wish to use the fact that $1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k} \approx \log k + \gamma$, where $\log$ is the natural logarithm (to base e), and $\gamma \approx 0.57721$ is a number.)

PROBLEM 12.10. 100 passengers are about to board an airplane with 100 seats. Everyone has a ticket with an assigned seat. However, the first passenger has lost eir ticket and sits in a random seat. Every subsequent passenger attempts to take eir assigned seat, but if it is already taken, then ey chooses a random unoccupied seat. You are the last passenger in line to board, so there is only one seat left when you get on the plane. What is the probability that you get to sit in your assigned seat?

PROBLEM 12.11. If you choose a random permutation of n , what is the probability that 1 and 2 are in the same cycle?

PROBLEM 12.12. If you choose a random permutation of n , what is the probability that 1, 2, and 3 are in the same cycle?

Further reading. The book [Bon12] contains material on permutation statistics. For more problems like the prisoner box problem (including that one), see [Win04].

¹This is an example of a *Spivak pronoun*, which is a third-person singular gender-neutral pronoun. To form these pronouns, take the third-person plural pronouns, which start with “th,” and then remove the “th,” producing “ey,” “em,” “eir,” and so forth. The only exception is “themselves,” which becomes “emself.” One of my missions is to make Spivak pronouns into a commonly used part of the English language. Please do your part to support the cause!

WORKSHEET 13

Invariants and Monovariants

PROBLEM 13.1. Suppose we start with an $m \times n$ bar of chocolate that we wish to divide into individual 1×1 squares. On every turn, we cut one of the pieces into two by breaking it along one of the horizontal or vertical grid lines. What is the smallest number of breaks we have to do to break the entire bar into 1×1 squares?

PROBLEM 13.2. The numbers from 1 to 1000 are written on the board. At every turn, we take two of the numbers, say a and b , erase them, and replace them with the number $a + b - 1$. After 999 steps, we are left with a single number. What are the possibilities for this number?

PROBLEM 13.3. The numbers from 1 to 1000 are written on the board. At every turn, we take two of the numbers, say a and b , erase them, and replace them with the number $|a - b|$. After 999 steps, we are left with a single number. Could this single number be 3?

PROBLEM 13.4. We have a 10×10 grid of lights. Initially, the light in the top left cell is on, and all the other lights are off. At each step, we pick a row or column of the grid and toggle all the lights in that row or column. Is it possible to perform some sequence of these steps such that all the lights are turned on at the same time?

PROBLEM 13.5. You have a quarter infinite (to the right and up) grid, with checkers initially placed at the three corner squares, as shown in Figure 1. At every step, if there is a checker at the cell (x, y) , you can remove it and replace it with a checker at $(x, y + 1)$ and a checker at $(x + 1, y)$, provided that neither of these two cells already has a checker. Is it possible to perform some sequence of moves such that the three cells that initially contained checkers are empty?

PROBLEM 13.6. Start with a finite collection $a_1, a_2, \dots, a_n$ of positive integers. On each turn, you choose two numbers a_j and a_k such that neither one is a multiple of the other, and then delete them both and replace them with $\gcd(a_j, a_k)$ and $\text{lcm}(a_j, a_k)$.¹ If there is no pair

¹ $\gcd(a, b)$ is the greatest common divisor: the largest number that divides both a and b . $\text{lcm}(a, b)$ is the least common multiple: the smallest number that is a multiple of both a and b .

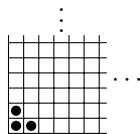


Figure 1. Three checkers placed at the bottom left of a quarter-infinite chessboard.

(a_j, a_k) such that neither is a multiple of the other, then the game ends. Is it possible to keep playing forever?

As you solved these problems, you probably noticed that they have something in common: some quantity that doesn't change when you perform an operation. This quantity is called an *invariant* for the process.

Sometimes, we find a quantity that doesn't stay the same at every step of the process, but it only changes in one direction: it can only increase, or only decrease. In that case, the quantity is called a *monovariant*.

PROBLEM 13.7. Suppose we have n red points and n blue points in the plane, with no three points collinear. Show that it is possible to draw n nonintersecting line segments, each connecting one red point and one blue point.

PROBLEM 13.8. We have a 10×10 grid, with each cell containing a strawberry. Initially, 9 of the strawberries are moldy. If at any time, a strawberry has two neighboring (either horizontal or vertical) moldy strawberries, then it becomes moldy as well. Is it possible for the mold to spread to all the strawberries?

We conclude with a walkthrough of a remarkable monovariant problem called *Conway's checkers* or *Conway's soldiers*. We have an infinite (in all directions) grid, with the center of each grid point having the form (m, n) , where m and n are integers. We initially have a checker at each point (m, n) where $n < 0$, as shown in Figure 2.

On each turn, if we have two checkers on adjacent squares, and the next square is empty, then we can make a capture, as shown in Figure 3. Capturing can be done in the horizontal direction or the vertical direction. The challenge is to perform a sequence of captures to get a checker as high up the board as possible.

PROBLEM 13.9. Find a sequence of captures to get a checker to the point $(0, 3)$. (That is, four rows above the top row of checkers in the initial position.)

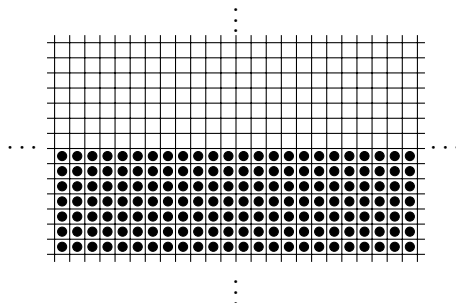


Figure 2. The initial position of Conway's checkers.

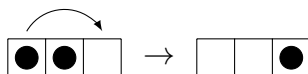


Figure 3. Capturing in Conway's checkers: if we have checkers on the two black squares on the left, and the next square is empty, then the leftmost checker can capture the one on the middle square, yielding the position on the right.

We will now show that it is not possible to reach the point $(0, 4)$. Let $f(x, y)$ be a function, to be determined later. We wish to choose f such that if we make a capture from (x, y) to $(x, y + 2)$ or $(x + 2, y)$ that gets closer to $(0, 4)$, then $f(x, y + 2) = f(x, y) + f(x, y + 1)$ or $f(x + 2, y) = f(x, y) + f(x + 1, y)$; in other words, getting closer to $(0, 4)$ doesn't change the sum of the values of the checkers on the board. However, if we capture away from $(0, 4)$, then the sum of the values should decrease.

PROBLEM 13.10. Is there a function of the form $f(x, y) = a^{x+y}$ with this property? If not, can you find a closely related function with this property?

PROBLEM 13.11. Now that you have found a suitable function f , find the sum of the values of all the checkers in the initial position, and the value of a hypothetical checker at $(0, 4)$.

PROBLEM 13.12. Conclude that it is not possible to reach $(0, 4)$ in finitely many captures.

PROBLEM 13.13. Suppose we consider Conway's checkers in 3 dimensions, with checkers initially at all integer points (x, y, z) with $z < 0$. What is the largest value of z for which it is possible to get a checker to $(0, 0, z)$?

Further reading. See [Zei16, §3.4] and [Eng98, Chapter 1] for many more problems involving invariants and monovariants.

WORKSHEET 14

Eulerian Paths and Circuits

A *graph* is a collection of dots, or *vertices*, with curves, or *edges*, connecting pairs of dots. A typical example is shown in Figure 1.

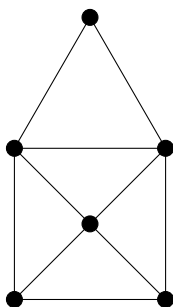


Figure 1. A graph.

Given a graph, it is natural to ask if it is possible to start at one of the vertices and travel along each edge exactly once. Such a path is called an *Eulerian path*. If this path also ends up at the starting vertex, it is called an *Eulerian circuit*.

PROBLEM 14.1. Does the graph shown in Figure 1 have an Eulerian path? What about an Eulerian circuit?

PROBLEM 14.2. Draw several other graphs with a small number of vertices, checking in each one whether it has an Eulerian path, an Eulerian circuit, or neither.

PROBLEM 14.3. Can you find a criterion for when a graph has an Eulerian path or an Eulerian circuit? Make sure to stress test your criterion by challenging it on examples where you think it is most likely to fail. This will help you to refine your criterion until it is exactly right.

In order to answer questions like this, it is helpful to be aware of the two types of criteria: *necessary* and *sufficient* conditions. Let P be a property that a graph might or might not have. (For instance, P could be the property that the graph has exactly 8 vertices, or that it has a vertex connected to more than half of the other vertices.) We

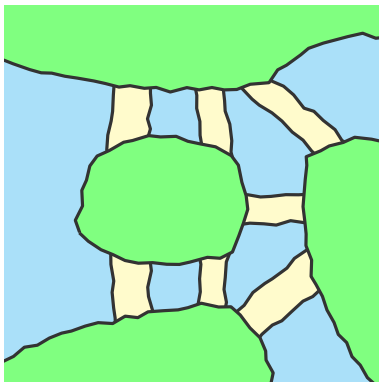


Figure 2. The seven bridges of Königsberg. The green areas represent land, the blue represents water, and the yellow regions are bridges.

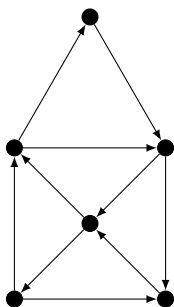


Figure 3. A directed graph.

say that P is a *necessary condition* for having an Eulerian path (say) if every graph with an Eulerian path has property P , and we say that P is a *sufficient condition* for having an Eulerian path if every graph with property P has an Eulerian path. The ideal situation is that we can find a condition that is both necessary and sufficient. In the case of both Eulerian paths and Eulerian circuits, such a necessary and sufficient condition exists.

PROBLEM 14.4. The seven bridges of Königsberg, shown in Figure 2, is a famous problem solved by Euler. Is it possible to travel over each bridge exactly once? Is it possible to do so while returning to the same region?

A *directed graph* is similar to a graph, except that each edge has an arrow on it, and it is only possible to travel in the direction of the arrow. See Figure 3 for an example.

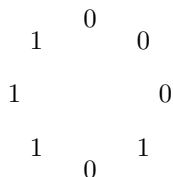


Figure 4. A de Bruijn sequence with $b = 2$ and $n = 3$.

PROBLEM 14.5. Find a necessary and sufficient condition for when a directed graph has an Eulerian path or an Eulerian circuit.

PROBLEM 14.6. By now, you should know when an Eulerian path or circuit exists in a (possibly directed) graph. But can you actually find it? Can you come up with an algorithm that will quickly find an Eulerian path or circuit when one exists? (There are several good answers to this problem.)

PROBLEM 14.7. If G is an (undirected) graph, let $2 \cdot G$ be the graph with the same vertices as G , but with two edges for each edge of G , connecting the same pair of vertices. For which graphs does $2 \cdot G$ have an Eulerian path or circuit?

PROBLEM 14.8. Suppose a graph has an Eulerian path that isn't an Eulerian circuit. Can it also have an Eulerian circuit? Could it have another Eulerian path with different starting and ending vertices?

PROBLEM 14.9. Let b and n be positive integers. The *de Bruijn graph* $DB(b, n)$ has b^{n-1} vertices, which are indexed by sequences of length $n - 1$, each of which is an integer from 0 to $b - 1$, and we place a directed edge from a vertex $(s_1, s_2, \dots, s_{n-1})$ to $(s_2, s_3, \dots, s_{n-1}, d)$ for each d with $0 \leq d \leq b - 1$. Show that the de Bruijn graph has an Eulerian circuit.

PROBLEM 14.10. A *de Bruijn sequence* of window length n is a way of arranging b^n numbers around a circle, each of which is an integer from 0 to $b - 1$ (each used b^{n-1} times), in such a way that every window of n consecutive numbers is different from every other window. (See Figure 4 for an example when $b = 2$ and $n = 3$.) Explain how to use an Eulerian circuit on the de Bruijn graph to construct a de Bruijn sequence.

Further reading. See [Wil10, §2.2] for more on Eulerian paths and circuits. For more on de Bruijn sequences, see [DG15, Chapters 2–4].

WORKSHEET 15

Vertices, Edges, and Faces

The data of a graph includes only vertices and edges. However, if we draw a graph on the plane with no edges crossing over other edges (i.e. a *planar graph*), we can also consider what is left of the plane once we remove the graph. The regions left over once we remove the graph are called the *faces*. We include the outer region as one of the faces. For instance, the planar graph in Figure 1 has 6 faces, as marked.

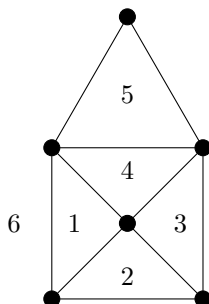


Figure 1. A planar graph with faces marked.

PROBLEM 15.1. Draw several planar graphs and count the number V of vertices, E of edges, and F of faces in each. Do you notice any patterns?

As you probably noticed (spoiler alert!), it appears that if a planar graph is connected, then $V - E + F = 2$. We will now explain this fact.

PROBLEM 15.2. Given a graph with one vertex and no edges, compute the values of V , E , F , and $V - E + F$.

Now, let's imagine adding edges to this graph, one at a time, seeing how this affects the value of $V - E + F$. We will assume throughout the rest of the worksheet that all graphs under consideration are connected and planar.

PROBLEM 15.3. Suppose we add an edge between two existing vertices that don't already share an edge, without adding crossings. How does this operation affect the values of V , E , F , and $V - E + F$?

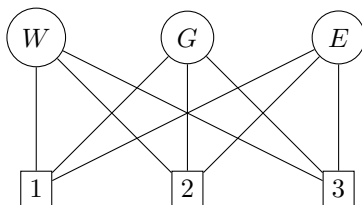


Figure 2. Connecting houses to utilities.

PROBLEM 15.4. Suppose we add a new vertex to the graph. To keep the graph connected, let's connect it to one other vertex with an edge. How does this operation affect the values of V , E , F , and $V - E + F$?

Now, let's go the other way, removing edges one by one.

PROBLEM 15.5. Suppose we start with a planar graph and remove an edge without disconnecting the graph. How does this operation affect the values of V , E , F , and $V - E + F$?

PROBLEM 15.6. The edge we remove might be the only edge adjacent to one of its vertices. When this happens, we also remove this vertex, known as a *leaf*. (If the graph contains just two vertices and one edge, then we only remove one of the leaves, so that the graph remains connected.) How does this operation affect the values of V , E , F , and $V - E + F$?

PROBLEM 15.7. Show that if a connected planar graph has at least two vertices, then there exists some edge that we can remove as in the previous two problems without disconnecting the graph.

PROBLEM 15.8. Show that $V - E + F = 2$ for any connected planar graph.

Let's now see a few classic applications of this formula.

PROBLEM 15.9. Suppose we have three houses, and we wish to connect each of them to the water, gas, and electricity providers, with a separate line from each house to each company. Is there a way to make all nine connections without any of these lines crossing over any others? Figure 2 shows a way to do it with several crossings. We call this graph $K_{3,3}$.

PROBLEM 15.10. Is there a planar graph with 5 vertices, with an edge between every pair of vertices? We call this graph K_5 .

For the remainder of the problems, we will assume that all graphs are simple. That is, there are no edges connecting a vertex to itself, and there is at most one edge between each pair of vertices.

PROBLEM 15.11. Show that if G is a connected planar graph with $V \geq 3$ vertices and E edges, then $E \leq 3V - 6$.

DEFINITION. The *degree* $\deg(v)$ of a vertex v of a graph is the number of edges containing it.

PROBLEM 15.12. Show that if G is a planar graph, then it has a vertex v with $\deg(v) \leq 5$.

If we take a plane and bunch it up so that the points far away meet at a point at ∞ , then we end up with a (possibly distorted) sphere, so we can think of a sphere as being a plane together with an extra point at ∞ . Thus any planar graph can be considered instead as a spherical graph, and again we have $V - E + F = 2$. For any spherical graph, we can think of it as being a planar graph by making sure that the point at ∞ is in the interior of some face, after which that face becomes the outer face of the corresponding planar graph.

Now, if we take the outside of any convex polyhedron, then we can inflate it to a sphere, in which case we have a spherical graph.

PROBLEM 15.13. Use the formula $V - E + F = 2$ for spherical graphs to find an expression for V, E, F for Platonic solids in terms of the Schläfli symbols $\{p, q\}$.

PROBLEM 15.14. Suppose we have a convex polyhedron made up of only pentagons and hexagons, with exactly three faces meeting at each vertex. How many pentagons must it contain?

Further reading. To learn more about planar graphs and their vertices, edges, and faces, see [Wil10, §4.2].

Identification Spaces and Surfaces

If you take a square and glue the opposite edges together, you get a cylinder, as shown in Figure 1.

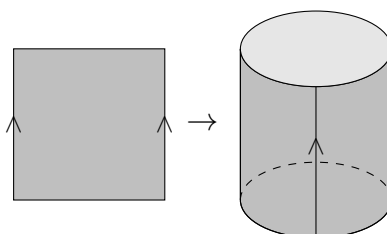


Figure 1. Gluing opposite edges of a square yields a cylinder.

PROBLEM 16.1. What happens when you glue the opposite edges of the cylinder? Equivalently, what happens when you glue the edges as marked in Figure 2? Note that you can't do this with paper, but you can do it using a stretchy material.

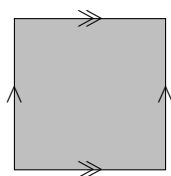


Figure 2. What happens when you glue the marked edges here?

PROBLEM 16.2. What happens when you glue the marked edges in Figure 3?

A figure like the one in Figure 3 is called an *identification space*. An identification space consists of a polygon, together with instructions for how to glue the edges.

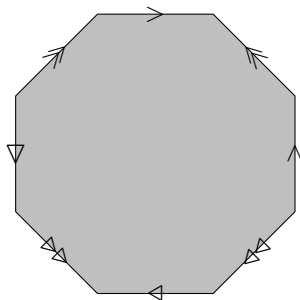


Figure 3. An octagon with edges ready to be glued.

When you draw a graph on a surface, an edge can cross over the boundary and continue across the other edge it's glued to.

PROBLEM 16.3. Show that it is possible to draw K_5 and $K_{3,3}$ with no crossing edges on the surface indicated by Figure 2, known as a *torus*.

PROBLEM 16.4. More generally, let K_n be the complete graph, which consists of n vertices and edges between every pair. Show that it is possible to draw K_6 and K_7 with no crossing edges on a torus.

The most natural graphs to draw on a torus are connected graphs such that each face can be continuously shrunk to a point, i.e. they don't go around any "hole" of the torus. We will call such graphs *good* graphs; this isn't a standard term, so its scope will be limited to this worksheet.

PROBLEM 16.5. Draw some good graphs on a torus and count the number of vertices V , edges E , and faces F . Do you notice any patterns?

PROBLEM 16.6. Go through the argument showing that for planar graphs, $V - E + F = 2$, but for good graphs on a torus. Which parts of the argument still hold? Which parts need modification? Can you supply the appropriate modifications? The value of $V - E + F$ should always be the same and match the one that you found in the previous problem, but the value will not be 2.

PROBLEM 16.7. Show that we cannot draw K_8 on a torus with no crossing edges.

Since the number $V - E + F$ of a good graph on a surface is independent of the choice of good graph, it is an invariant of the surface alone. We call it the *Euler characteristic* of a surface.

PROBLEM 16.8. What is the Euler characteristic of the surface shown in Figure 3? Work with the easiest good graph possible.

PROBLEM 16.9. Is it possible to glue the edges of the octagon differently, still in pairs, to get a different Euler characteristic? What values can you get?

PROBLEM 16.10. By now, you presumably know that the octagon in Figure 3 is a double-holed torus. Find an identification space for a triple-holed torus, and more generally for an n -holed torus. What is the Euler characteristic of an n -holed torus?

Further reading. For more on identification spaces, see [Wee02, Chapters 1–4].

WORKSHEET 17

Modular Arithmetic

Let m be a positive integer, and let a and b be integers. We say that a is congruent to b modulo m and write $a \equiv b \pmod{m}$ if $a - b$ is a multiple of m .

PROBLEM 17.1. Show that $a \equiv b \pmod{m}$ if and only if a and b leave the same remainder when divided by m .

PROBLEM 17.2. Show that there is a unique integer b between 0 and $m - 1$ for which $a \equiv b \pmod{m}$.

PROBLEM 17.3. Show that if $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$.

PROBLEM 17.4. Show that if $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$, $a - c \equiv b - d \pmod{m}$, and $ac \equiv bd \pmod{m}$.

PROBLEM 17.5. Use the previous problem to determine, relatively quickly, what day of the week it was on January 1, 1900. (Make sure you know all the rules for leap years!)

PROBLEM 17.6. Suppose $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, and $\frac{a}{c}$ and $\frac{b}{d}$ are both integers. Is it necessarily true that $\frac{a}{c} \equiv \frac{b}{d} \pmod{m}$?

PROBLEM 17.7. Suppose $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, where c and d are nonnegative integers. Is it necessarily true that $a^c \equiv b^d \pmod{m}$?

PROBLEM 17.8. Is $2222^{5555} + 5555^{2222}$ divisible by 7?

PROBLEM 17.9. You are likely familiar with some divisibility rules:

- A number is divisible by 3 if and only if the sum of its digits is divisible by 3.
- A number is divisible by 9 if and only if the sum of its digits is divisible by 9.
- A number is divisible by 11 if and only if the *alternating* sum of its digits is divisible by 11. (The alternating sum of the digits of $n = a_0a_1a_2 \cdots a_k$ is $a_0 - a_1 + a_2 - \cdots + (-1)^k a_k$.)

Explain why all of these are true.

PROBLEM 17.10. For each integer m from 3 to 8, determine those integers b with $0 \leq b \leq m - 1$ for which there exists an integer x such that $x^2 \equiv b \pmod{m}$.

PROBLEM 17.11. Show that if $n \equiv 3 \pmod{4}$, then there do not exist two integers x and y such that $x^2 + y^2 = n$.

PROBLEM 17.12. Show that if $n \equiv 7 \pmod{8}$, then there do not exist three integers x , y , and z such that $x^2 + y^2 + z^2 = n$.

For the remainder of this worksheet, let p be an *odd* prime, and let $h = \frac{p-1}{2}$.

PROBLEM 17.13. Suppose that $p \nmid c$ and $ac \equiv bc \pmod{p}$. Show that $a \equiv b \pmod{p}$.

PROBLEM 17.14. Let a be an integer not divisible by p . Use Bézout's Lemma (see Worksheet 7) to show that there is an integer b such that $ab \equiv 1 \pmod{p}$. Show that there is only one such b such that $1 \leq b \leq p - 1$. We call this value of b the *multiplicative inverse* of a modulo p , and we write it as a^{-1} .

PROBLEM 17.15. Suppose x and y are not divisible by p . Show that $x^2 \equiv y^2 \pmod{p}$ if and only if $x \equiv y \pmod{p}$ or $x \equiv -y \pmod{p}$.

PROBLEM 17.16. Show that if $1 \leq a, b \leq h$ and $a^2 \equiv b^2 \pmod{p}$, then $a = b$. In other words, the numbers $1^2, 2^2, \dots, h^2$ are pairwise incongruent modulo p .

PROBLEM 17.17. Show that if x is any integer not divisible by p , then there is a unique y with $1 \leq y \leq h$ such that $x^2 \equiv y^2 \pmod{p}$.

DEFINITION. Let p be a prime and let a be an integer. We say that a is a *quadratic residue* modulo p if a is not a multiple of p , and there exists an integer x such that $x^2 \equiv a \pmod{p}$. If there is no integer x such that $x^2 \equiv a \pmod{p}$, then we call a a *quadratic nonresidue* modulo p . We write the symbol $\left(\frac{a}{p}\right)$ which is 1 if a is a quadratic residue modulo p , -1 if a is a quadratic nonresidue modulo p , and 0 if a is a multiple of p .

PROBLEM 17.18. For how many integers a with $1 \leq a \leq p - 1$ is $\left(\frac{a}{p}\right) = 1$?

PROBLEM 17.19. Compute $4! \pmod{5}$, $6! \pmod{7}$, and $10! \pmod{11}$. Make a conjecture about $(p - 1)! \pmod{p}$.

PROBLEM 17.20. Let x be one of the numbers $1, 2, \dots, p - 1$. Show that $x \equiv x^{-1} \pmod{p}$ if and only if $x = 1$ or $x = p - 1$.

PROBLEM 17.21. Prove *Wilson's Theorem*, which says that $(p - 1)! \equiv -1 \pmod{p}$.

We will now compute $(p-1)!$ in a different way. Let a be an integer that is not divisible by p . For each $x \in \{1, 2, \dots, p-1\}$, let $T_a(x)$ be the unique integer in $\{1, 2, \dots, p-1\}$ such that $T_a(x) \equiv ax^{-1} \pmod{p}$.

PROBLEM 17.22. Show that $T_a(T_a(x)) = x$ for all $x \in \{1, 2, \dots, p-1\}$. Conclude that the numbers $\{1, 2, \dots, p-1\}$ are split into pairs by T_a , except that some numbers are paired with themselves. What is the product of the numbers in each pair?

PROBLEM 17.23. Identify those values of x for which $T_a(x) = x$. Relate the number of such values of x to a Legendre symbol.

PROBLEM 17.24. Show that $a^h \equiv \left(\frac{a}{p}\right) \pmod{p}$. This is known as *Euler's criterion*.

PROBLEM 17.25. Show that $\left(\frac{a}{p}\right)\left(\frac{b}{p}\right) = \left(\frac{ab}{p}\right)$.

PROBLEM 17.26. For which primes p is $\left(\frac{-1}{p}\right) = 1$? Can you explain why?

PROBLEM 17.27. Gather data and make a conjecture about which primes p satisfy $\left(\frac{2}{p}\right) = 1$. Can you explain why?

Further reading. For more on modular arithmetic, see [Sil13, Chapters 8–11]. For more on quadratic residues, see [Sil13, Chapters 20–23]. Another good source is [Dav08, Parts II and III].

WORKSHEET 18

Vieta's Formulae

There are two basic ways we can represent a polynomial: either in terms of its coefficients, or in terms of its roots. That is, if we have a polynomial $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, with complex roots $r_1, r_2, \dots, r_n$, then we can factor it as $f(x) = a_n(x - r_1)(x - r_2) \cdots (x - r_n)$. Note that f might have fewer than n distinct complex roots, because some of them could be repeated, such as in the case of $f(x) = x^2 - 2x + 1 = (x - 1)^2$. We need to count each root with the correct multiplicity.

Since we have two representations for the same polynomial, there ought to be some relations between the two of them.

PROBLEM 18.1. Suppose $f(x) = x^2 + ax + b = (x - r_1)(x - r_2)$. Express a and b in terms of r_1 and r_2 .

PROBLEM 18.2. Suppose $f(x) = x^3 + ax^2 + bx + c = (x - r_1)(x - r_2)(x - r_3)$. Express a , b , and c in terms of r_1 , r_2 , and r_3 .

PROBLEM 18.3. More generally, suppose $f(x) = x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1x + a_0 = (x - r_1)(x - r_2) \cdots (x - r_n)$. Express the a_i 's in terms of the r_j 's.

PROBLEM 18.4. Finally, do the general case, where $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = a_n(x - r_1)(x - r_2) \cdots (x - r_n)$. The formulae you should have ended up with are called *Vieta's formulae*.

As you can see, the case where the leading coefficient (a_n) is equal to 1 can be used to do the general case and leads to slightly cleaner formulae, so from now on we'll just assume that $a_n = 1$ unless otherwise noted. A polynomial with leading coefficient equal to 1 is said to be a *monic polynomial*.

PROBLEM 18.5. Let $f(x) = x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1x + a_0$ have roots $r_1, r_2, \dots, r_n$. Express $r_1^2 + r_2^2 + \cdots + r_n^2$ in terms of the a_i 's.

PROBLEM 18.6. With notation as in the previous problem, assume none of the r_i 's are equal to 0. Express $\frac{1}{r_1} + \frac{1}{r_2} + \cdots + \frac{1}{r_n}$ in terms of the a_i 's.

PROBLEM 18.7. Let $f(x) = x^3 + ax^2 + bx + c$ have roots α, β, γ . Find a polynomial whose roots are $\alpha + 2, \beta + 2, \gamma + 2$, in terms of a, b, c .

PROBLEM 18.8. With notation as in the previous problem, find a polynomial whose roots are $\alpha^2, \beta^2, \gamma^2$.

PROBLEM 18.9. With notation as in the last two problems, find a polynomial whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$. (Assume that none of α, β, γ are 0.)

PROBLEM 18.10. With notation as in the last three problems, find a polynomial whose roots are $\alpha\beta, \beta\gamma, \alpha\gamma$.

PROBLEM 18.11. Solve the system of equations

$$\begin{aligned}\alpha + \beta + \gamma &= 17, \\ \alpha\beta + \beta\gamma + \alpha\gamma &= 92, \\ \alpha\beta\gamma &= 160.\end{aligned}$$

Further reading. Vieta's formulae lead naturally into the theory of symmetric functions and symmetric polynomials. For more on Vieta's formulae and symmetric functions, see [Dic22, Chapter IX].

WORKSHEET 19

Arithmetic Functions

An *arithmetic function* is a function from the positive integers to the complex numbers, but frequently to the integers. A typical arithmetic function is the function $d(n)$, which is the number of (positive) divisors of n .

PROBLEM 19.1. Write down the divisors of 60 in an organized manner, so that it is very easy to count them. What is $d(60)$?

PROBLEM 19.2. Find a formula for $d(n)$, in terms of the prime factorization of n .

PROBLEM 19.3. For which values of n is $d(n)$ odd?

A related function is $\sigma(n)$, which is the sum of the divisors of n .

PROBLEM 19.4. Find a formula for $\sigma(n)$, in terms of the prime factorization of n .

PROBLEM 19.5. Let $\beta(n)$ be the number of divisors of n congruent to 1 (mod 4), minus the number of divisors of n congruent to 3 (mod 4). Find a formula for $\beta(n)$, in terms of the prime factorization of n .¹

The reason that there are such nice formulae for $d(n)$ and $\sigma(n)$ is that these functions are of a special type.

PROBLEM 19.6. Suppose that m and n are relatively prime, i.e. $\gcd(m, n) = 1$. Show that $d(mn) = d(m)d(n)$ and $\sigma(mn) = \sigma(m)\sigma(n)$. Show that neither of these is necessarily true if $\gcd(m, n) > 1$.

An arithmetic function f is said to be *multiplicative* if $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$. It is said to be *totally multiplicative* if $f(mn) = f(m)f(n)$ for all positive integers m and n .

PROBLEM 19.7. Show that if f is a multiplicative function, then the values of f on the powers of primes determine the values of f on all positive integers. Show that if f is totally multiplicative, then the values of f on the primes determine the values of f on all positive integers.

¹The notations $d(n)$ and $\sigma(n)$ as above are standard, whereas β is just a notation I made up for the sake of this problem.

Let n be a positive integer. A positive integer $a \leq n$ is said to be a *totative* for n if $\gcd(a, n) = 1$. Euler's *totient function* $\phi(n)$ is the number of totatives for n .

PROBLEM 19.8. Show that $\phi(n)$ is a multiplicative function, and use this to find a formula for $\phi(n)$ in terms of the prime factorization of n .

The *Möbius function* $\mu(n)$ is the multiplicative function defined on the prime powers by

$$\mu(p^k) = \begin{cases} 1 & k = 0, \\ -1 & k = 1, \\ 0 & k \geq 2. \end{cases}$$

We write $d \mid n$ to say that d is a divisor of n . If f is an arithmetic function, then we write $\sum_{d \mid n} f(d)$ for the sum of the f -values of the divisors of n .

PROBLEM 19.9. Let $g(n) = \sum_{d \mid n} f(d)$. Show that if f is multiplicative, then g is as well.

PROBLEM 19.10. Evaluate the sums

$$\sum_{d \mid n} \mu(d)$$

and

$$\sum_{d \mid n} \phi(d).$$

PROBLEM 19.11. If $g(n) = \sum_{d \mid n} f(d)$, can you recover f from g somehow? How do you do it?

Further reading. Arithmetic functions are a major topic in analytic number theory. For more on the sort of treatment of arithmetic functions given in this worksheet, see [Apo76, Chapter 2], and then see the rest of the book [Apo76] for much more you can do with arithmetic functions.

WORKSHEET 20

The Peano Axioms

In this worksheet, we will rigorously develop the set $\mathbb{N}$ of nonnegative integers from a set of axioms, known as the *Peano axioms*. The point is that, while you may be familiar using certain facts about the nonnegative integers, it is also important to know how to prove them from first principles and get used to the axiomatic approach to mathematics, so that in the future when you have to define more exotic objects, you are prepared to deal with the formalism that that entails. When you solve these problems, do not use anything you think you know from elsewhere about the nonnegative integers; you may only use the axioms and previous facts you have deduced from them.

Here are the Peano axioms for constructing the set $\mathbb{N}$ of nonnegative integers:

- (1) $\mathbb{N}$ contains an element called 0.
- (2) There is a function $S : \mathbb{N} \rightarrow \mathbb{N}$, called the *successor function*.
- (3) S is injective, i.e. if $S(a) = S(b)$, then $a = b$.
- (4) There is no element n in $\mathbb{N}$ such that $S(n) = 0$.
- (5) (Principle of Induction.) Suppose A is a subset of $\mathbb{N}$, i.e. a collection of some (possibly none, possibly all) of the elements of $\mathbb{N}$, such that 0 is in A , and whenever n is in A , then $S(n)$ is in A as well. Then $A = \mathbb{N}$.

The successor function S is the axiomatic version of the “add one” function, i.e. we should think of $S(n)$ as $n+1$. However, at the moment, addition has not yet been defined.

The general strategy for proving statements about $\mathbb{N}$ in terms of the Peano axioms is to follow this template:

- We begin by proving $P(0)$.
- We prove that for all $n \in \mathbb{N}$, $P(n)$ implies $P(S(n))$.

If the statement also contains parameters $a, b, \dots$, fix those parameters before beginning the induction. They are arbitrary but remain unchanged during the induction. If no special properties of the parameters are assumed, then the final choice holds for every choice of them.

PROBLEM 20.1. In this problem, we will show that for all n in $\mathbb{N}$ other than $n = 0$, there exists some a in $\mathbb{N}$ such that $n = S(a)$. Let $A = \{0\} \cup \{S(a) : a \text{ in } \mathbb{N}\}$.

- Show that 0 is in A .
- Show that if n is in A , then $S(n)$ is in A .
- Use the Principle of Induction to show that $A = \mathbb{N}$.
- Conclude that every $n \in \mathbb{N}$ other than $n = 0$ is $S(a)$ for some a .
- Show that this a is unique.

We next define addition, again directly in terms of the axioms. We begin with addition by 0: for all a in $\mathbb{N}$, we define $a + 0$ to be a . Next, if a and b are in $\mathbb{N}$, then we define

$$a + S(b) = S(a + b).$$

PROBLEM 20.2. Define $1 = S(0)$ and $2 = S(1)$. Verify that you can express $a + 1$ and $a + 2$ in terms of the successor function, directly from the definition of addition.

PROBLEM 20.3. In this problem, we will show that $0 + b = b$ for all b in $\mathbb{N}$. Let $A = \{b \text{ in } \mathbb{N} : 0 + b = b\}$.

- Show that 0 is in A .
- Show that if b is in A , then $S(b)$ is in A .
- Conclude that $A = \mathbb{N}$, hence that $0 + b = b$ for all b in $\mathbb{N}$.

PROBLEM 20.4. Fix an arbitrary a in $\mathbb{N}$. Show that $S(a) + b = S(a + b)$ for all b in $\mathbb{N}$.

Since a is arbitrary, this shows that $S(a) + b = S(a + b)$ for all a and b in $\mathbb{N}$. Note that we did not need to perform a separate induction on a .

This is the typical model for showing a statement involving two or more variables.

PROBLEM 20.5. Show that if a and b are in $\mathbb{N}$, then $a + b = b + a$.

PROBLEM 20.6. Show that if a, b, c are in $\mathbb{N}$, then $a + (b + c) = (a + b) + c$.

PROBLEM 20.7. Show that if a, b, c are in $\mathbb{N}$ and $a + c = b + c$, then $a = b$.

PROBLEM 20.8. Show that if $a + b = 0$, then $a = b = 0$.

Next, we define multiplication. As with addition, we start with multiplication by 0: if a is in $\mathbb{N}$, we define $a \cdot 0 = 0$. Next, if a and b are in $\mathbb{N}$, we define

$$a \cdot S(b) = a \cdot b + a.$$

PROBLEM 20.9. Verify that you can express $a \cdot 1$ and $a \cdot 2$ in terms of addition, directly from the definition of multiplication.

PROBLEM 20.10. Show that $0 \cdot b = 0$ for all b .

PROBLEM 20.11. Show that $S(a) \cdot b = a \cdot b + b$.

PROBLEM 20.12. Show that if a, b, c are in $\mathbb{N}$, then $a \cdot (b + c) = a \cdot b + a \cdot c$.

PROBLEM 20.13. Show that if a, b, c are in $\mathbb{N}$, then $a(bc) = (ab)c$.

PROBLEM 20.14. Show that if a and b are in $\mathbb{N}$, then $ab = ba$.

We also need to talk about the ordering on $\mathbb{N}$. If a and b are in $\mathbb{N}$, we say that $a \leq b$ if there is some c in $\mathbb{N}$ such that $a + c = b$. We say that $a < b$ if $a \leq b$ and $b \not\leq a$.

PROBLEM 20.15. Show that $0 \leq a$ for all a in $\mathbb{N}$.

PROBLEM 20.16. Show that $a \leq a$ for all a in $\mathbb{N}$.

PROBLEM 20.17. Show that $a < S(a)$ for all a in $\mathbb{N}$.

PROBLEM 20.18. Show that if $a \leq b$ and $b \leq c$, then $a \leq c$.

PROBLEM 20.19. Show that if $a \leq b$ and $b \leq a$, then $a = b$.

PROBLEM 20.20. Show that $a \leq b$ if and only if $S(a) \leq S(b)$.

PROBLEM 20.21. Show that if a and b are in $\mathbb{N}$, then exactly one of the following is true:

- $a < b$,
- $a = b$,
- $b < a$.

Further reading. For more on the Peano axioms in the same spirit as this worksheet, see [Whi17]. For a more detailed look, see [Kör19, Chapter 6].

Basic Infinite Set Theory

We say that two sets A and B have the same *cardinality* if there is a way of matching up every element of A with a unique element of B , and vice versa. See Figure 1 for an example. We write $|A| = |B|$ if A and B have the same cardinality. The matching function giving us an element of B for each element of A is called a *bijection*.

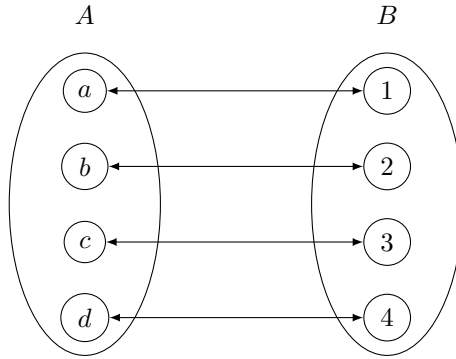


Figure 1. A one-to-one correspondence between two sets.

PROBLEM 21.1. Show that if A and B are finite sets, then $|A| = |B|$ if and only if A and B have the same number of elements.

PROBLEM 21.2. Show that if $|A| = |B|$ and $|B| = |C|$, then $|A| = |C|$.

If A and B are finite sets, then the concept of cardinality agrees with what you already know about counting. But when they are infinite, things get weirder!

PROBLEM 21.3. Let $\mathbb{N} = \{0, 1, 2, \dots\}$ be the set of nonnegative integers, and let $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$ be the set of positive integers. Do $\mathbb{N}$ and $\mathbb{Z}^+$ have the same cardinality? If so, give a bijection between them.

PROBLEM 21.4. Let $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ be the set of integers. Do $\mathbb{N}$ and $\mathbb{Z}$ have the same cardinality? If so, give a bijection between them.

PROBLEM 21.5. Let $\mathbb{Q}$ be the set of rational numbers, i.e. numbers of the form $\frac{a}{b}$ where a and b are integers and $b \neq 0$, where two such expressions are considered the same if they are equal as fractions. (So, for instance, $\frac{1}{2}$ and $\frac{3}{6}$ are the same, even though they are written differently.) Do $\mathbb{N}$ and $\mathbb{Q}$ have the same cardinality? If so, give a bijection between them.

So far (spoiler alert!), all the infinite sets we've discussed have the same cardinality. We call this cardinality $\aleph_0$. The letter $\aleph$ is *aleph*, the first letter of the Hebrew alphabet. You may find it surprising at this point that, say, $\mathbb{N}$ and $\mathbb{Q}$ have the same cardinality, because $\mathbb{Q}$ seems to be much larger than $\mathbb{N}$, so you would be forgiven for suspecting at this point that all infinite sets have the same cardinality. However, this is not the case!

One infinite set that does *not* have cardinality $\aleph_0$ is the set $\mathbb{R}$ of real numbers. The idea to showing this is start with *any* function $f : \mathbb{N} \rightarrow \mathbb{R}$ and then show that it cannot possibly be a bijection because some element of $\mathbb{R}$ will not be of the form $f(n)$ for any n in $\mathbb{N}$. It is technically easier, in fact, to show that there cannot be a bijection from $\mathbb{N}$ to $[0, 1)$, the set of real numbers x such that $0 \leq x < 1$.

PROBLEM 21.6. Suppose $f : \mathbb{N} \rightarrow [0, 1)$ is any function. Find a number whose first digit after the decimal point is not equal to that of $f(0)$, whose second digit after the decimal point is not equal to that of $f(1)$, whose third digit after the decimal point is not equal to that of $f(2)$, and so on.

We would like to conclude that the resulting number is not of the form $f(n)$ for any n . (Why?) But there is a slight issue, namely that some numbers have more than one decimal expansion. For instance, $0.09999\ldots = 0.1$. However, the only numbers with two decimal expansions are ones where one expansion ends with all 9's, and the other ends with all 0's.

PROBLEM 21.7. Modify the construction from the previous problem to avoid this issue.

PROBLEM 21.8. Conclude that $[0, 1)$ does not have the same cardinality as $\mathbb{N}$.

PROBLEM 21.9. Show that $[0, 1)$ has the same cardinality as $\mathbb{R}$.

PROBLEM 21.10. Conclude that $\mathbb{N}$ does not have the same cardinality as $\mathbb{R}$.

We call the cardinality of $\mathbb{R}$ *the continuum* and denote it either $\mathfrak{c}$ or $2^{\aleph_0}$.

Another interesting construction is the *power set*. Given any set A , its *power set* $\mathcal{P}(A)$ is the set of all subsets of A . For instance, if $A = \{1, 2\}$, then

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}.$$

Thus the cardinality of $\mathcal{P}(A)$ is 4. Note that $\emptyset = \{\}$ denotes the empty set, i.e. the set containing no elements.

PROBLEM 21.11. Suppose A is a finite set with n elements. Show that $\mathcal{P}(A)$ is a finite set. How many elements does it have?

PROBLEM 21.12. Show that if A is any set, then A and $\mathcal{P}(A)$ do not have the same cardinality. (You can use a very similar technique to the one we used to show that $\mathbb{N}$ and $[0, 1)$ do not have the same cardinality.)

PROBLEM 21.13. Show that the set of infinite binary sequences has the same cardinality as the set of infinite ternary (i.e. base 3) sequences.

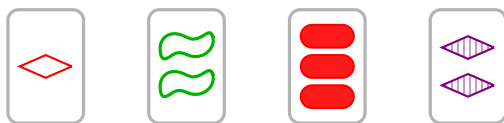
PROBLEM 21.14. Show that $\mathbb{R}$ and $\mathcal{P}(\mathbb{N})$ have the same cardinality.

Further reading. See [Vel06, Chapter 7] and [RS23, Chapters 23 and 24] for more about infinite set theory.

WORKSHEET 22

The Game of Set

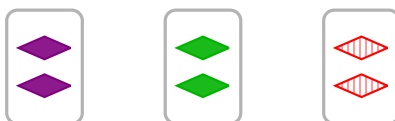
In this worksheet, we'll look at some of the mathematical properties of the popular game of SET. The game of SET is played with a deck of cards. Each card has four attributes: number, color, shape, and shading. There are three possibilities for each attribute. For number, the possibilities are 1, 2, and 3. For color, they are red, green, and purple. For shape, they are diamonds, ovals, and squiggles. For shading, they are hollow, shaded, and solid. Some sample cards are



A *set* consists of three cards such that, in every attribute, all the cards are either all the same or all different. For instance, the cards



form a set. They all have the same number and the same color, but they all have different shape and shading. On the other hand, the cards



do not form a set, because two of them are solid and one is shaded.

PROBLEM 22.1. In a SET deck, there is exactly one card with each type for each attribute. How many cards does the deck contain?

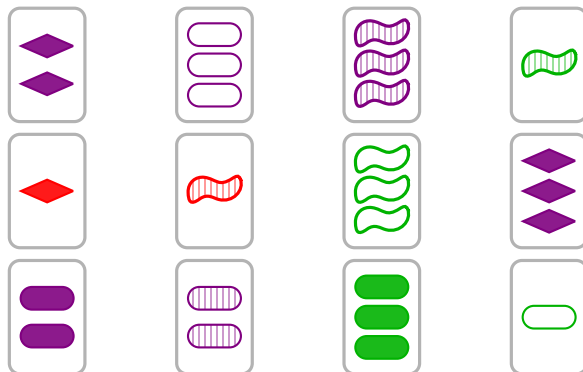
PROBLEM 22.2. Show that, for any two cards, there is exactly one card in the deck forming a set with the first two.

PROBLEM 22.3. What is the total number of sets?

PROBLEM 22.4. We can further subdivide based on the number of attributes where all three cards are the same. How many sets are there with 0, 1, 2, and 3 common attributes?

PROBLEM 22.5. We can represent a SET card as an ordered quadruple (a, b, c, d) , where each a, b, c, d is 1, 2, or 3. The first coordinate represents number, the second coordinate color, the third coordinate shape, and the fourth coordinate shading. If we use the ordering red, green, purple for color, diamond, oval, squiggle for shape, hollow, shaded, and solid for shading, then the quadruple $(1, 1, 2, 3)$ represents the card with one red solid oval. Suppose we have three cards represented by (a_1, b_1, c_1, d_1) , (a_2, b_2, c_2, d_2) , and (a_3, b_3, c_3, d_3) . What is the condition on these numbers that guarantees that they form a set?

Here is how to play SET as a game. First, twelve cards are dealt face up on the table in a 3×4 grid, like this:



Then, anyone who finds a set calls out the word “set!” and points to the three cards forming the set. Then the cards are removed, and the player who found the set collects those cards, to get points at the end of the game. Then three new cards are dealt to replace that set.

PROBLEM 22.6. Find a set in the arrangement above. Is there more than 1? How many are there?

Sometimes, however, there are no sets. In that case, when the players agree that there are no sets on the table, then three more cards are dealt, and the game continues. If there are 15 cards on the table, then the players try to find two sets before more cards are dealt, if possible, so that by default there are 12 cards on the table at any time. However, the number can dip below 12 if there are no more cards left in the deck.

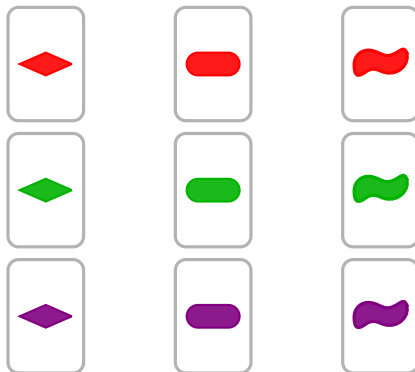
PROBLEM 22.7. Find an arrangement of 12 cards with no set. Can you find more than 12 cards with no set? What is the largest set-free collection you can find? (Finding, with proof, the largest possible set-free collection is challenging, so find one that is as large as you are able to.)

PROBLEM 22.8. A common convention is to deal the last card in the deck face down. It can still be used as part of a set, but the players

must determine what the card would be. Without memorizing all the cards that have been removed, how can they determine the last card?

PROBLEM 22.9. Suppose at the end of the game, there are three cards left on the table. Must they form a set?

Suppose we restrict to just two attributes, so we have 9 cards. For example, if we fix the shading as solid and number as 1, and vary only color and shape, we have the following 9 cards:



PROBLEM 22.10. How many ways are there to partition up these nine cards into three disjoint “parallel” sets?

PROBLEM 22.11. Are there other collections of 9 cards for which there are the same number of ways of partitioning them into three disjoint parallel sets? Of course, the answer is yes, because we can (for instance) make 2 of each shape, or make them all hollow, or various other things. But are there any substantially different such collections? (Figure out first what you think that should mean.)

Further reading. See [MGGG16] for a book about the mathematics of SET.

WORKSHEET 23

A Taste of Knot Theory

A knot, in mathematics, is a way of taking a string, placing it into three-dimensional space $\mathbb{R}^3$, and then gluing the edges together. Because our paper is only 2-dimensional, we cannot draw knots fully, but only projections of them onto the plane $\mathbb{R}^2$. But when we project them, we may have two (or more) points on the knot projecting to the same point on the plane. We will always consider projections such that only at most two points ever get projected to the same point on the plane, and there are only finitely many pairs of points getting projected to the same point. Thus we need only indicate which point is higher and which is lower. We indicate this by making a gap in the lower strand. For instance, in Figure 1, the red strand is above the blue strand. See Figure 2 for a full knot with all the crossings indicated in this way.



Figure 1. The red strand is above the blue strand.

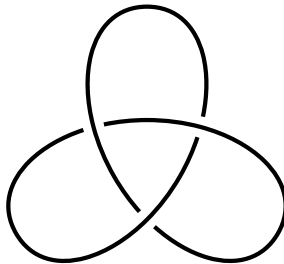


Figure 2. A trefoil knot.

A knot projection splits a knot into *strands*. At every undercrossing, we cut the underpassing part of the curve at the crossing. The

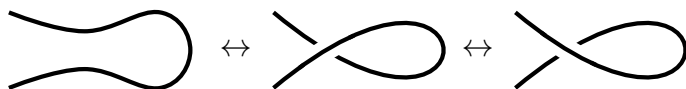


Figure 3. Reidemeister 1.

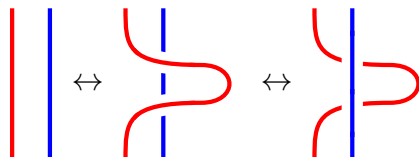


Figure 4. Reidemeister 2.

connected pieces that remain are the strands. A strand may pass over several crossings, but it may not pass under any crossings.

When we have two knots, or more precisely knot projections, we would like to know whether it is possible to deform the string continuously to get from one to the other. One question in particular is to determine when we can unknot a knot: can we deform it so that it becomes a circle (or an *unknot* in knot theory speak)? There exist methods to solve this problem in general, but all of them involve some rather advanced machinery.

Still, there are some things we can do. The first thing to do is to work out some elementary “moves” to get from one knot diagram to another equivalent knot diagram, i.e. one that can be obtained from deforming the string. There are three basic moves, called the *Reidemeister moves* and denoted R1, R2, and R3. See Figures 3, 4, and 5. All of these figures are intended to be embedded in larger knot diagrams, such that the rest of the diagram is unchanged by the Reidemeister moves. Reidemeister’s Theorem says that these are the only moves needed: given any two equivalent knot diagrams, it is possible to get from one to the other via a sequence of Reidemeister moves.

Let us now use Reidemeister moves to construct some invariants of knots, i.e. functions of knot projections such that two equivalent knots have the same value under the function.

PROBLEM 23.1. Suppose that I is a function on knot projections, say to $\mathbb{R}$, i.e. if K is any knot projection, then $I(K)$ is a real number. Show that if whenever K_1 and K_2 are related to each other by a Reidemeister move we have $I(K_1) = I(K_2)$, then I is a knot invariant.

One such function is the number of *Fox 3-colorings*. Note that at every crossing in a knot projection, there are three strands emanating

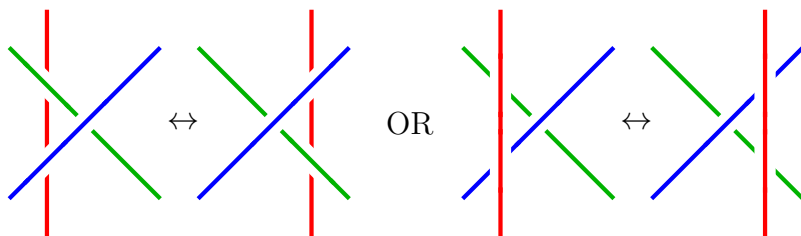


Figure 5. Reidemeister 3.

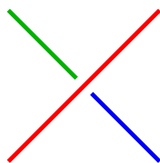


Figure 6. The three strands emanating from a crossing, marked in different colors.

from it: the overcrossing strand, and the two broken pieces of the undercrossing strand, as shown in Figure 6. A Fox 3-coloring is a way of coloring all the strands in a knot diagram in such a way that at each crossing, either all three strands are the same color, or else all three strands are different colors; this is called the *crossing condition*. Note that a Fox 3-coloring is a complete color assignment to all strands of the knot diagram simultaneously such that every crossing satisfies the crossing condition.

REMARK. Note that, while the crossing condition is a local test, looking at a single crossing at a time, a Fox 3-coloring is a global object, coloring the entire diagram at once. We are not counting crossings; we are counting complete assignments of colors to all the strands. If a coloring satisfies the crossing condition at every crossing, then it contributes 1 to the count of Fox 3-colorings. If it fails the crossing condition even at a single crossing, then it contributes 0.

PROBLEM 23.2. Show that the number of Fox 3-colorings is a knot invariant.

PROBLEM 23.3. Show that the trefoil knot is not equivalent to the unknot.

PROBLEM 23.4. Show that the Borromean rings, shown in Figure 7, cannot be unlinked, even though any two of them are unlinked. (Note that the Borromean rings are not a knot, but rather a *link*, which consists of several nonintersecting knots.)

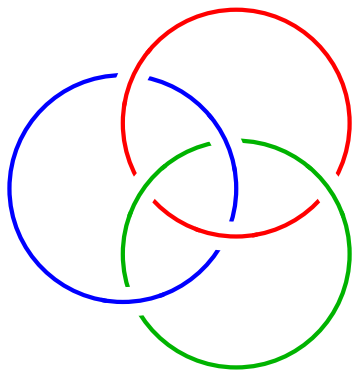


Figure 7. Borromean rings. Note that here we are coloring the individual links with different colors; this does *not* represent a Fox 3-coloring.

PROBLEM 23.5. Show that the number of Fox 3-colorings of a knot is always divisible by 3.

It turns out that the number of Fox 3-colorings is always a power of 3. The usual way of showing this involves linear algebra, but we will give a self-contained argument that unpacks the linear algebra over the course of the next several problems.

PROBLEM 23.6. Replace the colors with the numbers 0, 1, and 2, in some order. Find a simple interpretation for the condition that when three strands meet at a crossing, the colors must either all be the same or all different, in terms of modular arithmetic.

We can now let each strand be a variable, taking on the values 0, 1, and 2, so if there are n strands, then the condition of being a Fox 3-coloring is a system of equations (one for each crossing) in n variables. The number of Fox 3-colorings is the number of solutions of this system of equations.

PROBLEM 23.7. Suppose that, in a system of equations modulo 3, if there is some variable, say x_m , that doesn't appear in any of the equations. Show that once we have solved for all the other variables, we can assign x_m any of the 3 values, independently of the other variables.

PROBLEM 23.8. Suppose x_m occurs in some equation, with coefficient either 1 or 2 (or -1). Show that this equation determines x_m uniquely once we know the values of all the other variables.

PROBLEM 23.9. In the setting of the previous problem, substitute this expression for x_m into all the other equations. If we consider the new system of equations without x_m in it, then show that any solution

of this new system corresponds to a unique solution of the old system, and vice versa. Thus it suffices to count the number of solutions to this new system.

PROBLEM 23.10. What happens when you have repeated the processes in the previous problems as many times as you can? (Try your procedure with some of the knots and links we have seen so far, so that you can see concretely what happens at the end.)

PROBLEM 23.11. Show that the number of Fox 3-colorings of a knot is always a power of 3.

PROBLEM 23.12. How do you generalize from Fox 3-colorings to Fox n -colorings? There are many possible ideas, but make sure that your way is actually a knot invariant; many attempts here won't work!

PROBLEM 23.13. Find a knot that cannot be distinguished from the unknot by the number of Fox 3-colorings, but that can be distinguished by Fox n colorings for some $n > 3$.

Further reading. For more on Fox 3-colorings and other knot colorings, see [Liv93, Chapter 3]. The algebraic interpretation of Fox 3-colorings in terms of modular arithmetic and the rules that they are forced to satisfy by the Reidemeister moves give rise to objects called *quandles*. For accessible introductions to quandles, see [Aus16] and [Prz19], or see [EN15] for a book on the subject.

WORKSHEET 24

Introduction to Game Theory

The most basic type of game in (economic) game theory is the *matrix game*. It is played by two players, who will be named Rose and Colin. There is a matrix, and each entry of the matrix contains two numbers. The first number is Rose's payout, and the second number is Colin's payout. Both players simultaneously make their moves: Rose chooses one of the rows of the matrix, and Colin chooses one of the columns of the matrix. Both players wish to get as large of a payout as possible.

For instance, consider the game

		Colin	
		L	R
Rose	T	2, 3	0, 2
	B	3, 0	1, 1

Rose can choose the top row (T) or the bottom row (B), whereas Colin can choose the left column (L) or the right column (R). If Rose chooses T and Colin chooses L , then Rose gets a payout of 2, and Colin gets a payout of 3.

PROBLEM 24.1. If both players are playing the above matrix game optimally, what should they do, and what are the payouts?

PROBLEM 24.2. What are the optimal plays in this game?

		Colin	
		L	R
Rose	T	3, 2	1, 1
	B	0, 0	2, 3

PROBLEM 24.3. What about this one?

		Colin	
		<i>L</i>	<i>R</i>
Rose	<i>T</i>	1, -1	-1, 1
	<i>B</i>	-1, 1	1, -1

The rows (for Rose) and columns (for Colin) are called the *pure strategies* of the game. As you have seen, sometimes there is an optimal selection of pure strategies for each player, and sometimes there isn't. We call a strategy profile (i.e. a strategy for each player) a *Nash equilibrium* if neither player can improve their payoff, assuming that the other player's strategy remains unchanged.

PROBLEM 24.4. Find all the pure strategy Nash equilibria in the above games.

The players are not required to play pure strategies: they can use randomness. For instance, Rose could elect to play *T* with probability $\frac{2}{3}$ and *B* with probability $\frac{1}{3}$ in some matrix game. We call strategies like this that involve randomness *mixed strategies*. Just like pure strategies, mixed strategies can also be Nash equilibria. The payoff for a mixed strategy is the weighted average of the pure strategies involved in it. For instance, if Rose's payoffs for *T* and *B* are t and b , respectively, then in the mixed strategy where she plays *T* with probability $\frac{2}{3}$ and *B* with probability $\frac{1}{3}$, her payoff is $\frac{2}{3}t + \frac{1}{3}b$.

We say that a player is *indifferent* between two (pure or mixed) strategies if they yield the same payoff, when the other player's strategy is fixed.

PROBLEM 24.5. Show that in a mixed strategy Nash equilibrium, both players must be indifferent among all the pure strategies that are played with positive probability.

PROBLEM 24.6. Find all the Nash equilibria in the above games, allowing mixed strategies.

PROBLEM 24.7. Suppose that Rose (say) has two pure strategies *A* and *B* such that, for any pure strategy by Colin, Rose's payoff in *A* is strictly greater than that of *B*. Show that *B* cannot be played with positive probability in any Nash equilibrium. Furthermore, show that if we eliminate *B* from Rose's options, the Nash equilibria are the same as in the original game.

We call a strategy that is always worse than another strategy a *strictly dominated* strategy. In the matrix, we can cross out all strictly dominated strategies and ignore them. This operation may be performed multiple times. One possibility is that, by using this technique

of eliminating dominated strategies, we end up with just a single option for each player. We call this technique iterated elimination of dominated strategies.

PROBLEM 24.8. Can any of the above games be solved by the technique of iterated elimination of dominated strategies?

PROBLEM 24.9. Construct an example to show that it is not always possible to eliminate *weakly* dominated strategies without eliminating some of the Nash equilibria, where by a weakly dominated strategy we mean the same as a strictly dominated one, but with $\geq$ instead of $>$.

PROBLEM 24.10. Suppose there is a matrix entry where the payoff for each player is higher than that of any other entry. Show that this is a Nash equilibrium. Is it necessarily the only Nash equilibrium?

PROBLEM 24.11. What is the maximum number of Nash equilibria a 2×2 matrix game with all the entries distinct can have? What are all possible such values? Can you classify the circumstances under which each number appears?

PROBLEM 24.12. We can consider ROCK PAPER SCISSORS to be a matrix game, with matrix

		Colin		
		<i>R</i>	<i>P</i>	<i>S</i>
Rose	<i>R</i>	0, 0	-1, 1	1, -1
	<i>P</i>	1, -1	0, 0	-1, 1
	<i>S</i>	-1, 1	1, -1	0, 0

What are all the Nash equilibria?

Further reading. For more on game theory and Nash equilibria, see [Str93] and [Os04, Chapters 1–4].

Gerrymandering

In the US House of Representatives, each state is allotted a certain number of representatives, based on population. Each state then divides itself up into districts of (roughly) equal population, and then the people in each district vote for a representative, and the winner represents that district in the House of Representatives.

Clearly, the way of dividing the state up into regions can influence the winner, and in particular the political party of the winner, and that can have a significant impact on laws passed. We'll assume that there are two political parties, which we'll call the green party and the purple party.¹

Here are some things that we might hope for in good districts:

Equipopulous: Each region should have the same population.

Connected: Each region should be in one piece.

Compactness: The shapes of regions are fairly nice and geographically concentrated.

Fairness: The people who are likely to win an election somehow represent the will of the people, and not just the will of the people drawing the maps.

Equitable representation: The number of seats each party wins is somehow related to the number of people who vote for that party. (This is deliberately vague.)

Exchangeability: Each voter should have roughly an equal amount of say in the outcome of the election.

There are also some other constraints that have to be observed, most notably the Voting Rights Act, but since this is more of a political and historical point than a mathematical one, we won't discuss that here. Note that we aren't promising that these goals are easy or attainable, or even that we know precisely how to define them; these are just aspirational at the moment.

¹I wish to avoid red and blue, since those have political connotations. I am trying to avoid any political biases here; we're just discussing the underlying mathematical issues.

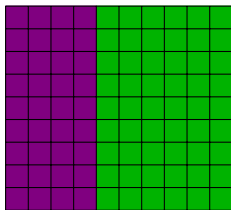


Figure 1. A sample state to split into districts.

It would be nice to have a clear understanding of what we mean by compactness and fairness and some of the other terms mentioned above, but this is complicated.

PROBLEM 25.1. How would *you* give a measure of compactness of a region? Discuss your answer with some friends and see if their measures of compactness are the same as yours or not. If they are different, how different are they? Are there maps such that, under one measure of compactness, the regions all have low compactness scores, and under the other one, some of the regions have high compactness scores? (Let us assume that a low score is compact and a high score is noncompact.)

PROBLEM 25.2. Same question, but now for fairness.

To test your measures, consider the toy state shown in Figure 1. We suppose that each square represents a person, with the parties being indicated by colors. Thus there are 90 people in the state, of whom 54 are in the green party and 36 are in the purple party. We'll also assume that the state is allotted 10 districts, and due to the equal population constraint, each district must contain 9 people.

PROBLEM 25.3. Split the state into 10 connected² districts so as to optimize for compactness. Then split it to optimize for fairness. Are these the same splitting? Can you optimize for both simultaneously?

Let's now look at some examples of compactness measures.

Skew measure: $\frac{L}{W}$, where L is the length and W is the width of the rectangle with the smallest area containing the region (chosen such that $L \geq W$).

Isoperimetric measure: $\frac{P^2}{A}$, where P is the perimeter and A is the area.

Total perimeter: The total perimeter of all the regions.

²For the grid districts we are considering in this worksheet, each region must consist of grid cells. Cells that only meet at a vertex are not considered to be adjacent.

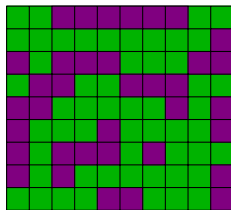


Figure 2. A more realistic state to split into districts.

For the skew measure and isoperimetric measure, perhaps we want to take the sum over all regions in the state; small numbers are more compact than large ones.

PROBLEM 25.4. Compute these measures for your splittings in the previous problem.

Now let's consider the somewhat more realistic toy state shown in Figure 2. Again, there are 54 green voters and 36 purple voters, and they have to be split into 10 districts of 9 people each.

PROBLEM 25.5. Split this state into 10 districts that are as compact as possible. How many green districts are there, and how many purple districts are there? Which measure(s) of compactness are you using?

PROBLEM 25.6. Same thing, but now optimize for proportionality: the number of green and purple districts should approximate the proportion of people in each party in the state.

PROBLEM 25.7. You work for the green party and are hired to draw districts so that green wins as many as possible. How do you do it? How many green districts can you create?

PROBLEM 25.8. Same thing, but this time you work for the purple party.

PROBLEM 25.9. If the purple party has only a fraction of $x < \frac{1}{2}$ of the population, show that the purple party cannot gerrymander the map into equipopulous districts in such a way as to win more than a fraction of $2x$ of the seats.

Let's now test your intuition a bit.

PROBLEM 25.10. Suppose we have a state with 10 districts where 60% of the population is in the green party, and 40% of the population is in the purple party, and the population is fairly large. If the state is fairly homogeneous in that it doesn't contain green "pockets" and purple "pockets," and the map is drawn randomly, how many districts would you expect each party to win in an election?

If you answered the previous question correctly, then you can imagine that this is probably not what we want, and it also isn't what usually happens. But most of the reason that it isn't what happens in practice most of the time is because real states will tend to have purple-majority pockets, even though there are considerably more green people. Still, it suggests that maybe proportional representation isn't a reasonable goal, since that won't happen unless maps are specifically chosen for that goal; random maps don't give proportional representation.

Further reading. See [DW22] for a book about the mathematics of districting and gerrymandering, and [HK18, Chapter 12] for a shorter look at the subject.

Probability Paradoxes

Sometimes in mathematics, your intuition will fail you and lead you to believe false things. This is one reason why it is important to have proofs and not just intuition; intuition is a great first step, but it should never be considered fully satisfactory on its own.

Perhaps the area of mathematics where our intuition is most frequently wrong is in probability. We'll take a look at some of these surprising paradoxes in probability, where many people have faulty intuition. Knowing that many people, myself included, often have faulty intuition when it comes to probability, my rule is that I should not allow myself to believe anything in probability without calculating. Ideally, I will deny myself the right to have an intuition at all. In math we trust!

Here's the first example, known as *Bertrand's Paradox*. Consider an equilateral triangle inscribed in a circle. If we select a random chord of the circle, what is the probability that its length is longer than a side of the triangle?

There are several approaches to this problem, since there are several ways to select a random chord.

PROBLEM 26.1. Suppose we select a random chord by choosing two random points on the circle. Then what is the probability that it is longer than a side of the triangle?

PROBLEM 26.2. Another way to select a chord is to choose a direction, and then select a random chord perpendicular to that direction. By symmetry, the answer is independent of the direction, so we can assume that the chord will be horizontal. Then we select a random point on the vertical diameter of the circle for the chord to pass through. With this construction, what is the probability that the chord is longer than a side of the triangle?

PROBLEM 26.3. Another possibility is to select a random point on the inside of the circle, and then choose a chord with that point as its midpoint. (Other than the center, there will always only be one chord having that midpoint.) Then what is the probability that the chord is longer than a side of the triangle?

Let's see another probability paradox! This one is known as *Simpson's paradox*. This is a real example coming from batting averages in

baseball, looking at the batting averages of two players, Derek Jeter and David Justice from 1995 and 1996. They had the following results:

	1995	1996
Derek Jeter	12/48	183/582
David Justice	104/411	45/140

PROBLEM 26.4. Compute each player's batting average in each year. Which one was better in each year?

PROBLEM 26.5. Look at each player's combined results. Which one was better overall?

PROBLEM 26.6. What can you conclude? Which of the two players do you consider to be better?

The next example is known as *intransitive dice*. Given two dice A and B , we say that A is better than B if, when we roll A and B together, the result of A 's roll is higher than the result of B 's roll more than half the time. Note that we're considering the possibility of dice to have nonstandard labelings.

PROBLEM 26.7. Consider the following dice:

- Die A has sides 2, 2, 4, 4, 9, 9.
- Die B has sides 1, 1, 6, 6, 8, 8.
- Die C has sides 3, 3, 5, 5, 7, 7.

For each pair of dice, determine which one is better, or if they are equal.

The next example is the *Monty Hall problem*, which is based on the game show *Let's Make a Deal*.

PROBLEM 26.8. Suppose you are on a game show, with the following setup: There are three doors, and you are told that behind one of the doors, there is a car, and behind the other two doors, there are goats. The host knows what is behind each door. You pick a door, and then the host opens one of the other two doors containing a goat. You are then asked if you would like to stick to your original guess, or whether you would like to switch to the remaining closed door. You get to keep whatever is behind the door. Assuming that you prefer to win a car than a goat, what should you do?

This is a typical problem where you get the right answer by calculation, but many people get the wrong answer by intuition. However, in this case, the probability of winning if you keep your initial guess is clearly either $\frac{1}{2}$ or $\frac{1}{3}$. I think that in this situation, many people have the wrong intuition because these numbers are too close together. What if we force them further apart?

PROBLEM 26.9. The setup is similar, but now there are 1000000 doors. There is a goat behind 999999 of them, and a car behind one of

them. You pick a door, and then the host opens 999998 of the other doors, all of which contain goats. Now do you want to switch? What should this tell you about the original problem?

PROBLEM 26.10. How many people do you need so that the probability is more than $\frac{1}{2}$ that two share a birthday? Assume that we consider only 365 days in a year, and all days are equally likely to be someone's birthday.

PROBLEM 26.11. At your first meeting with a person, ey tells you that ey has two children, at least one of which is a boy. What is the probability that both are boys? (This is the usual phrasing of the problem. However, it is ambiguous as stated, because the context under which you are being given this piece of information is not provided in the problem statement. So, the way you should interpret it is a two-child family is chosen uniformly at random from all families with two children in which at least one is a boy.)

PROBLEM 26.12. At your first meeting with a person, ey tells you that ey has two children, at least one of which is a boy who was born on a Tuesday. What is the probability that both are boys?

PROBLEM 26.13. Why are the answers different?

Further reading. See [Hav07] for a book about counterintuitive probability, and [Mos65] for a book of challenging probability problems. For a good book on probability theory in general, see [GS97].

WORKSHEET 27

Inequalities of Means

Given two positive numbers a and b , their *arithmetic mean* is $\frac{a+b}{2}$, whereas their *geometric mean* is $\sqrt{ab}$.

PROBLEM 27.1. Show that for any two positive numbers a and b , the arithmetic mean of a and b is greater than or equal to their geometric mean, i.e. $\frac{a+b}{2} \geq \sqrt{ab}$. Under what condition do we get equality?

This result is (a simple case of) one of the most important inequalities, called the *AM-GM inequality*.

PROBLEM 27.2. Show that if $a > 0$, then $a + \frac{1}{a} \geq 2$.

PROBLEM 27.3. Show that if a, b, c are positive, then $a^2 + b^2 + c^2 \geq ab + bc + ca$.

PROBLEM 27.4. Show that if a, b, c are positive, then $(a+b)(b+c)(c+a) \geq 8abc$.

The arithmetic mean and the geometric mean are the two most important types of means in mathematics, but there are some others that occasionally show up. The next two most important are called the *harmonic mean* and the *quadratic mean* (also known as the *root mean square*):

$$HM = \frac{2}{\frac{1}{a} + \frac{1}{b}}, \quad QM = \sqrt{\frac{a^2 + b^2}{2}}.$$

PROBLEM 27.5. Show that for any positive numbers a and b , we have

$$QM \geq AM \geq GM \geq HM.$$

PROBLEM 27.6. Explain the pictorial proof of these inequalities shown in Figure 1.

These inequalities also hold for more numbers, if we generalize the means in the right way. If we have n positive numbers $a_1, \dots, a_n$, then

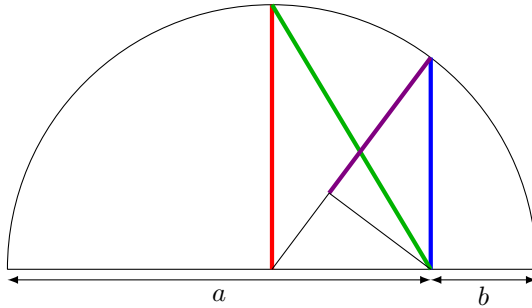


Figure 1. A pictorial proof of the inequalities $QM \geq AM \geq GM \geq HM$.

we define

$$QM = \sqrt{\frac{a_1^2 + a_2^2 + \cdots + a_n^2}{n}},$$

$$AM = \frac{a_1 + a_2 + \cdots + a_n}{n},$$

$$GM = \sqrt[n]{a_1 a_2 \cdots a_n},$$

$$HM = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}}.$$

PROBLEM 27.7. Show that the AM-GM inequality holds for $n = 4$ by using the AM-GM inequality for $n = 2$ several times.

PROBLEM 27.8. Deduce the AM-GM inequality for $n = 3$ from the AM-GM inequality for $n = 4$ by making a clever choice for the fourth variable.

PROBLEM 27.9. Show that the AM-GM inequality holds when n is a power of 2.

PROBLEM 27.10. Show that the AM-GM inequality holds for any number of variables.

PROBLEM 27.11. Show that $GM \geq HM$ for any number of variables.

PROBLEM 27.12. Show that $QM \geq AM$ for any number of variables.

There is a great generalization of all these inequalities, called the *power mean inequality*. Given positive numbers $a_1, \dots, a_n$ and a nonzero real number k , we define the k^{th} power mean to be

$$P_k(a_1, \dots, a_n) = \left(\frac{a_1^k + a_2^k + \cdots + a_n^k}{n} \right)^{1/k}.$$

The power mean inequality says that if $m \geq k$, then

$$P_m(a_1, \dots, a_n) \geq P_k(a_1, \dots, a_n).$$

Since the quadratic, arithmetic, and harmonic means are the power means for $k = 2$, $k = 1$, and $k = -1$, respectively, this implies the inequalities among their respective means. As for the geometric means, note that the 0^{th} power mean is not defined, but we can still take the k^{th} power mean for k close to 0. It turns out that the geometric mean is the limit of the k^{th} power means as k tends to 0. However, the proof of these facts is a bit beyond the scope of this book, as it relies on more advanced inequalities (in particular, Jensen's inequality).

Further reading. See [Ste04] for a book about inequalities, including the ones discussed in this worksheet but also many others.

Deterministic Finite Automata and Regular Languages

In this worksheet, we will study a very primitive model of a computer, called a *deterministic finite automaton*, or *DFA* for short. To start with, we need an *alphabet* $\mathcal{A}$, which is a finite collection of symbols we'll call letters. The alphabet could be the usual English alphabet, consisting of 26 letters, for instance, or it could be the binary alphabet, consisting of just two "letters," called 0 and 1. We'll often work with the binary alphabet due to its simplicity.

Given an alphabet $\mathcal{A}$, a DFA on $\mathcal{A}$ consists of a directed graph with a finite collection of nodes, called the *states*, with edges labeled by letters in $\mathcal{A}$ such that for every state q and every letter a , there is a unique edge out of q labeled with the letter a . Next, one state is designated as the start state, and some states (possibly all of them, possibly none of them), are considered *accepting states*.

We typically represent a DFA by means of a diagram. An example is shown in Figure 1. The states marked with double circles are the accepting states.

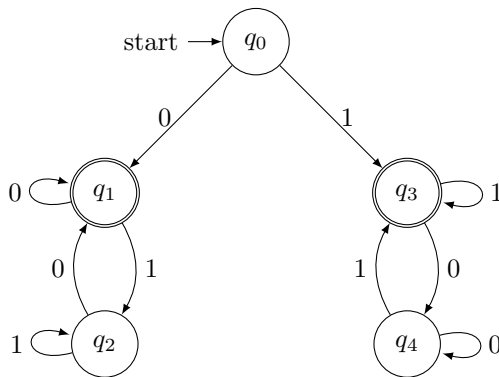


Figure 1. The diagram of a DFA on the alphabet $\{0, 1\}$.

Given an alphabet $\mathcal{A}$, we write $\mathcal{A}^*$ for the set of all finite strings of any length with letters in $\mathcal{A}$, including the empty string. Thus $\{0, 1\}^* = \{\varepsilon, 0, 1, 00, 01, 10, 11, 000, 001, \dots\}$, where ε denotes the empty string. Given a string w in $\mathcal{A}^*$ and a DFA M on $\mathcal{A}$, we let M compute on w as follows: we begin in the starting state. Then, for each letter in w , starting from the left we follow the arrow corresponding to that letter from the current state. If the final state we reach when the string ends is an accepting state, then we say that M *accepts* w . Otherwise, M *rejects* w . For instance, if $w = 0110$ and M is the DFA from Figure 1, then we start at state q_0 and the move to states q_1 , q_2 , q_2 , and q_1 , in that order. Since q_1 is an accepting state, M accepts w .

Given a DFA M , the *language* $\mathcal{L}(M)$ of M is the set of all strings in $\mathcal{A}^*$ that M accepts.

PROBLEM 28.1. For the DFA M above, describe $\mathcal{L}(M)$ concretely, i.e. without reference to a DFA.

PROBLEM 28.2. Find a DFA on $\{0, 1\}$ whose language consists of all finite strings consisting of a bunch (which could be none) of 0's, followed by a bunch of 1's. (That is, no 0 can come after any 1.)

PROBLEM 28.3. Find a DFA on $\{0, 1\}$ whose language consists of all finite strings containing exactly 3 1's, and any number of 0's.

PROBLEM 28.4. Find a DFA on $\{0, 1\}$ whose language consists of all finite strings that contain three 1's in a row somewhere in the string.

PROBLEM 28.5. Find a DFA on $\{0, 1\}$ whose language consists of all finite strings that contain three 1's in a row, but not four 1's in a row.

A language L is said to be *regular* if $L = \mathcal{L}(M)$ for some DFA M .

PROBLEM 28.6. Suppose L_1 and L_2 are regular languages. Show that $L_1 \cup L_2$ is regular. (That is, the set of strings that are in L_1 or L_2 .¹)

PROBLEM 28.7. Suppose L_1 and L_2 are regular languages. Show that $L_1 \cap L_2$ is regular. (That is, the set of strings that are in both L_1 and L_2 .)

PROBLEM 28.8. Suppose L_1 and L_2 are regular languages. Show that $L_1 \triangle L_2$ is regular, i.e. the set of strings that are in exactly one of L_1 and L_2 .

Let us now investigate a language that is *not* regular. Let $L = \{0^n 1^n\}$, i.e. the set of strings with some number of 0's, followed by the same number of 1's.

¹Recall our convention on “or” in mathematics: that includes strings in both L_1 and L_2 .

PROBLEM 28.9. Suppose M is a DFA with s states. Show that if w is a string of length at least s , then in the computation of M on w , some state (possibly the starting state) must be visited twice.

PROBLEM 28.10. Suppose state q is visited twice in the computation of M on w . Let x be the substring of w up to the first visit to q , let y be the substring starting after the first visit to q up to the second visit to q , and let z be the substring following the second visit to q . Thus w decomposes as xyz , i.e. first x , then y , then z . Show that M accepts w if and only if it accepts $xyyz$. Conclude that if M accepts w , then it accepts xy^kz for all $k \geq 0$.

This is a special case of the first significant result in the theory of automata, namely the *pumping lemma*.

PROBLEM 28.11. Show that $L = \{0^n 1^n\}$ is not a regular language.

Further reading. For more on DFAs and other similar automata, see [Sip12, Chapter 1], and then see the rest of [Sip12] for more powerful types of computing machines. See [AS03, Chapters 4 and 5] for another interpretation of DFAs, in terms of automatic sequences.

WORKSHEET 29

Voting Systems

Suppose we have an election, and there are several candidates. How should we design a voting system to pick the winner?

The most common voting system is known as *first past the post*. In this voting system, every voter votes for exactly one candidate, and the candidate who collects the most votes wins.

When there are two candidates, first past the post is a great system. But we quickly find problems with it when there are three or more candidates, in the sense that it does not necessarily capture the will of the voters effectively.

What exactly does “the will of the voters” mean? Defining this precisely is a tricky problem, but here is a definition that will certainly be useful:

DEFINITION. A (*strict*) *preference ranking* is a ranking of all the candidates from most preferred to least preferred, with no ties. We write $A \succ_i B$ or $B \prec_i A$ to mean that voter i prefers candidate A to candidate B . The collection of all the voters’ preference rankings is called a *preference profile*.

We will always choose our examples so that ties do not occur. However, a voting rule ought to include a fixed tie-breaking mechanism in advance. One possibility is a coin flip.

For the first part of this worksheet, we assume that every ballot represents the voter’s actual preference ranking. Later on, we will consider *strategic voting*, where a voter may cast a ballot that doesn’t represent eir preferences, in order to attempt to engineer a better outcome.

Let us now suppose that there are three candidates. We’ll distinguish them by colors: we have orange, green, and purple candidates. Suppose that there are 9 voters, and here is their preference profile:

Voter	1	2	3	4	5	6	7	8	9
First	●	●	●	●	●	●	●	●	●
Second	●	●	●	●	●	●	●	●	●
Third	●	●	●	●	●	●	●	●	●

PROBLEM 29.1. Who wins this election under the first past the post system?

PROBLEM 29.2. Now, suppose that orange weren't part of the election, so it's just between green and purple. Now who wins, with the same preference rankings?

PROBLEM 29.3. Which candidate is most deserving of winning the election, given these preference rankings?

Before we move on and look at other voting systems and the issues they can suffer from, let me offer a bit of advice. It may be tempting to vote for your favorite candidate, and perhaps doing so makes you feel good about voting for the candidate you support. But instead, you should treat voting not as a way of displaying support, but as a game, and when you play a game you should try to work out the best strategy, so as to lead to as good of an outcome as you can. Vote in such a way as to maximize the expected outcome (or some other metric), and this might not be the same as voting for your favorite candidate.

Suppose we have three candidates, and again we'll call them orange, green, and purple. Given the voters' preference profile, we might imagine that one candidate ought to win if that candidate is preferred to each of the other two in a two-candidate election. We'll call such a candidate the *Condorcet winner*.

PROBLEM 29.4. Which candidate is the Condorcet winner in the election above?

PROBLEM 29.5. Does there exist a preference profile for which there is no Condorcet winner? Assume that the number of voters is odd, so that there cannot be ties.

We say that a voting system satisfies the *Condorcet criterion* if the system always picks the Condorcet winner when one exists.

Another possible voting method is called the *Borda count*. Suppose there are n candidates. Then each voter ranks the candidates from 1 to n , with 1 being the top choice and n being the last choice. Then we give each candidate n points for each voter who ranked em in first place, $n - 1$ points for each voter who ranked em in second place, and so on.¹

PROBLEM 29.6. Does the Borda count satisfy the Condorcet criterion?

Another voting system that has gained popularity recently, and is now being used in some places in the United States, is *instant runoff voting*, which is a popular form of *ranked choice voting*. In instant runoff voting, voters rank the candidates from 1 to n . If any candidate gets the majority of the first place votes, then that candidate is declared the

¹There are other conventions for how to award points based on rankings as well.

winner. If no candidate gets the majority of the first-place votes, then the candidate who received the smallest number of first-place votes is eliminated, and all votes for that candidate are transferred to the voters' next pick; in particular, the voters who ranked that candidate first now move on to their second-place pick. This elimination process repeats until one candidate gets the majority of the first-place votes.

PROBLEM 29.7. Suppose we have five candidates, called A, B, C, D, E , and 20 voters. Here are the number of voters with their preference profile:

Number	3	4	4	6	2	1
1 st	B	C	B	D	B	E
2 nd	C	A	D	C	E	A
3 rd	A	D	C	A	A	D
4 th	D	B	A	E	C	B
5 th	E	E	E	B	D	C

Who wins this election under instant runoff voting?

PROBLEM 29.8. Does instant runoff voting satisfy the Condorcet criterion?

PROBLEM 29.9. For each of the voting systems we have looked at so far, find an example of a preference profile such that one voter can achieve a more preferable outcome to the election by using a preference ranking that does not represent his true preferences. That is, assuming that everyone else submits the same ballot, that one voter can elect a more preferred candidate by submitting a ballot that does not represent his true preferences.

By now, we have seen a number of voting systems, as well as some flaws with all of them. So, you might wonder if there are any voting systems that are free from the sorts of defects we have seen. Unfortunately, this is not the case. Let's make this more precise.

A *winner rule* is a function that takes in a voting profile and outputs a winning candidate. We write $F(R_1, \dots, R_n)$ for the winner when voter i submits ranking B_i . A voter's true preference need not be the ranking he submits. Suppose voter i 's true ranking is R_i , and the other voters' ballots are fixed. Then perhaps voter i can submit a different ranking R'_i such that voter i prefers candidate $F(R_1, \dots, R'_i, \dots, R_n)$ to candidate $F(R_1, \dots, R_i, \dots, R_n)$. (That is, everyone else submits the same ranking in both cases, but voter i achieves a better outcome by submitting a ranking that doesn't match his preferences.) We say that a winner rule is *strategy-proof* if this is not possible: every voter always achieves a result that is as good as possible by submitting a ranking that matches his preferences, in every possible collection of rankings and preferences.

Another property that a winner rule might have is being a *winner dictator*: voter i is a winner dictator for voting rule F if $F(R_1, \dots, R_n)$ is always the top-ranked candidate in R_i , for all possible rankings of all the voters. Naturally, this is an undesirable property of a winner rule!

A positive property that a winner rule might have is *unanimity*: if every voter ranks candidate A first, then A wins.

The Gibbard–Satterthwaite theorem gives us the following unfortunate conclusion:

THEOREM (Gibbard–Satterthwaite, simplified form). *A winner rule that is unanimous and strategy-proof must have a winner dictator.*

The proof of the Gibbard–Satterthwaite Theorem is beyond the scope of this book. However, we can look at some ingredients that go into the proof.

PROBLEM 29.10. Let F be a strategy-proof winner rule. Suppose that $F(R_1, \dots, R_n) = A$. Now, suppose one voter changes her submitted ranking by moving candidate A upward, without changing the relative order of any of the other candidates. All other voters keep the same ranking. Show that now the winner is either A or B .

PROBLEM 29.11. Suppose that $F(R_1, \dots, R_n) = A$ in a strategy-proof winner rule F . Suppose that one voter moves A up in the ranking, and no other changes are made. Show that A still wins.

So far, we have looked at voting systems that select a single winning candidate. We can also consider voting systems that produce a ranking of all the candidates.

A *social ranking rule* is a function that assigns, to every collection of rankings, a complete ranking of all the candidates, called the *social ranking*. The Borda count is a social ranking rule, for instance.

One desirable property that a social ranking rule might have is the *Pareto condition*: if every voter ranks A above B , then the social ranking places A above B .

A social ranking rule satisfies *independence of irrelevant alternatives* (IIA) if the relative ranking of A and B does not depend on the rankings of any other candidates. That is, if voters change their rankings without modifying the relative positions of A and B , then the social ranking rule does not change whether or not A finishes ahead of B .

PROBLEM 29.12. Construct an example to show that the Borda count does not satisfy IIA.

A voter v is a *ranking dictator* for a social ranking rule if, for every collection of rankings, the social ranking rule produces the same ranking as voter v submitted.

THEOREM (Arrow). *Fix a finite set of at least three candidates and a fixed collection of voters. Suppose that a social ranking rule satisfies the Pareto condition and IIA. Then the rule has a ranking dictator.*

Here is one idea that goes into the proof of Arrow's Theorem:

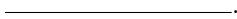
PROBLEM 29.13. Let F be a social ranking rule satisfying the Pareto condition and IIA. Call a candidate A *polarizing* if every voter ranks A either first or last. (It can be the case that some voters rank A first, and others rank A last.) Show that the social ranking must rank A either first or last.

Further reading. For a detailed look at the mathematics of voting, see [HK18].

WORKSHEET 30

The Koch Curve and Its Friends

Consider the following process: start with a line segment:



Now, every time you see a segment, replace the middle third of it by the other two sides of an equilateral triangle with that segment as one of its sides, pointing outward. Thus this line segment becomes



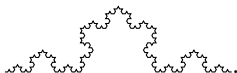
If we repeat the process, replacing every segment with the figure above, we get the following figure:



If we do it again, we get



And again:



The limiting figure we obtain in this way is called the *Koch curve*.

PROBLEM 30.1. Suppose the initial line segment has length 1. What is the total length of the n^{th} iterate of this process? The line segment is considered to be iteration 0.

PROBLEM 30.2. What is the area of the region between the initial line segment and the n^{th} iterate of the Koch curve? In the case of $n = 2$, this means to find the area of the shaded region below.



If, instead of starting with a line segment, we start with an equilateral triangle, we get a figure known as the *Koch snowflake*, shown in Figure 1.

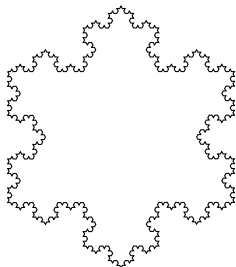


Figure 1. A Koch snowflake.

PROBLEM 30.3. What is the area of the n^{th} iterate of the Koch snowflake? What is the area of the final limiting version of it? Assume the original triangle has sidelength 1.

PROBLEM 30.4. Note that the Koch snowflake has infinite perimeter but finite area. This may seem counterintuitive. For instance, imagine filling the inside of a Koch snowflake with paint. It seems that we would need an infinite amount of paint to paint the boundary curve, because it is infinitely long, but if we fill the inside using only a finite amount of paint, then we should also have painted the boundary. How can you resolve this paradox?

PROBLEM 30.5. What happens when you put the Koch curves on the inside of the triangle instead of the outside?

PROBLEM 30.6. It is possible to tile the plane with Koch snowflakes, as indicated in Figure 2. However, this is only a finite iterate of the construction. Verify that the tiling extends to the limiting Koch snowflake.

PROBLEM 30.7. In the tiling shown in Figure 2, extended to the full plane, what is the ratio of the area of a black snowflake to a white snowflake? What proportion of the plane is covered with each color of snowflake?

The next thing we'll do is to calculate the dimension of the Koch curve. One of the many ways to define dimension is to look at the amount of stuff we get when we scale the figure. For instance, if we take a line segment of length 1 and we magnify it by a factor of 3, then we can break down the resulting figure into three versions of the original line segment, as shown in Figure 3. But if we magnify a square by a factor of 3, then we can break the resulting figure down into nine versions of the original square, as shown in Figure 4. If we triple a cube, we get 27 copies of the original cube.

This suggests the following definition of the dimension of a self-similar figure: suppose that when we magnify a figure by a ratio of r , we get r^d copies of the original figure. Then the dimension is d .

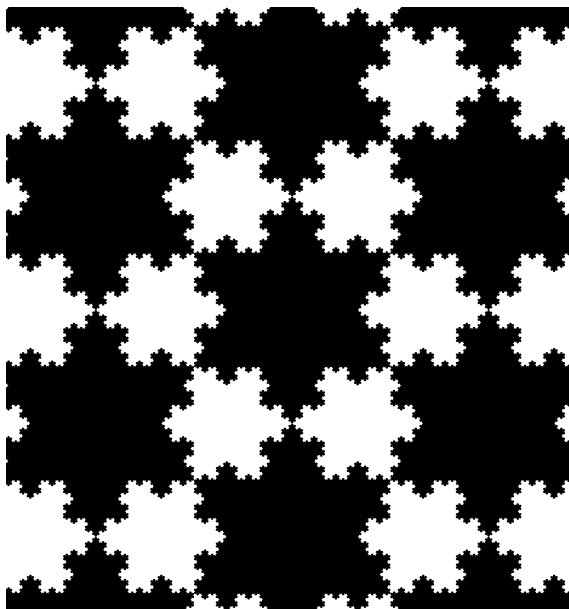


Figure 2. Koch snowflakes tiling the plane.



Figure 3. Tripling a line segment.

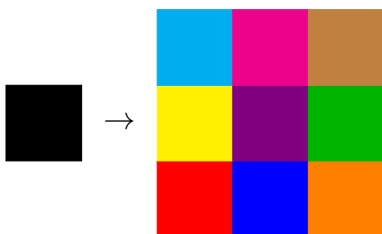


Figure 4. Tripling a square.

PROBLEM 30.8. Using this definition, what is the dimension of the Koch curve or the Koch snowflake? (Note that this is referring to the boundary curve, not the interior of the snowflake.)

Note that the dimension of the Koch curve is not an integer, but is somewhere between 1 and 2. This makes sense: it seems that it takes up somewhat more room than a 1-dimensional curve, but somewhat less than a 2-dimensional figure.



Figure 5. The first few iterates of the Sierpiński triangle.

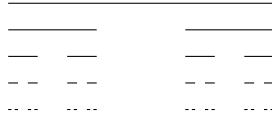


Figure 6. The first few iterates of the Cantor set.

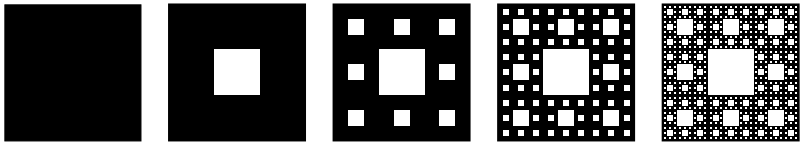


Figure 7. The first few iterates of the Sierpiński carpet.

Let's now look at some other related objects. The first is the *Sierpiński triangle*. To form a Sierpiński triangle, start with a filled-in equilateral triangle, and then delete the interior of the central equilateral triangle. Then, we do this process iteratively, as shown in Figure 5. The final Sierpiński triangle is what remains filled in after infinitely many steps of this process.

PROBLEM 30.9. What is the dimension of the Sierpiński triangle?

The *Cantor set* is defined as follows: start with a line segment, and then at each iterate, remove the middle third of each line segment. The first few iterates are shown in Figure 6. The final Cantor set is what remains after infinitely many steps of this process.

PROBLEM 30.10. What is the dimension of the Cantor set?

Another related object is the Sierpiński carpet, which starts with a square and removes the middle square (of one third the sidelength) at each step, as shown in Figure 7.

PROBLEM 30.11. What is the dimension of the Sierpiński carpet?

There is also a 3-dimensional version, called the Menger sponge. To form the Menger sponge, start with a cube and divide it into 27 smaller cubes. Then remove the central cube and the cube in the middle of each face, leaving 20 of the initial 27 cubes. Then iterate this process.

PROBLEM 30.12. What is the dimension of the Menger sponge?

Further reading. For a more thorough mathematical introduction to fractals, see [Edg08, Chapter 1].

WORKSHEET 31

Finite Differences

A great way of analyzing sequences is to look at the differences between successive terms. For instance, if we start with the squares and write down the differences between successive terms below them, we have the following:

1	4	9	16	25	36	49	64	81.
	3	5	7	9	11	13	15	17

We find that the differences between the consecutive square numbers are the consecutive odd numbers.

PROBLEM 31.1. Explain why this is the case.

PROBLEM 31.2. What are the differences between the successive cubes and fourth powers?

However, this is not where we find the nicest behavior. Define the *rising and falling factorials*, respectively, to be

$$n^{\overline{k}} = n(n+1)(n+2) \cdots (n+k-1)$$

and

$$n^{\underline{k}} = n(n-1)(n-2) \cdots (n-k+1),$$

which are the product of k consecutive numbers going up and down, respectively, starting from n .

PROBLEM 31.3. What are the differences between successive terms of the sequences $n^{\overline{2}}$, $n^{\overline{3}}$, $n^{\overline{4}}$, $n^{\underline{2}}$, $n^{\underline{3}}$, and $n^{\underline{4}}$? Do you see any interesting patterns? Can you explain them in general?

Given a function $f(n)$, defined on the nonnegative integers, define a new function $\Delta f(n)$ by $\Delta f(n) = f(n+1) - f(n)$, which is the function of successive differences of the values of f . Also, if a and b are nonnegative integers with $a \leq b$, define $\sum_{n=a}^b f(n)$ to be $f(a) + f(a+1) + f(a+2) + \cdots + f(b-1) + f(b)$.

PROBLEM 31.4. Simplify $\sum_{n=a}^b \Delta f(n)$ and $\Delta \sum_{k=a}^n f(k)$.

Problem 31.4 is a discrete version of the Fundamental Theorem of Calculus.

PROBLEM 31.5. Find formulae for $\sum_{n=a}^b n^2$, $\sum_{n=a}^b n^3$, $\sum_{n=a}^b n^4$, $\sum_{n=a}^b n^2$, $\sum_{n=a}^b n^3$, and $\sum_{n=a}^b n^4$. Generalize.

PROBLEM 31.6. Use what you have learned to evaluate $\sum_{n=1}^b n^2$, $\sum_{n=1}^b n^3$, and $\sum_{n=1}^b n^4$.

PROBLEM 31.7. What are $\Delta 2^n$ and $\Delta 3^n$? Use this to evaluate $\sum_{n=0}^b 2^n$ and $\sum_{n=0}^b 3^n$. Generalize!

PROBLEM 31.8. Evaluate $\sum_{n=0}^b n2^n$.

PROBLEM 31.9. If f and g are two functions, evaluate $\Delta(f(n)g(n))$ in terms of $f(n)$, $g(n)$, $\Delta f(n)$, $\Delta g(n)$, and the shift operator $E f(n) = f(n+1)$. (There are many ways of doing this, but some of them are a lot nicer than others.)

PROBLEM 31.10. Show that

$$\sum_{n=a}^b f(n)\Delta g(n) = E(fg)(b) - fg(a) - \sum_{n=a}^b \Delta(f(n))Eg(n).$$

Here, $fg(x)$ means $f(x)g(x)$.

The formula in Problem 31.10 is called *summation by parts*. It is a discrete analogue of integration by parts.

PROBLEM 31.11. Use the previous problem to give another solution to Problem 31.8.

Further reading. For more on the method of finite differences, see [GKP94, §2.6].

WORKSHEET 32

The Stern–Brocot Tree

Start with the “fractions” $\frac{0}{1}$ and $\frac{1}{0}$, written next to each other. Then, iteratively, perform the following operation: between each pair $\frac{a}{b}$ and $\frac{c}{d}$ of consecutive fractions, write down the number $\frac{a+c}{b+d}$, known as the *mediant*. The first few steps of this process yield the following sequences:

$$\begin{array}{l} 0 \ 1 \\ \overline{1}, \overline{0} \\ 0 \ 1 \ 1 \\ \overline{1}, \overline{1}, \overline{0} \\ 0 \ 1 \ 1 \ 2 \ 1 \\ \overline{1}, \overline{2}, \overline{1}, \overline{1}, \overline{0} \\ 0 \ 1 \ 1 \ 2 \ 1 \ 3 \ 2 \ 3 \ 1 \\ \overline{1}, \overline{3}, \overline{2}, \overline{3}, \overline{1}, \overline{2}, \overline{1}, \overline{1}, \overline{0} \\ 0 \ 1 \ 1 \ 2 \ 1 \ 3 \ 2 \ 3 \ 1 \ 4 \ 3 \ 5 \ 2 \ 5 \ 3 \ 4 \ 1 \\ \overline{1}, \overline{4}, \overline{3}, \overline{5}, \overline{2}, \overline{5}, \overline{3}, \overline{4}, \overline{1}, \overline{3}, \overline{2}, \overline{3}, \overline{1}, \overline{2}, \overline{1}, \overline{1}, \overline{0}. \end{array}$$

PROBLEM 32.1. Show that the numbers in each step are listed in increasing order.

PROBLEM 32.2. Suppose $\frac{a}{b}$ and $\frac{c}{d}$ are two consecutive terms in any step of this construction, with $d > 0$. What is $\frac{c}{d} - \frac{a}{b}$?

PROBLEM 32.3. Show that all fractions that appear are in lowest terms.

We can build a tree out of the terms other than $\frac{0}{1}$ and $\frac{1}{0}$ of this construction, as shown in Figure 1.

We call the level of the tree at which a number appears its *depth*. For instance, $\frac{1}{1}$ has depth 1, and $\frac{3}{5}$ has depth 4. The tree is called the *Stern–Brocot tree*.

Given two numbers a and b in the tree, write $a \wedge b$ for their common ancestor with the greatest depth. For instance, $\frac{4}{3} \wedge \frac{3}{1} = \frac{2}{1}$.

Each number a in the tree has a left child, which we call $\ell(a)$ and a right child, which we call $r(a)$. For instance, $\ell(\frac{2}{3}) = \frac{3}{5}$ and $r(\frac{2}{3}) = \frac{3}{4}$.

PROBLEM 32.4. Suppose that a and b are numbers in the tree, and let $c = a \wedge b$. Suppose that $c \neq a$ and $c \neq b$. Show that $a < b$ if and only if a is a descendant of $\ell(c)$ and b is a descendant of $r(c)$.

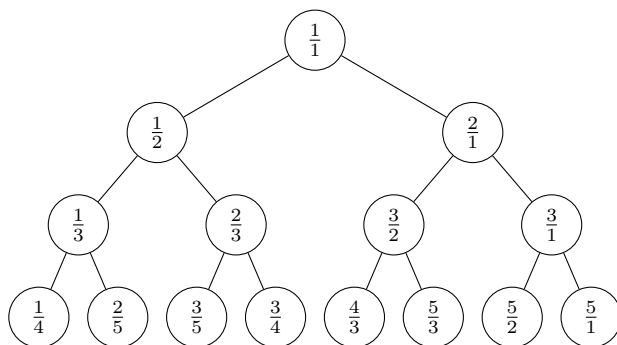


Figure 1. The fourth stage of the Stern–Brocot tree.

PROBLEM 32.5. Suppose that a, b, c are three consecutive terms in some stage of the Stern–Brocot tree, such as $\frac{2}{5}, \frac{1}{2}, \frac{3}{5}$. Show that b is the rational number with the smallest denominator strictly between a and c .

PROBLEM 32.6. Show that every positive rational number appears in the Stern–Brocot tree.

PROBLEM 32.7. Show that if $\frac{a}{b}$ has depth n in the Stern–Brocot tree, then $\frac{a}{a+b}$ and $\frac{b}{a+b}$ have depth $n+1$.

PROBLEM 32.8. What is the sum of the numerators of all the numbers in the Stern–Brocot tree of depth n ? Can you explain why?

PROBLEM 32.9. For a number $\frac{a}{b}$, define $w(\frac{a}{b}) = \frac{1}{ab}$. What is the sum of $w(\frac{a}{b})$, where $\frac{a}{b}$ ranges over all the numbers in the Stern–Brocot tree of depth n ? Can you explain why?

We can encode an arbitrary positive rational number by a sequence of L 's and R 's, indicating whether we take the left branch or the right branch of the tree at each step.

PROBLEM 32.10. How can you determine this encoding for a positive rational number $\frac{a}{b}$ without drawing the entire tree, or a substantial portion of it? Test out your method by working out the encoding of $\frac{57}{106}$.

Further reading. For more on the Stern–Brocot tree, see [GKP94, §4.5].

Chromatic Polynomials

In this worksheet, we assume that all graphs are simple, i.e. there are no edges connecting a vertex to itself, and there is at most one edge between each pair of vertices.

Given a finite graph G , we would like to color its vertices using n colors, in such a way that no two vertices sharing an edge have the same color. We do not require that all the colors be used. We write $\chi_G(n)$ for the number of such colorings.

PROBLEM 33.1. The *path graph* P_k (shown for $k = 8$ in Figure 1) consists of k vertices in a line. More formally, we put a vertex at the point $(i, 0)$ for each i with $1 \leq i \leq k$, and we connect the vertices at $(i, 0)$ and $(j, 0)$ if and only if $|i - j| = 1$. What is $\chi_{P_k}(n)$?

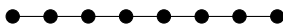


Figure 1. The path graph P_8 .

PROBLEM 33.2. The *complete graph* K_k consists of k vertices, and edges between every pair of vertices. What is $\chi_{K_k}(n)$?

PROBLEM 33.3. The *cycle graph* C_k (shown in Figure 2 for $k = 8$) is the same as the path graph, but with the vertices arranged in a circle. What is $\chi_{C_k}(n)$?

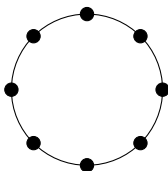


Figure 2. The cycle graph C_8 .

Computing $\chi_{C_k}(n)$ was probably much harder than computing $\chi_{P_k}(n)$ or $\chi_{K_k}(n)$. But if you did it, you might notice that in all cases, $\chi_G(n)$ is a polynomial function of n . Is $\chi_G(n)$ a polynomial function

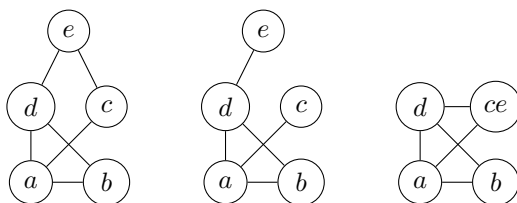


Figure 3. Left: A graph G . Middle: The deletion $G \setminus \{c, e\}$ of the edge $\{c, e\}$ between c and e . Right: The contraction $G / \{c, e\}$ of the edge $\{c, e\}$ between c and e .

of n for every finite graph G ? That's one of the main questions we'll answer in this worksheet.

PROBLEM 33.4. Are there other interesting families of graphs G for which you can compute $\chi_G(n)$?

Given a graph G and an edge e , we can form two new graphs out of G that come from performing some operation on e . The first operation is called *deletion*: just remove the edge e . We write $G \setminus e$ for this graph. The second operation, called *contraction*, is a bit more subtle. Suppose e is an edge between two vertices v and w . Then we merge vertices v and w into a single new vertex. For any vertex u in G , other than v and w , we connect u to the merged v - w vertex if and only if u is connected to at least one of v and w in G . Note that this may cause pairs of vertices to have multiple edges between them; if this happens, collapse them into a single edge. We write G/e for this contracted graph. See Figure 3 for an example of a graph with a deletion and a contraction.

PROBLEM 33.5. Find a simple relation between $\chi_G(n)$, $\chi_{G/e}(n)$, and $\chi_{G \setminus e}(n)$.

PROBLEM 33.6. Suppose G is a graph with d vertices and no edges. What is $\chi_G(n)$?

PROBLEM 33.7. Show that for any finite graph G , $\chi_G(n)$ is a polynomial function of n .

Now that we know that $\chi_G(n)$ is always a polynomial, we can see how certain properties of the graph G are encoded in the polynomial. We call $\chi_G(n)$ the *chromatic polynomial* of G .

PROBLEM 33.8. What is the degree of $\chi_G(n)$?

PROBLEM 33.9. Some of the coefficients of $\chi_G(n)$ are positive, some of them are negative, and some are zero. Can you find a rule to determine which are which? Can you explain why?

PROBLEM 33.10. For which graphs G is $\chi_G(2) > 0$?

It is not really possible to move beyond that with a comparably simple criterion for arbitrary graphs. In particular, given a graph G , it is a famously hard problem to determine whether $\chi_G(3) > 0$ in general. In particular, it is an example of an NP-complete problem, which is one that is computationally intractable in practice, unless a widely believed conjecture is false.

PROBLEM 33.11. What is the leading coefficient of $\chi_G(n)$? (That is, if $\deg \chi_G(n) = d$, that's the coefficient of the n^d term.)

PROBLEM 33.12. Suppose $\deg \chi_G(n) = d$. What is the coefficient of n^{d-1} ?

PROBLEM 33.13. What other structural properties of the graph can you easily read off from $\chi_G(n)$?

Further reading. For more on the chromatic polynomial of graphs, see [Wil10, §5.2].

Rotation Numbers and the Doubling Map

Let $[0, 1)$ be the set of real numbers that are at least 0 and less than 1: $[0, 1) = \{x : 0 \leq x < 1\}$. We define the *doubling map* $D : [0, 1) \rightarrow [0, 1)$ by setting $D(x)$ to be the fractional part of $2x$, i.e.

$$D(x) = \begin{cases} 2x & 0 \leq x < \frac{1}{2}, \\ 2x - 1 & \frac{1}{2} \leq x < 1. \end{cases}$$

We can apply the doubling map multiple times. For instance, if we start with $x = \frac{1}{10}$, iterates of the doubling map are

$$\frac{1}{10} \mapsto \frac{1}{5} \mapsto \frac{2}{5} \mapsto \frac{4}{5} \mapsto \frac{3}{5} \mapsto \frac{1}{5} \mapsto \frac{2}{5} \mapsto \cdots.$$

For some x in $[0, 1)$, let $\text{Orb}(x)$, called the *orbit* of x , be the set of all numbers obtained by iterating D some number of times, including x itself (which arises from 0 iterates of D). Thus we have

$$\text{Orb}\left(\frac{1}{10}\right) = \left\{\frac{1}{10}, \frac{1}{5}, \frac{2}{5}, \frac{4}{5}, \frac{3}{5}\right\}.$$

But we can also write all the numbers in the orbit in increasing order:

$$\text{Orb}\left(\frac{1}{10}\right) = \left\{\frac{1}{10}, \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}\right\}.$$

PROBLEM 34.1. For which values of x is $\text{Orb}(x)$ a finite set?

PROBLEM 34.2. Note that for $x = \frac{1}{10}$, while the orbit is finite, we never get back to the starting point. For which values of x is there an iterate of D equal to x , i.e. when do we go around in a cycle starting from the very beginning? More generally, if we don't go around in a cycle immediately, how can you quickly determine the smallest iterate of x under D for which it does go in a cycle? Note that this is $\frac{1}{5}$ for $x = \frac{1}{10}$.

When there is some iterate of x under D that is equal to x , we say that x is *periodic*. The length of the cycle to get back to x is called the *period*. For instance, $\frac{1}{5}$ is periodic with period 4.

PROBLEM 34.3. If x is periodic, how can you quickly determine the period of x ?

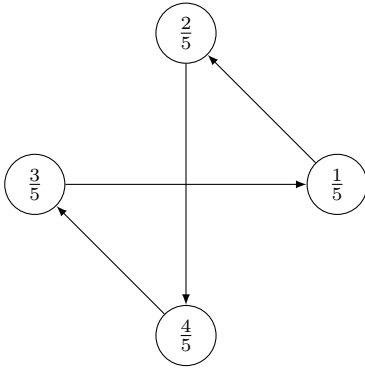


Figure 1. The action of D on $\text{Orb}(\frac{1}{5})$.

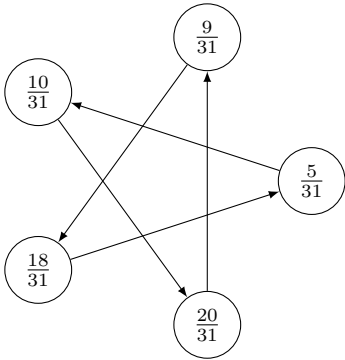


Figure 2. The action of D on $\text{Orb}(\frac{5}{31})$.

Let's draw a picture of D on $\text{Orb}(\frac{1}{5})$, by writing the elements of $\text{Orb}(\frac{1}{5})$ in a circle in increasing order, and drawing arrows between x and $D(x)$. See Figure 1. Note that we go forward by different amounts on the cycle each time: we go forward by 1 to get from $\frac{1}{5}$ to $\frac{2}{5}$, but we go forward by 2 to get from $\frac{2}{5}$ to $\frac{4}{5}$, for instance. By contrast, consider the orbit of $\frac{5}{31}$, shown in Figure 2. Here, we go forward by 2 each time, so the picture looks much nicer. When this happens, we say that the orbit forms a *rotation set*.

When we have a number x such that the orbit under the doubling map forms a rotation set, let q denote the period, and let p be the number we move forward in the orbit. In the case of $x = \frac{5}{31}$, we have $q = 5$ and $p = 2$. We call the number $\frac{p}{q}$ the *rotation number* of x . Thus $\frac{5}{31}$ has rotation number $\frac{2}{5}$.

PROBLEM 34.4. Must a rotation number $\frac{p}{q}$ be a fraction in lowest terms?

PROBLEM 34.5. Find numbers with rotation numbers $\frac{1}{3}$, $\frac{2}{3}$, $\frac{1}{4}$, $\frac{1}{5}$, $\frac{1}{6}$, and $\frac{3}{7}$.

PROBLEM 34.6. Write $\frac{1}{5}$ and $\frac{5}{31}$ as repeating binary expansions, i.e. as $.d_1d_2\cdots$, meaning $\sum_{n=1}^{\infty} \frac{d_n}{2^n}$. Interpret $D(x)$ in terms of the binary expansion of x .

PROBLEM 34.7. Describe the elements of $\text{Orb}(x)$ in terms of the binary expansion of x . How can you determine the increasing order of $\text{Orb}(x)$?

PROBLEM 34.8. Start with many numbers x and determine whether their orbits form rotation sets. Can you find a simple criterion on x , or $\text{Orb}(x)$, for when $\text{Orb}(x)$ is a rotation set?

PROBLEM 34.9. If x and y are in the same orbit, then $\text{Orb}(x)$ is a rotation set if and only if $\text{Orb}(y)$ is, and if they are in the same orbit and the orbit is a rotation set, then they have the same rotation number. Is it possible to find two numbers x and y with different orbits such that $\text{Orb}(x)$ and $\text{Orb}(y)$ are both rotation sets, with the same rotation number?

Further reading. See [BFM15] and [Gol92] for discussions on the dynamics of the doubling map.

Sharkovskii's Theorem

Let I be an interval in $\mathbb{R}$, possibly bounded or unbounded, and with its endpoints included or excluded in any combination. In this worksheet, we'll investigate and prove some cases of a remarkable theorem due to Sharkovskii about the periods of periodic points of continuous functions from I to I . The star of the show is the Intermediate Value Theorem. I'm not going to ask you to prove it, because that would rely on a considerable amount of real analysis (but if you come back for the Transition to Proofs class, we will prove it in that class), but I want you to convince yourself that it is true. I'm not going to give you the precise definition of a continuous function from $\mathbb{R}$ to $\mathbb{R}$, but you should think of it as one whose graph you can draw without lifting your pen off the paper. This is not the correct definition, but it will suffice for our purposes here. You may also freely use, without proof, the fact that sums, differences, and compositions of continuous functions are continuous.

PROBLEM 35.1 (Intermediate Value Theorem). Suppose $f : I \rightarrow \mathbb{R}$ is a continuous function, and suppose that a , b , and y are real numbers with $a < b$, and $f(a) \leq y$ and $f(b) \geq y$. Convince yourself that there is some c with $a \leq c \leq b$ such that $f(c) = y$.

Here is a typical example of its use:

PROBLEM 35.2. Show that there is some real number x such that $1 - x^5 = x$.

If $f : I \rightarrow I$ is a function and S is a subset of I , then we write $f(S)$ for the set of all numbers of the form $f(s)$ for s in S . Also, the notation $A \subseteq B$ means that A is a *subset* of B , i.e. every element of A is an element of B .

For the rest of this worksheet, we make the convention that a closed interval is nonempty and bounded; that is, it is an interval of the form $[s, t]$, where s and t are real numbers with $s < t$. This interval is the set of all real numbers u such that $s \leq u \leq t$.

Let I be an interval, and let $f : I \rightarrow I$ be a continuous function. Let J and K be closed intervals contained in I . We say that J *covers* K under f if $f(J) \supseteq K$, i.e. every point in K is of the form $f(x)$ for some x in J .

PROBLEM 35.3. Suppose that J covers itself, i.e. $f(J) \supseteq J$. Show that there is some x in J such that $f(x) = x$.

If $f(x) = x$, then we say that x is a *fixed point* of f .

PROBLEM 35.4. Show that if J covers K , then there is a closed subinterval L of J such that $f(L) = K$. (A fully rigorous proof uses some facts from analysis about closed intervals that we have not yet developed. Give as convincing a geometric argument as you can.)

PROBLEM 35.5. Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, a, b, c are real numbers with $a < b < c$, and $f(a) = b$, $f(b) = c$, and $f(c) = a$. Let $I_0 = [a, b]$ and $I_1 = [b, c]$. Show that $I_1 \subseteq f(I_0)$ and $I_0 \cup I_1 \subseteq f(I_1)$.¹ Show that f has a fixed point in I_1 .

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function, let $f^2(x) = f(f(x))$, and more generally, let $f^n(x)$ be the result of applying f n times to x . Let us also set $f^0(x) = x$.

PROBLEM 35.6. Under the hypotheses of the previous problem, show that f^2 has a fixed point in I_0 .

We say that a point x is *periodic* if $f^n(x) = x$ for some $n \geq 1$. If n is the *smallest* positive integer for which $f^n(x) = x$, then we say that x has period n .

PROBLEM 35.7. Strengthen the result of the previous problem to show that f has a periodic point of period exactly 2 in I_0 .

PROBLEM 35.8. Conclude that if f has a periodic point of period 3, then f has a fixed point and also a periodic point of period 2. (Note that the preceding problems handle one of the two possible cyclic orders of the three points: $a \mapsto b \mapsto c \mapsto a$, where $a < b < c$. We could also have $a \mapsto c \mapsto b \mapsto a$. Make sure you understand how to handle this case as well.)

This is already quite impressive, since why should a function having a periodic point of some period imply anything about having periodic points of other periods? But in fact, we're just getting started!

PROBLEM 35.9. Let $f : I \rightarrow I$ be continuous, let n be a positive integer, and let $J_0, J_1, \dots, J_n$ be closed intervals contained in I , where $J_n = J_0$. Suppose that $f(J_i) \supseteq J_{i+1}$ for every i with $0 \leq i < n$. Show there is a point p in J_0 such that $f^i(p)$ is in J_i for every i with $0 \leq i \leq n$, and $f^n(p) = p$.

The previous problem shows that $f^n(p) = p$, but it does not show that p has period n , since there may be some $m < n$ such that $f^m(p) = p$.

¹The notation $S \cup T$ means the *union* of S and T , i.e. the set of elements that are in at least one of S and T .

PROBLEM 35.10. Show that if f has a periodic point of period 3, then for every positive integer n , f has a periodic point of period exactly n . Note that it does not suffice to find a point p such that $f^n(p) = p$, since its period could be smaller than n .

PROBLEM 35.11. Show that if f has a periodic point of period 5, then it has periodic points of all prime periods other than possibly 3.

PROBLEM 35.12. Show that if f has a periodic point of period 7, then it has periodic points of all prime periods other than possibly 3 and 5.

The full version of Sharkovskii's Theorem involves the following ordering of the positive integers:

$$\begin{aligned} &3, 5, 7, 9, 11, 13, \dots, \\ &2 \cdot 3, 2 \cdot 5, 2 \cdot 7, 2 \cdot 9, 2 \cdot 11, 2 \cdot 13, \dots, \\ &2^2 \cdot 3, 2^2 \cdot 5, 2^2 \cdot 7, 2^2 \cdot 9, 2^2 \cdot 11, 2^2 \cdot 13, \dots, \\ &2^3 \cdot 3, 2^3 \cdot 5, 2^3 \cdot 7, 2^3 \cdot 9, 2^3 \cdot 11, 2^3 \cdot 13, \dots, \\ &\dots, \\ &\dots, 2^5, 2^4, 2^3, 2^2, 2^1, 1. \end{aligned}$$

In other words, we list all the odd numbers other than 1 in increasing order, then all numbers divisible by 2 but not 4 other than 2 in increasing order, then all numbers divisible by 4 but not 8 other than 4 in increasing order, and so forth, and then finally we list all powers of 2 in decreasing order.

Sharkovskii's Theorem says that if $f : I \rightarrow I$ is a continuous function with a periodic point of period n , then it must also have a periodic point of period m , for all m occurring after n in the above list. In particular, if f has a periodic point of period 3, then it must have periodic points of every period, and if f does not have a fixed point, then it does not have any periodic points at all.

Furthermore, for every positive integer n , there exists a continuous function $f : I \rightarrow I$ having a periodic point of period n but no period occurring earlier in the list.

PROBLEM 35.13. Find a continuous function $f : I \rightarrow I$ having a periodic point of period 1 (i.e. a fixed point) but no periodic point of period 2.

PROBLEM 35.14. Find a continuous function $f : I \rightarrow I$ having a periodic point of period 2 but no periodic point of period 4.

PROBLEM 35.15. Find a continuous function $f : I \rightarrow I$ having a periodic point of period 5 but no periodic point of period 3.

PROBLEM 35.16. Find a continuous function $f : I \rightarrow I$ having a periodic point of period 7 but no periodic point of period 5.

Further reading. See [LY75] for an accessible paper showing that if f has a periodic point of period 3, then it has periodic points of every period. See [BH11] for the full Sharkovskii Theorem.

WORKSHEET 36

Billiards

Consider a rectangular billiard table. When a ball rolling on the table hits one of the sides, it bounces off at an equal angle, as shown in Figure 1.

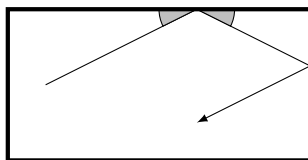


Figure 1. The ball bounces at an equal angle when it hits a side of the table.

We will assume that the table is frictionless, so it never stops moving once it gets started. We also assume that the ball never hits a corner of the table.

PROBLEM 36.1. Find several starting points and directions for the ball that cause it to repeat its path.

When this happens, we say that the ball follows a *periodic trajectory*. The *period* is the number of bounces before it repeats itself.

PROBLEM 36.2. Find periodic trajectories with period 2, 4, 6, 8, and 10. Does there exist a periodic trajectory with odd period?

A very useful way of thinking about billiards is by “unfolding” the table so as to tile the plane. Then, instead of bouncing off the side, just move on to the next table tile in a straight line, as shown in Figure 2.

PROBLEM 36.3. Do there exist aperiodic trajectories, i.e. trajectories that are not periodic?

PROBLEM 36.4. Ponder the connection between the tiling of the plane by reflected billiard tables and the identification space of a torus. Is there a way of thinking of billiards as being played on a torus?

Now, we don’t have to consider rectangular billiard tables; we can consider any polygonal table.

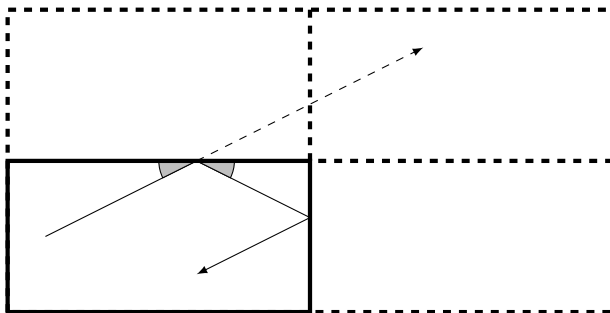


Figure 2. Reflections of the original table tile the plane.

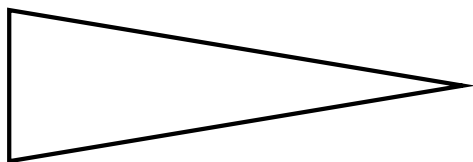


Figure 3. A triangular billiard table.

PROBLEM 36.5. Consider a triangular billiard table, like the one shown in Figure 3. Is it possible for the ball to bounce around on the sides of the triangle, so that it gets stuck near the sharp vertex, getting closer and closer to it?

Given a trajectory, periodic or not, on a rectangular table, we can encode it as a sequence of A 's and B 's, by recording an A every time it hits the left or right edge, and a B every time it hits the top or bottom edge. We call this sequence of A 's and B 's the *cutting sequence* of the trajectory.¹

For the next three problems, suppose the table is square.

PROBLEM 36.6. Suppose we have a trajectory of slope $\frac{2}{5}$. What are the possible cutting sequences?

PROBLEM 36.7. Is it possible for the cutting sequence to have two A 's in a row and also two B 's in a row?

PROBLEM 36.8. The cutting sequence is an infinite sequence of A 's and B 's, but we can truncate it after N letters, for some large value of N . If the slope of the trajectory is s , then roughly what proportion of the first N letters are A 's?

¹Technically, this is not quite the cutting sequence, but it is very closely related, and it is easy to convert from one to the other.

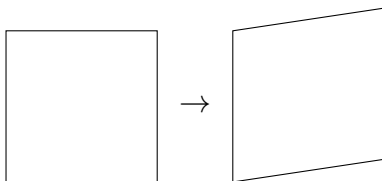


Figure 4. Deforming the square into a parallelogram.

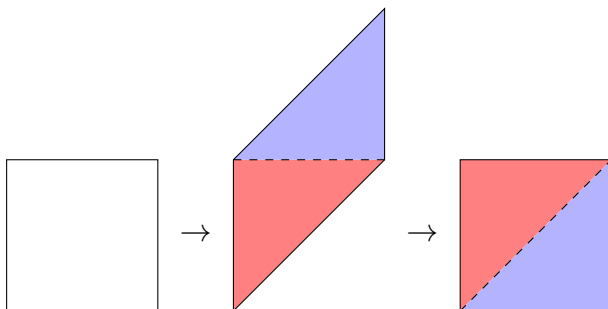


Figure 5. If we deform the square enough into a parallelogram, then we can cut the parallelogram into two triangles and translate one of them to give us back the original square.

If we start with a square billiard table, we can deform the sides a bit to turn it into a parallelogram, as shown in Figure 4. If we deform it into just the right parallelogram, as shown in Figure 5, then we can cut the parallelogram into two triangles and rearrange them back into the original square.

PROBLEM 36.9. Draw some trajectories on the original square, and watch what happens when we drag the trajectory along with the deformation into a parallelogram. When we deform the square into the parallelogram shown in Figure 5 and then cut it up and rearrange the parallelogram into a square, we get a new trajectory on the square. How is this new trajectory related to the original one? Can you easily work out the cutting sequence of the new trajectory in terms of the old one?

PROBLEM 36.10. What operation does this cut and paste procedure correspond to on a torus?

Further reading. For an accessible introduction to rational billiards, see [Dav16].

WORKSHEET 37

The Cubic and Quartic Formula

You are presumably familiar with the quadratic formula, for solving equations of the form $ax^2 + bx + c = 0$. But the higher-degree analogues are not as well known. Let's see how to derive formulae for solving equations of degrees 3 and 4.

We begin with the cubic formula, i.e. the formula for solving equations of the form $ax^3 + bx^2 + cx + d = 0$. We will see Cardano's method for solving cubics.

PROBLEM 37.1. Show that any cubic equation can be converted into one of the form $x^3 + px + q = 0$, in the sense that if you know how to solve all equations of this form, then you can easily solve all cubic equations. We call a cubic of the form $x^3 + px + q$ a *depressed cubic*.

PROBLEM 37.2. Consider a depressed cubic equation $x^3 + px + q = 0$. Make the substitution $x = \alpha + \beta$. Note that there are many choices for α and β ; this will be useful later. The resulting equation has the form $\alpha^3 + \beta^3 + q + f(\alpha, \beta) = 0$, for some f . Write down f explicitly, and show that f can be factored into the product of two nonconstant polynomials in α and β .

PROBLEM 37.3. Put an extra condition on α and β that forces $f(\alpha, \beta) = 0$, thanks to the factorization.

PROBLEM 37.4. You should now have a system of two equations: $\alpha^3 + \beta^3 = -q$ and $\alpha\beta = -\frac{p}{3}$. Reduce this system of equations into a single quadratic equation in the variable α^3 .

PROBLEM 37.5. Finish finding one root of the cubic equation.

PROBLEM 37.6. Let $\omega = \frac{-1 + \sqrt{-3}}{2}$. Observe that $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$. Show that if (α, β) is one solution of the equations from the previous problems, so that $\alpha + \beta$ is a root of the cubic, then $\omega\alpha + \omega^2\beta$ and $\omega^2\alpha + \omega\beta$ are also roots.

PROBLEM 37.7. Use Cardano's method to find all the roots of the equation $x^3 - 6x - 9 = 0$.

Recall that a quadratic polynomial $ax^2 + bx + c$ has a *discriminant* $b^2 - 4ac$, which dictates whether its roots are real or not.

PROBLEM 37.8. Show that if the roots of the quadratic polynomial $ax^2 + bx + c$ are r and s , then the discriminant is equal to $a^2(r - s)^2$.

A cubic polynomial also has a discriminant. The discriminant for the general cubic polynomial $ax^3 + bx^2 + cx + d$ is the somewhat unwieldy expression

$$18abcd - 4b^3d + b^2c^2 - 4ac^3 - 27a^2d^2.$$

But in the case of the “depressed” cubic $x^3 + px + q$, the discriminant simplifies to $-4p^3 - 27q^2$.

PROBLEM 37.9. In terms of the discriminant, can you determine how many of the roots are real, and whether there are repeated roots?

PROBLEM 37.10. Show that if the roots of the depressed cubic $x^3 + px + q$ are r, s, t , then the discriminant is equal to $(r - s)^2(r - t)^2(s - t)^2$.

We now turn to quartic equations, which are ones of the form $x^4 + ax^3 + bx^2 + cx + d = 0$. (We can also allow the case where the coefficient of x^4 is not 1, but then we can divide by this coefficient without changing the roots, so we will assume that the coefficient of x^4 is 1.) This time our strategy is quite different from the cubic case. We’ll look at Ferrari’s method.

PROBLEM 37.11. Move the quadratic, linear, and constant terms to the other side of the equation, then complete the square on the quartic and cubic terms.

PROBLEM 37.12. At this point, one side of the equation is a square, but the other side probably isn’t. So, add a new term to both sides to keep the side that is already a square a square, by introducing a new variable.

At this point, you should have reached the equation

$$\left(x^2 + \frac{1}{2}ax + \frac{1}{2}y\right)^2 = \left(\frac{1}{4}a^2 - b + y\right)x^2 + \left(\frac{1}{2}ay - c\right)x + \left(\frac{1}{4}y^2 - d\right),$$

or something very similar to that, where the coefficients may differ from the ones above, but in an easily recoverable manner. The left side is a square for any value of y , but the right side is generally not a square.

PROBLEM 37.13. Are there values of y that would make the right side a square? (What is the easy method to determine if a quadratic polynomial is a square?)

You should find, at this point, that the condition on the right side being a perfect square is a cubic equation in y , which you now know how to solve. Thus, at least in theory, you now know how to solve quartic polynomials.

PROBLEM 37.14. Show that you can also solve quartic equations in practice by using Ferrari's method to find the roots of $x^4 - x^3 + 5x - 3 = 0$.

You might hope to continue, by solving quintic and higher-degree equations as well. However, this is impossible to do in general, at least using addition, subtraction, multiplication, division, and roots! This is a famous theorem known as the Abel–Ruffini Theorem. While Abel originally proved this theorem, it was famously proven a bit later in a much more enlightening way by Galois. Note, however, that while *general* quintics cannot be solved in radicals, there are many specific quintics, and even families of quintics, that can be solved in radicals.

Further reading. This topic is very rich, and there are many excellent sources presenting relevant aspects of it. See [Irv13] for a problem-based book on many techniques for solving low-degree polynomials. See [HH10] for a historical survey of solving polynomial equations. See [Bro07] for a derivation of cubic resolvents, describing the information about what resolvents do. See [Pes03] for a book describing Cardano's method and Ferrari's method, resolvents, and a historical outline of Abel's proof of the Abel–Ruffini Theorem. See [Ram22] for a self-contained proof of the Abel–Ruffini Theorem.

When Unique Factorization Fails

The most important theorem in number theory is the Fundamental Theorem of Arithmetic, which says that every positive integer has a unique factorization into a product of primes. For instance, the prime factorization of 168 is $2^3 \times 3 \times 7$, and there are no other ways of factoring 168 as a product of primes, other than by reordering the factors.

This may seem obvious, but it isn't really. To see this, we'll look at an object that is quite similar to the integers, but where we do not have unique factorization. This object is $\mathbb{Z}[\sqrt{-5}]$, which consists of all numbers of the form $a + b\sqrt{-5}$, where a and b are integers.

PROBLEM 38.1. Show that the sum, difference, or product of any two elements in $\mathbb{Z}[\sqrt{-5}]$ is still in $\mathbb{Z}[\sqrt{-5}]$.

Just to remain consistent with the rest of the literature, we use the term *irreducible* for what is typically thought of as a prime, but we have to be a bit careful.

DEFINITION. A number x in $\mathbb{Z}[\sqrt{-5}]$ is said to be a *unit* if there is some y in $\mathbb{Z}[\sqrt{-5}]$ such that $xy = 1$.

The norm function is very useful when it comes to finding units.

DEFINITION. For a number $a + b\sqrt{-5}$ in $\mathbb{Z}[\sqrt{-5}]$, its *norm* $N(a + b\sqrt{-5})$ is defined as $N(a + b\sqrt{-5}) = a^2 + 5b^2$.

PROBLEM 38.2. Suppose x and y are in $\mathbb{Z}[\sqrt{-5}]$. Show that $N(xy) = N(x)N(y)$.

PROBLEM 38.3. What are all the units in $\mathbb{Z}[\sqrt{-5}]$?

DEFINITION. A number x in $\mathbb{Z}[\sqrt{-5}]$ is said to be *irreducible* if x is not a unit, and whenever y and z are in $\mathbb{Z}[\sqrt{-5}]$ and $x = yz$, then one of y and z is a unit.

PROBLEM 38.4. Show that if $N(x)$ is prime, then x is irreducible.

The reason we don't use the term *prime* is that it has a slightly different meaning:

DEFINITION. A number x that isn't 0 or a unit in $\mathbb{Z}[\sqrt{-5}]$ is said to be *prime* if whenever y and z are numbers in $\mathbb{Z}[\sqrt{-5}]$ such that yz is a multiple of x , then at least one of y and z is itself a multiple of x .

In $\mathbb{Z}$, primes and irreducibles are the same thing, but this is not the case more generally.

DEFINITION. We say that a number x in $\mathbb{Z}[\sqrt{-5}]$ has *unique factorization* if whenever we have two factorizations

$$x = p_1 p_2 \cdots p_k = q_1 q_2 \cdots q_\ell,$$

where all the p_i 's and q_i 's are irreducibles, then $k = \ell$, and after rearranging all the p_i 's and q_i 's if necessary, there exist units $u_1, \dots, u_k$ such that $p_i = u_i q_i$ for all i .

PROBLEM 38.5. Show that 6 does not have unique factorization in $\mathbb{Z}[\sqrt{-5}]$ by finding two distinct factorizations into irreducibles. Show that all numbers you claim to be irreducibles are, in fact, irreducibles.

Another famous counterexample to unique factorization is the set of even integers. Let $2\mathbb{Z}$ be the set of even integers. Since $2\mathbb{Z}$ does not contain 1, we cannot use the same definition of irreducibility as before, so here we declare that a positive even is irreducible if it cannot be written as a product of two positive even integers.

PROBLEM 38.6. What are the irreducibles in $2\mathbb{Z}$?

PROBLEM 38.7. What is the smallest number in $2\mathbb{Z}$ that has two distinct factorizations into irreducibles? As before, this is up to signs and reordering: for instance 2×6 , 6×2 , and $(-2) \times (-6)$ do not count as distinct factorizations.

PROBLEM 38.8. Can you find a formula for the number of distinct factorizations of a number n in $2\mathbb{Z}$?

PROBLEM 38.9. Do the numbers in $\mathbb{Z}[\sqrt{2}]$, $\mathbb{Z}[\sqrt{-1}]$, and $\mathbb{Z}[\sqrt{-3}]$ have unique factorization?

Further reading. For a detailed look at factorization in $\mathbb{Z}[\sqrt{-5}]$, see [Dia56]. For a broader survey of unique factorization and its failures, see [Sam68]. Both of these references go on to discuss the way of repairing unique factorization in objects like $\mathbb{Z}[\sqrt{-5}]$, using *ideals*.

WORKSHEET 39

Tropical Algebra

Tropical algebra and, especially, tropical geometry is an up-and-coming area of mathematics. We'll just barely scratch the surface.

Tropical algebra concerns the set $\mathbb{R} \cup \{\infty\}$, together with a pair of operations that we denote $\oplus$ and $\odot$ that share certain properties with normal addition and multiplication, but not all of them. If x and y are real numbers, then we define

$$x \oplus y = \min(x, y)$$

and

$$x \odot y = x + y.$$

That is, the tropical sum is the minimum, and the tropical product is normal addition. We also need to allow for the possibilities that at least one of x and y is ∞ , so here is how we complete the definitions of the tropical sum and product: for any x in $\mathbb{R} \cup \{\infty\}$, we have $x \oplus \infty = x$ and $x \odot \infty = \infty$.

PROBLEM 39.1. Write down the tropical addition and multiplication tables for $x \oplus y$ and $x \odot y$ for $0 \leq x, y \leq 5$.

Given an operation $\star$ on a set X , we say that it is *commutative* if $x \star y = y \star x$ for all x and y , and we say that it is *associative* if $x \star (y \star z) = (x \star y) \star z$ for all x, y, z .

PROBLEM 39.2. Show that the tropical sum and product are both commutative and associative.

PROBLEM 39.3. Show that the tropical product distributes over the tropical sum, i.e.

$$x \odot (y \oplus z) = (x \odot y) \oplus (x \odot z).$$

PROBLEM 39.4. In normal arithmetic, we have a 0 element with the property that $0 + x = x$ for all x . Does there exist a similar 0 element for tropical addition? That is, does there exist some $z \in \mathbb{R} \cup \{\infty\}$ such that $z \oplus x = x$ for all $x \in \mathbb{R} \cup \{\infty\}$?

PROBLEM 39.5. In normal arithmetic, we have a 1 element with the property that $1 \cdot x = x$ for all x . Does there exist a similar 1 element for tropical multiplication?

PROBLEM 39.6. In normal arithmetic, we can define *subtraction* in terms of addition: $x - y$ is the number z such that $y + z = x$. Is there an analogous tropical subtraction $x \ominus y$?

We can define tropical exponentiation in the same way that we define usual exponentiation: as repeated multiplication. We'll just write it the same way as we do in normal arithmetic: in the context of tropical arithmetic, x^2 means $x \odot x$, for instance. I hope that there will be no confusion.

PROBLEM 39.7. It would be nice in normal arithmetic if $(x + y)^2$ were equal to $x^2 + y^2$, but alas this is usually not the case. Show, however, that $(x \oplus y)^2 = x^2 \oplus y^2$.

PROBLEM 39.8. More generally, show that $(x \oplus y)^n = x^n \oplus y^n$.

We can also talk about tropical polynomials. If x is a variable, then we can write something like $5 \odot x^3$, which means $5 \odot x \odot x \odot x$; this is a *monomial*. A *polynomial* is a finite sum of monomials.

More generally, we can consider polynomials in several variables. If x, y, z are variables, for instance, then $5 \odot x^2 y z^3$ means $5 \odot x \odot x \odot y \odot z \odot z \odot z$. A (multivariate) polynomial is a finite (tropical) sum of such monomials.

PROBLEM 39.9. Draw the graph of the tropical polynomial $1 \odot x^3 \oplus 2 \odot x^2 \oplus 4 \odot x \oplus 7$. Note that the order of operations is analogous to the usual order of operations: perform tropical multiplication before tropical addition.

Note that, when we write something like this, we obey the usual order of operations: we perform (tropical) multiplications before (tropical) additions.

PROBLEM 39.10. Show that the graph of a tropical cubic polynomial $a \odot x^3 \oplus b \odot x^2 \oplus c \odot x \oplus d$ has at most four line segments. Under what conditions does it have exactly four line segments?

PROBLEM 39.11. More generally, show that if p is a single-variable tropical polynomial of degree n , then the graph of p consists of only line segments, with the maximum number of segments being $n + 1$.

PROBLEM 39.12. Graph the tropical polynomials $0 \odot x^2 \oplus 10 \odot x \oplus 2$, $0 \odot x^2 \oplus 2 \odot x \oplus 2$, and $0 \odot x^2 \oplus 1 \odot x \oplus 2$. What is going on? What about $0 \odot x^2 \oplus 0 \odot x \oplus 2$?

As you saw in the previous problem, it is possible for two different tropical polynomials to have the same graph, and thus the same values for all inputs of the variable x .

PROBLEM 39.13. Can you find a factorization of the polynomial $p(x) = 0 \odot x^2 \oplus 10 \odot x \oplus 2$ into linear factors, i.e. $a \odot (x \oplus b) \odot (x \oplus c)$ for

some a, b, c ? By this, we mean that this product has the same graph as $p(x)$, not that it expands to p .

PROBLEM 39.14. Show that every single-variable tropical polynomial can be uniquely factored into linear factors as a tropical polynomial function. That is, suppose that $p(x) = a_0 \oplus a_1 \odot x \oplus \cdots \oplus a_d \odot x_d$ is a tropical polynomial, where $a_0, a_d \neq \infty$. Show that there are unique real numbers $r_1 \leq r_2 \leq \cdots \leq r_d$ such that

$$p(x) = a_d \odot (x \oplus r_1) \cdot (x \oplus r_2) \cdots \odot (x \oplus r_d)$$

for all real numbers x .

Further reading. See [SS09] for an elementary introduction to tropical algebra and tropical geometry.

Complementary Sequences

Recall the notation $\lfloor x \rfloor$, the *floor* of x , which means the greatest integer $\leq x$. For instance, $\lfloor \pi \rfloor = 3$, and $\lfloor 2 \rfloor = 2$.

Consider the two sequences $a_n = \lfloor \phi n \rfloor$ and $b_n = \lfloor \phi^2 n \rfloor$, where $\phi = \frac{1+\sqrt{5}}{2}$ is the golden ratio. The a_n sequence begins

$$1, 3, 4, 6, 8, 9, 11, 12, 14, 16, 17, 19, 21, \dots,$$

whereas the b_n sequence begins

$$2, 5, 7, 10, 13, 15, 18, 20, \dots$$

Observe that it appears that every positive integer appears in exactly one of the two sequences. The sequences a_n and b_n are called the *lower Wythoff sequence* and the *upper Wythoff sequence*, respectively.

PROBLEM 40.1. Let N be a large positive integer. Roughly what proportion of the positive integers up to N belong to the lower Wythoff sequence? What about the upper Wythoff sequence? Conclude that it is plausible that every positive integer appears in exactly one of the two sequences.

PROBLEM 40.2. More generally, suppose that α and β are positive irrational numbers such that $\frac{1}{\alpha} + \frac{1}{\beta} = 1$. Show that it is plausible that every positive integer can be written as $\lfloor \alpha n \rfloor$ or $\lfloor \beta n \rfloor$ for some positive integer n , but not both.

It is indeed true that, with the hypotheses as above, every positive integer has the form $\lfloor \alpha n \rfloor$ or $\lfloor \beta n \rfloor$ but not both. These two sequences are called *Beatty sequences*, and this fact is called *Rayleigh's Theorem*. To show this, we must show that there are two things that do not happen, which we'll call *collisions* and *anticollisions*. A positive integer j is a *collision* if there exist positive integers k and m such that $j = \lfloor k\alpha \rfloor = \lfloor m\beta \rfloor$, and j is an *anticollision* if there are no positive integers k or m such that $j = \lfloor k\alpha \rfloor$ or $j = \lfloor m\beta \rfloor$.

PROBLEM 40.3. Suppose j is a collision. Write out inequalities for what this means, without referring to floors.

PROBLEM 40.4. Rearrange the inequalities you got in the previous problem, using the fact that $\frac{1}{\alpha} + \frac{1}{\beta} = 1$, to show that $j < k + m < j + 1$. Conclude that there are no collisions.

PROBLEM 40.5. Write the condition that j is an anticollision in terms of inequalities, without using floors.

PROBLEM 40.6. Rearrange the inequalities you got in the previous problem to show that if j is an anticollision, then there exist integers k and m such that $k + m < j < k + m + 1$. Conclude that there are no anticollisions.

You have now completed the proof of Rayleigh's Theorem!

The notation $\lceil x \rceil$ means the *ceiling* of x , i.e. the least integer $\geq x$. (In other words, floor means round down, and ceiling means round up.)

PROBLEM 40.7. Show that if α and β are positive real numbers, which need not be irrational, with $\frac{1}{\alpha} + \frac{1}{\beta} = 1$, then every integer (positive, negative, or zero) has the form $\lfloor n\alpha \rfloor$ or $\lceil n\beta \rceil - 1$ for some integer n , but not both.

PROBLEM 40.8. Let $\alpha = \frac{12}{7}$ and $\beta = \frac{12}{5}$ in the previous problem and write down the sequences. What does this have to do with the keys on a piano?

PROBLEM 40.9. Let a_n and b_n denote the lower and upper Wythoff sequences, respectively. Show that $b_n = a_{a_n} + 1$.

Call two sequences a_n and b_n *complementary* if $a_1 < a_2 < a_3 < \dots$, $b_1 < b_2 < b_3 < \dots$, and every positive integer occurs in exactly one of the sequences.

PROBLEM 40.10. Show that the upper and lower Wythoff sequences are the *only* solution to $b_n = a_{a_n} + 1$ where the a_n 's and b_n 's form complementary sequences.

PROBLEM 40.11. What are the solutions to the equation $b_n = a_{n+a_n} + 1$ in complementary sequences?

PROBLEM 40.12. For a set S of nonnegative integers, let $r_S(n)$ be the number of ordered pairs (s_1, s_2) such that s_1, s_2 are in S , $s_1 \neq s_2$, and $s_1 + s_2 = n$. Is it possible to partition the nonnegative integers into two sets A and B in such a way that $r_A(n) = r_B(n)$ for all nonnegative integers n ?

Further reading. See [GY12] for another proof of Rayleigh's Theorem, and some generalizations. For more on complementary sequences and equations in complementary sequences, see [Kim07] and [Kim08].

Inversion

Given a point P and a line AB , we can reflect P across AB : its reflection P' is the same distance from AB as P is, but on the opposite side, i.e. the lines AB and PP' are perpendicular. See Figure 1.

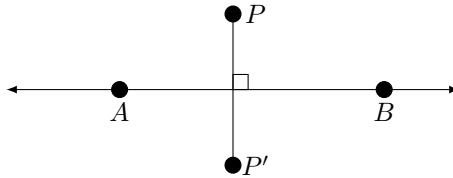


Figure 1. Reflecting P across AB .

Inversion is the analogue of reflection where we replace lines with circles. If we invert with respect to a circle, then the outside of the circle becomes the inside, and vice versa. Here's how we set it up more precisely: Suppose we have a circle Ω centered at O with radius r . Then if we invert a point $P \neq O$ about Ω to get P' , then P' should lie on the ray $\overrightarrow{OP}$, and we should have $OP \cdot OP' = r^2$. See Figure 2.

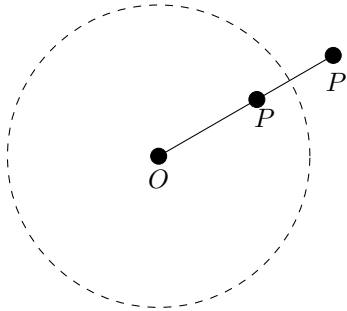


Figure 2. Inverting P about a circle.

PROBLEM 41.1. Let Ω be a circle centered at O , and let P and Q be two distinct points, neither of which is O . Let P' and Q' be

the inverses of P and Q , respectively, with respect to Ω . Show that $\triangle OPQ \sim \triangle OQ'P'$.

PROBLEM 41.2. Let ℓ be a line through O . What is the inverse of ℓ ? (That is, invert every point on ℓ other than O itself. What do these points look like?)

PROBLEM 41.3. Let $\mathcal{C}$ be a circle passing through O . What is the inverse of $\mathcal{C}$?

PROBLEM 41.4. Let ℓ be a line not passing through O . What is the inverse of ℓ ?

PROBLEM 41.5. Let $\mathcal{C}$ be a circle not passing through O . What is the inverse of $\mathcal{C}$?

PROBLEM 41.6. Which circles invert to themselves?

A *generalized circle* means either a circle or a line. The tangent line to a line at any of its points is simply the line itself. Suppose two generalized circles intersect at a point P . Their *angle* at P is the smaller angle between their tangent lines at P .

Here is one geometric description of the tangent line. If a point Q moves along a generalized circle $\mathcal{C}$ toward a point P on $\mathcal{C}$, then the secant line PQ approaches the tangent line to $\mathcal{C}$ at P .

We will now show that inversion preserves angles. We assume the following setup: Let $\mathcal{C}$ be a generalized circle, let $P \neq O$ be a point on $\mathcal{C}$, and let $\mathcal{C}'$ be the inverse of $\mathcal{C}$. Let P' be the inverse of P . Let Q be another point on $\mathcal{C}$ with inverse Q' . By Problem 41.1, $\angle OQP = \angle OP'Q'$.

PROBLEM 41.7. Let Q move along $\mathcal{C}$ toward P . Explain why Q' approaches P' , and why the secant lines PQ and $P'Q'$ approach the tangent lines to $\mathcal{C}$ and $\mathcal{C}'$, respectively. Conclude that the tangent to $\mathcal{C}'$ at P' is the reflection across the line OP of the tangent to $\mathcal{C}$ at P .

PROBLEM 41.8. Suppose that two generalized circles $\mathcal{C}_1$ and $\mathcal{C}_2$ intersect at a point $P \neq O$. Show that the angle between $\mathcal{C}_1$ and $\mathcal{C}_2$ at P is equal to the angle between $\mathcal{C}'_1$ and $\mathcal{C}'_2$ at P' .

Distances are not preserved under inversion. However, there is one ratio of distances that is preserved. Suppose A, B, C, D are four points in the plane. We define their *cross ratio* $(A, B; C, D)$ to be

$$(A, B; C, D) = \frac{AC \cdot BD}{BC \cdot AD}.$$

PROBLEM 41.9. Suppose A, B, C, D are four distinct points, none of which is O , and let Ω be a circle centered at O . Let A', B', C', D' be the inverses of A, B, C, D , respectively, with respect to O . Show that $(A, B; C, D) = (A', B'; C', D')$.

We will now see an example of how to use inversion to prove a classical theorem in geometry: Ptolemy's Theorem. Ptolemy's Theorem says that if $ABCD$ is a (non-intersecting) quadrilateral inscribed in a circle (i.e. a *cyclic quadrilateral*), then

$$AB \cdot CD + AD \cdot BC = AC \cdot BD.$$

Let us write $\mathcal{C}$ for the circle containing points A, B, C, D .

PROBLEM 41.10. Let Ω be a circle centered at D . What is the inverse of $\mathcal{C}$ under Ω ?

PROBLEM 41.11. Let A', B', C' be the inverses of A, B, C with respect to Ω . Show that $A'B' + B'C' = A'C'$.

PROBLEM 41.12. Use the result of the previous problem to get a relation among the distances between A, B, C, D .

PROBLEM 41.13. Show that you can rearrange the relation you got from the last problem into Ptolemy's equation, thus proving Ptolemy's Theorem.

PROBLEM 41.14. Show that if A, B, C, D do not lie on a generalized circle, then $AB \cdot CD + AD \cdot BC > AC \cdot BD$.

Further reading. See [CG67, Chapter 5] and [Che16, Chapter 8] for more on inversion.

Mascheroni Constructions

A family of classical problems in geometry is to construct various objects using a compass and straightedge. A lesser known challenge is to construct them using only a compass.

Before we do that, let's state the rules for compass-and-straightedge constructions:

- Given two points A and B , we can draw a line through A and B .
- Given two points A and B , we can draw a circle centered at A passing through B .
- Given two nonparallel lines AB and CD , construct the intersection point of AB and CD .
- Given a line AB and a circle $\mathcal{C}$, construct the intersection point(s) of AB and $\mathcal{C}$, if there are any.
- Given two circles, construct the intersection point(s) of them, if there are any.

With only a compass, we clearly can't draw a line, so it is not possible to draw the line through two points A and B . So, instead, we treat lines as being determined by two points, without needing to draw the full line. The Mohr–Mascheroni Theorem claims that anything we can construct using a compass and straightedge can be constructed using a compass alone. We assume that our compass is rigid, which means that if we have constructed points A , B , and C , then we can construct a circle centered at A with radius BC . This is not necessary though; it is also possible to use a collapsible compass in the Mohr–Mascheroni Theorem, which cannot be used to transfer distances.

PROBLEM 42.1. Construct an equilateral triangle with a compass alone. That is, given two distinct points A and B , construct a point C using a compass alone such that $\triangle ABC$ is equilateral.

PROBLEM 42.2. Construct a regular hexagon with a compass alone. That is, given two distinct points A and B , construct points C, D, E, F such that $\odot ABCDEF$ is a regular hexagon.

Let us now turn to the proof of the Mohr–Mascheroni Theorem. In order to prove this theorem, we need to explain how to simulate a

straightedge. It suffices to show how we can perform all five of the basic operations using just a compass, since those are the only fundamental things we can do with a straightedge and a compass.

Of these basic operations, three of them are straightforward with just a compass. The first one, constructing a line through two points, is moot, because we're claiming that a line simply *is* two points on it, so there's nothing to do here. With a compass, we can clearly draw a circle centered at A passing through B , and we can also intersect two circles with a compass. So, we have only two tasks left: finding the intersection of two lines, and finding the intersection of a line and a circle. In order to do those, we'll need to start with some preliminary constructions.

PROBLEM 42.3. Given a point C and a line AB , show that it is possible to reflect C across AB using only a compass.

PROBLEM 42.4. Given a line segment $\overline{AB}$, show that we can extend it to a line segment $\overline{AC}$ such that B is the midpoint of $\overline{AC}$, using only a compass.

PROBLEM 42.5. Given a point O , a circle Ω centered at O , and a point $A \neq O$, show that we can construct the inverse A' of A with respect to Ω using only a compass. (See Figure 1 for the construction, but you must still explain why this construction works. Note also that the construction might fail, if the circle centered at A through O does not intersect Ω at two distinct points. How do you repair the argument in that case, using some other constructions we have already seen?)

PROBLEM 42.6. Show that we can trisect a line segment with a compass alone.

Our next task is to construct the circumcenter of a triangle with only a compass. That is, given three points A, B, C that aren't collinear, we want to construct the center O of $\triangle ABC$. The construction is hard to come up with from scratch.

Let Γ be the circle centered at A passing through B . Let C' be the inverse of C with respect to Γ . Let X be the reflection of A across the line BC' . Let O be the inverse of X with respect to Γ .

PROBLEM 42.7. Show that $OA = OB = OC$. Thus O is the center of the circle passing through A, B , and C .

PROBLEM 42.8. Construct the intersection of two nonparallel lines using only a compass. (Start by drawing a random point O not on either line and a random circle Ω centered at O , and then inverting with respect to Ω .)

PROBLEM 42.9. Suppose $\mathcal{C}$ is a circle centered at O , and AB is a line such that O lies on the line AB . Construct the intersection points of AB and $\mathcal{C}$ using only a compass.

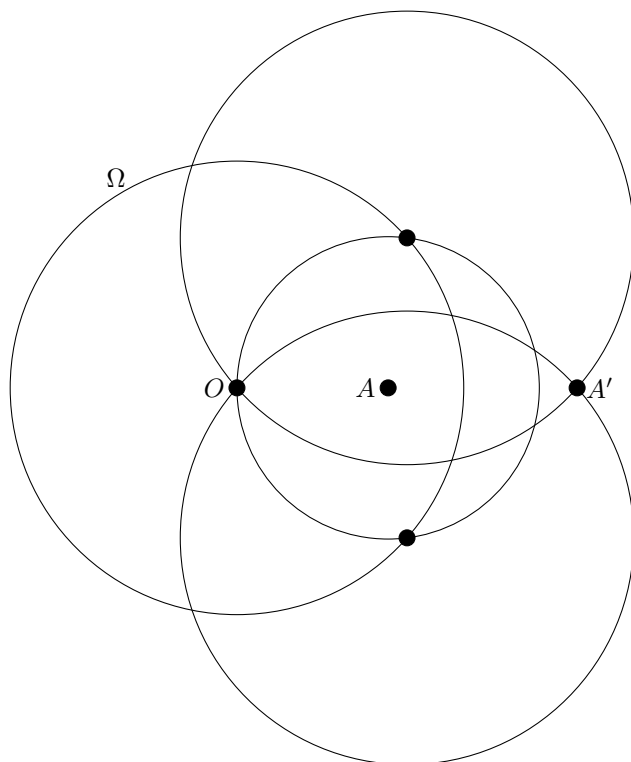


Figure 1. Inversion with a compass alone.

PROBLEM 42.10. Suppose Γ is a circle centered at O , and ABC is a line such that O does not lie on the line ABC . Construct the intersection points of AB and Γ using only a compass.

This completes the proof of the Mohr–Mascheroni Theorem.

PROBLEM 42.11. Construct a square with a compass alone.

You might wonder what you can do with just a straightedge. It turns out that with a straightedge alone, you cannot do very much, but if you are given just one circle and its center drawn on the page in advance, then you can simulate an arbitrary compass-and-straightedge construction with just a straightedge. This is known as the *Poncelet–Steiner Theorem*.

Further reading. See [Dör65, Chapter 33] for the Mohr–Mascheroni Theorem and [Dör65, Chapter 34] for the Poncelet–Steiner Theorem.

WORKSHEET 43

Information Entropy

The idea of information entropy is to determine the minimum number of yes/no questions you need, on average over a long run of questions, to tell someone else a piece of information.

PROBLEM 43.1. You are thinking of a number from 1 to 16. What is the minimum number of yes/no questions I have to ask you so that I can determine the number you are thinking of?

Note that I can ask any yes/no question, such as “Is the number 12?” or “Is the number one of the following numbers: 1, 3, 8, 14?”

In general, if you pick a number from 1 to 2^n , then I need to ask you n yes/no questions in order to learn what your number is. But what happens if the maximum is not a power of 2? In that case, the number of questions I need to ask may depend on the number chosen, and my goal is to make the *average* number of questions I have to ask as small as possible.

PROBLEM 43.2. You are thinking of a number from 1 to 100. What is the minimum average number of yes/no questions I have to ask you to determine your number, where the average is taken over all possible choices of your number?

Suppose you have a message x to deliver to me, that could take on one of several different values, say $x_1, \dots, x_n$. These need not be equally likely, so let's say that the probability that your message is x_i is p_i , where $0 \leq p_i \leq 1$. The *information entropy* is the average number of yes/no questions I have to ask to learn your message. The formula for the information entropy of x is

$$H(x) = \sum_{i=1}^n p_i \log_2 \frac{1}{p_i}.$$

PROBLEM 43.3. Show that the information entropy in problem 43.1 matches the answer you got for that problem.

The answer doesn't match in problem 43.2, because on average you can do better if you have to determine random numbers from 1 to 100 many times, and that's what the formula for information entropy is

getting at. That is, if you have to describe many numbers from 1 to 100 at the same time, you can do so in fewer questions than you would need to describe them all individually, because the questions can be about several of the numbers at once.

PROBLEM 43.4. Suppose you roll an unfair die, which rolls the following with different probabilities:

Number	1	2	3	4	5	6
Probability	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{32}$

How can I ask yes/no questions to determine the value of the die roll with the fewest average number of questions? Calculate the information entropy, and see if the answers agree.

PROBLEM 43.5. Same question, but now you roll two normal dice and wish to report their sum.

In general, we see that the information entropy provides a lower bound on the average number of yes/no questions we have to ask to learn a piece of information. The next challenge is to determine the actual minimum number of yes/no questions we need. This is accomplished by means of the *Huffman code*. If you found the optimal answer to the previous problem, you would have constructed a Huffman code.

PROBLEM 43.6. Suppose that I wish to learn a piece of information by means of asking yes/no questions. There are three possibilities for the information, say A , B , and C , which occur with probability $\frac{2}{5}$, $\frac{3}{10}$, and $\frac{3}{10}$, respectively. On my first question, I could ask if it's A , or I could ask if it's B , or if it's C . Which one allows me to learn the information in the fewest average number of guesses?

Given a sequence of yes/no questions, we can form a decision tree of our guesses, for instance as shown in Figure 1. We consider the left branch of a node as an answer of “no,” and a right branch as an answer of “yes.” The leaves of the tree, which have no children, are the possible messages. For instance, if we receive the answers yes, then yes, then no to three questions, then the message must have been e .

The most efficient decision tree, called the *Huffman tree*, is constructed as follows. We list all possible messages by their probabilities, with the highest probability messages on top and the lowest probability messages on bottom. Then we combine the two lowest probability messages and intend to ask one question at the end to distinguish them. Then we combine these two messages into a single message and then consider it as a new message whose probability is the sum of the two we merged. Then we repeat this process.

For instance, suppose we have five messages with the following probabilities:

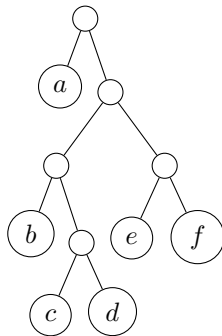
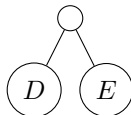


Figure 1. A decision tree.

Message	A	B	C	D	E
Probability	.4	.2	.3	.06	.04

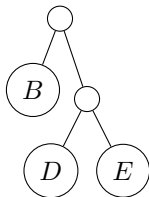
The two lowest probability messages are D and E , so we create a partial tree from them:



Now we have the following probabilities:

Message	A	B	C	DE
Probability	.4	.2	.3	.1

The two lowest probability messages are now B and DE , so we create a partial tree like this:



Then we continue in this way until we have constructed the full tree, which is the Huffman tree. The sequence of yes/no answers corresponding to each answer is the Huffman code.

PROBLEM 43.7. Construct the Huffman tree for the following sequence of messages and probabilities:

Message	A	B	C	D	E	F	G	H	I	J	K	L
Probability	.2	.15	.12	.25	.06	.03	.01	.12	.03	.01	.01	.01

PROBLEM 43.8. Suppose we wish to convert a string of text to a binary string, in such a way that each letter corresponds to a binary string, and the full binary string can be unambiguously decomposed

into individual letters. What is the shortest possible length of such a binary encoding of the word “ABRACADABRA”?

Further reading. See [Mac03, Chapters 1–7] and [Say06, Chapters 2 and 3] for more on entropy and Huffman codes.

WORKSHEET 44

Large Numbers

PROBLEM 44.1. Which number is larger: 2^{3000} or 3^{2000} ?

PROBLEM 44.2. What is the largest number that is a product of positive integers whose sum is 10000?

PROBLEM 44.3. The number called a *googol* is defined to be 10^{100} , and a *googolplex* is 10^{googol} . Is a googolplex larger than $2^{2^{2^{2^2}}}$? How many 2's do we need in the tower before it exceeds a googolplex?

Suppose we define an n -plex as 10^n . Then one way to create large numbers is to write something like “a googolplexplexplexplex... with a googol plexes.” Let's call this number G . But there are relatively compact expressions that describe much larger numbers.

If $f(x)$ is a function and n is a positive integer, let us write $f^{(n)}(x)$ for $f(f(\cdots(f(x))))$, where there are n f 's. For example, if $f(x) = x + 1$, then $f^{(n)}(x) = x + n$.

PROBLEM 44.4. If $f(x) = 2x$, what is $f^{(n)}(x)$. What about if $f(x) = 2^x$?

PROBLEM 44.5. If x is a nonnegative integer, define a sequence of function $f_k(x)$ by setting $f_1(x) = 2x$ and $f_{k+1}(x) = f_k^{(x)}(1)$. Describe $f_3(x)$ and $f_4(x)$ as concretely as you can.

PROBLEM 44.6. Can you determine the smallest positive integer k for which $f_k(k) > G$?

We now define a function called the *Ackermann function*. It is a function $A(m, n)$ of two variables m and n , defined recursively via the rule

$$A(m, n) = \begin{cases} n + 1 & m = 0, \\ A(m - 1, 1) & m > 0 \text{ and } n = 0, \\ A(m - 1, A(m, n - 1)) & m > 0 \text{ and } n > 0. \end{cases}$$

PROBLEM 44.7. Find a formula for $A(1, n)$ as a function of n .

PROBLEM 44.8. Find a formula for $A(2, n)$ as a function of n .

PROBLEM 44.9. Find a formula for $A(3, n)$ as a function of n .

PROBLEM 44.10. Find a formula for $A(4, n)$ as a function of n .

PROBLEM 44.11. Find the smallest number n such that $A(n, n) > G$.

Knuth's *up arrow notation* is a convenient way to express some rather large numbers. It is defined as follows:

$$a \uparrow^n b = \begin{cases} a^b & n = 1, \\ 1 & n > 1 \text{ and } b = 0, \\ a \uparrow^{n-1} (a \uparrow^n (b-1)) & \text{otherwise.} \end{cases}$$

We sometimes write $a \uparrow\uparrow b$ for $a \uparrow^2 b$, $a \uparrow\uparrow\uparrow b$ for $a \uparrow^3 b$, and so forth.

PROBLEM 44.12. Express $a \uparrow\uparrow b$ in terms of usual operations, without any reference to up arrows or recurrences.

PROBLEM 44.13. Express $A(m, n)$ in terms of up arrow notation.

Further reading. See [Knu76] for up arrow notation and some problems where it arises naturally. See [Kno81] for some large numbers of the sorts we have considered in this worksheet, and [Spe83] for some much larger ones.

Bibliography

- [AF04] Titu Andreescu and Zuming Feng. *A Path to Combinatorics for Undergraduates*. Birkhauser, 2004.
- [ANW19] Michael H. Albert, Richard J. Nowakowski, and David Wolfe. *Lessons in Play. An Introduction to Combinatorial Game Theory*. Boca Raton, FL: CRC Press/A K Peters, 2nd edition, 2019.
- [Apo76] Tom M. Apostol. *Introduction to Analytic Number Theory*. Springer, 1976.
- [Arm88] M. A. Armstrong. *Groups and Symmetry*. Springer, 1988.
- [AS99] Jean-Paul Allouche and Jeffrey Shallit. The ubiquitous Prouhet-Thue-Morse sequence. In *Sequences and their applications. Proceedings of the international conference, SETA '98, Singapore, December 14–17, 1998*, pages 1–16. London: Springer, 1999.
- [AS03] Jean-Paul Allouche and Jeffrey Shallit. *Automatic Sequences: Theory, Applications, Generalizations*. Cambridge University Press, 2003.
- [Aus16] David Austin. Knot quandaries quelled by quandles. American Mathematical Society, *Feature Column*, <https://www.ams.org/publicoutreach/feature-column/fc-2016-03>, March 2016. Accessed 13 July 2026.
- [AZ18] Martin Aigner and Günter M. Ziegler. *Proofs from THE BOOK. Including illustrations by Karl H. Hofmann*. Berlin: Springer, revised and enlarged 6th edition, 2018.
- [BFM15] David Bowman, Ross Flek, and Greg Markowsky. Rotation numbers and symbolic dynamics. *Ann. Acad. Sci. Fenn., Math.*, 40(1):227–234, 2015.
- [BH11] Keith Burns and Boris Hasselblatt. The Sharkovsky theorem: a natural direct proof. *Am. Math. Mon.*, 118(3):229–244, 2011.
- [Bon12] Miklos Bona. *Combinatorics of Permutations*. CRC Press, 2nd edition, 2012.
- [BQ03] Arthur T. Benjamin and Jennifer J. Quinn. *Proofs That Really Count: The Art of Combinatorial Proof*. Mathematical Association of America, 2003.
- [Bro07] Gary Brookfield. Factoring quartic polynomials: A lost art. *Mathematics Magazine*, 80(1):67–70, 2007.
- [CG67] H. S. M. Coxeter and S. L. Greitzer. *Geometry Revisited*. Mathematical Association of America, 1967.
- [Che16] Evan Chen. *Euclidean Geometry in Mathematical Olympiads*. Mathematical Association of America, 2016.
- [Cro97] Peter R. Cromwell. *Polyhedra*. Cambridge University Press, 1997.
- [Dav08] Harold Davenport. *The Higher Arithmetic: An Introduction to the Theory of Numbers*. Cambridge University Press, 8th edition, 2008.
- [Dav16] Diana Davis. Lines in positive genus: an introduction to flat surfaces. In *Dynamics done with your bare hands. Lecture notes by Diana Davis*,

- Bryce Weaver, Roland K. W. Roeder, Pablo Lessa*, pages 1–55. Zürich: European Mathematical Society (EMS), 2016.
- [DG15] Persi Diaconis and Ron Graham. *Magical mathematics. The mathematical ideas that animate great magic tricks. With a foreword by Martin Gardner*. Princeton, NJ: Princeton University Press, reprint of the 2012 hardback edition edition, 2015.
- [Dia56] Louis E. Diamond. Integers, unique factorization and ideals. *Mathematics Magazine*, 30(1):29–40, 1956.
- [Dic22] Leonard Eugene Dickson. *First Course in the Theory of Equations*. John Wiley & Sons, New York, 1922.
- [Dör65] Heinrich Dörrie. *100 Great Problems of Elementary Mathematics: Their History and Solution*. Dover, 1965.
- [DW22] Moon Duchin and Olivia Walch. *Political Geometry: Rethinking Redistricting in the US with Math, Law, and Everything In Between*. Birkhauser, 2022.
- [Edg08] Gerald Edgar. *Measure, topology, and fractal geometry*. Undergraduate Texts Math. New York, NY: Springer, 2nd edition, 2008.
- [Edw87] A. W. F. Edwards. *Pascal's Arithmetical Triangle*. Johns Hopkins University Press, 1987.
- [EN15] Mohamed Elhamdadi and Sam Nelson. *Quandles. An introduction to the algebra of knots*, volume 74 of *Stud. Math. Libr.* Providence, RI: American Mathematical Society (AMS), 2015.
- [Eng98] Arthur Engel. *Problem-Solving Strategies*. Springer, 1998.
- [GKP94] Ronald L. Graham, Donald E. Knuth, and Oren Patashnik. *Concrete Mathematics: A Foundation for Computer Science*. Addison-Wesley, 2nd edition, 1994.
- [Gol92] Lisa R. Goldberg. Fixed points of polynomial maps. I: Rotation subsets of the circles. *Ann. Sci. Éc. Norm. Supér. (4)*, 25(6):679–685, 1992.
- [GS97] Charles M. Grinstead and J. Laurie Snell. *Introduction to Probability*. American Mathematical Society, 1997.
- [GY12] Yuval Ginosar and Ilan Yona. A model for pairs of Beatty sequences. *The American Mathematical Monthly*, 119(8):636–645, 2012.
- [Hav07] Julian Havil. *Nonplussed! Mathematical Proof of Implausible Ideas*. Princeton University Press, 2007.
- [HH10] Harald Helfgott and Michel Helfgott. A modern vision of the work of Cardano and Ferrari on quartics. *Convergence*, February 2010.
- [HK18] Jonathan K. Hodge and Richard E. Klima. *The Mathematics of Voting and Elections: A Hands-On Approach*. American Mathematical Society, 2 edition, 2018.
- [Irv13] Ronald S. Irving. *Beyond the Quadratic Formula*, volume 43 of *Classroom Resource Materials*. Mathematical Association of America, Washington, DC, 2013.
- [Kim07] Clark Kimberling. Complementary equations. *J. Integer Seq.*, 10(1):article 07.1.4, 14, 2007.
- [Kim08] Clark Kimberling. Complementary equations and Wythoff sequences. *Journal of Integer Sequences*, 11, 2008. Article 08.3.3.
- [Kno81] R. Arthur Knoebel. Exponentials reiterated. *American Mathematical Monthly*, 88(4):235–252, 1981.
- [Knu76] Donald E. Knuth. Mathematics and computer science: Coping with finiteness. *Science*, 194(4271):1235–1242, 1976.
- [Kör19] T. W. Körner. *Where Do Numbers Come From?* Cambridge University Press, Cambridge, 2019.

- [Kos01] Thomas Koshy. *Fibonacci and Lucas Numbers with Applications*. Wiley, 2001.
- [Liv93] Charles Livingston. *Knot Theory*. Mathematical Association of America, 1993.
- [LY75] T. Y. Li and James A. Yorke. Period three implies chaos. *American Mathematical Monthly*, 82(10):985–992, 1975.
- [Mac03] David J. C. MacKay. *Information Theory, Inference, and Learning Algorithms*. Cambridge University Press, 2003.
- [MGGG16] Liz McMahon, Gary Gordon, Hannah Gordon, and Rebecca Gordon. *The Joy of SET: The Many Mathematical Dimensions of a Seemingly Simple Card Game*. Princeton University Press, 2016.
- [Mos65] Frederick Mosteller. *Fifty Challenging Problems in Probability with Solutions*. Dover, 1965.
- [Osb04] Martin J. Osborne. *An Introduction to Game Theory*. Oxford University Press, 2004.
- [Pes03] Peter Pesic. *Abel’s Proof: An Essay on the Sources and Meaning of Mathematical Unsolvability*. MIT Press, Cambridge, MA, 2003.
- [Prz19] Józef H. Przytycki. Knot theory: From Fox 3-colorings of links to Yang–Baxter homology and Khovanov homology. In Colin C. Adams, Cameron McA. Gordon, Vaughan F. R. Jones, Louis H. Kauffman, Sofia Lambropoulou, Kenneth C. Millett, Józef H. Przytycki, Renzo Ricca, and Radmila Sazdanovic, editors, *Knots, Low-Dimensional Topology and Applications: Knots in Hellas, International Olympic Academy, Greece, July 2016*, volume 284 of *Springer Proceedings in Mathematics & Statistics*, pages 115–145. Springer, Cham, 2019.
- [Ram22] Paul Ramond. The Abel–Ruffini theorem: Complex but not complicated. *The American Mathematical Monthly*, 129(3):231–245, 2022.
- [Ric08] David S. Richeson. *Euler’s Gem: The Polyhedron Formula and the Birth of Topology*. Princeton University Press, 2008.
- [RS23] Simon Rubinstein-Salzedo. *Transition to proofs*. World Scientific, 2023.
- [Sam68] Pierre Samuel. Unique factorization. *The American Mathematical Monthly*, 75(9):945–952, 1968.
- [Say06] Khalid Sayood. *Introduction to Data Compression*. Morgan Kaufmann, 3 edition, 2006.
- [Sil13] Joseph H. Silverman. *A Friendly Introduction to Number Theory*. Pearson, 4th edition, 2013.
- [Sip12] Michael Sipser. *Introduction to the Theory of Computation*. Cengage Learning, 3rd edition, 2012.
- [Spe83] Joel Spencer. Large numbers and unprovable theorems. *Am. Math. Mon.*, 90:669–675, 1983.
- [SS09] David Speyer and Bernd Sturmfels. Tropical mathematics. *Mathematics Magazine*, 82(3):163–173, 2009.
- [Ste04] J. Michael Steele. *The Cauchy–Schwarz Master Class*. Cambridge University Press, 2004.
- [Str93] Philip D. Straffin. *Game Theory and Strategy*. Mathematical Association of America, 1993.
- [Vel06] Daniel J. Velleman. *How to Prove It: A Structured Approach*. Cambridge University Press, 2nd edition, 2006.
- [Wee02] Jeffrey R. Weeks. *The Shape of Space*. CRC Press, 2nd edition, 2002.
- [Whi17] Jonathan J. White. The Peano axioms: An IBL unit constructing the natural numbers. *PRIMUS*, 27(7):725–735, 2017.
- [Wil10] Robin J. Wilson. *Introduction to Graph Theory*. Pearson Education Limited, Harlow, England, 5th edition, 2010.

- [Win04] Peter Winkler. *Mathematical Mind-Benders*. A K Peters, 2004.
- [Zei16] Paul Zeitz. *The Art and Craft of Problem Solving*. Wiley, 3rd edition, 2016.